

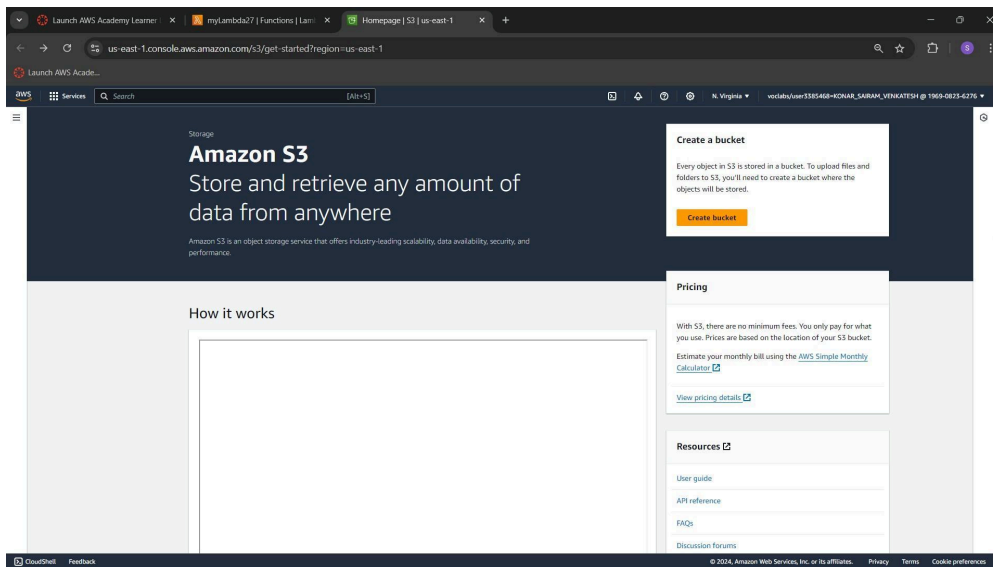
Aim: To create a Lambda function which will log “An Image has been added” once you add an object to a specific bucket in S3.

Prerequisites:

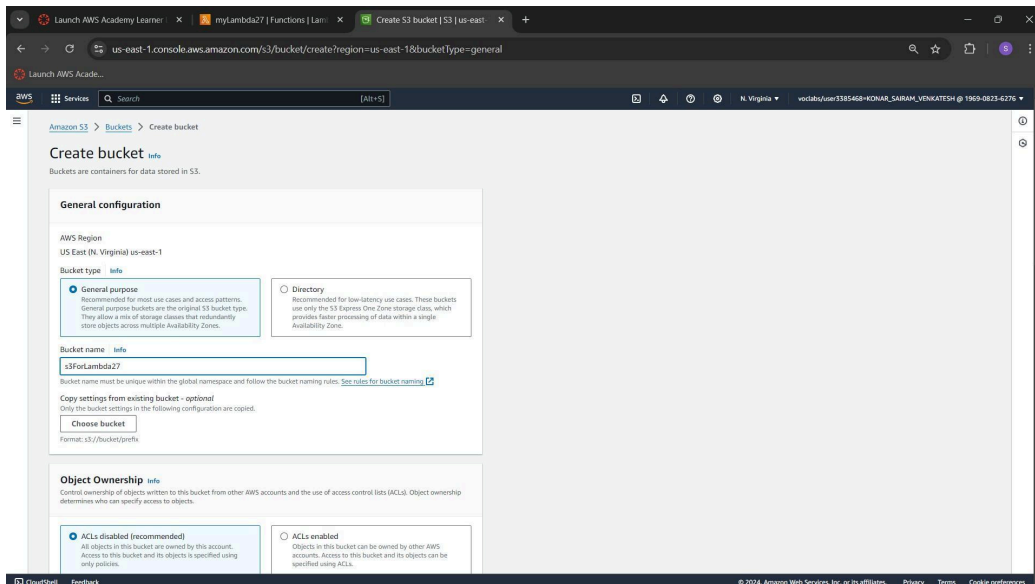
- 1) AWS account (academy preferable)
- 2) Lambda function (created in the previous experiment).

Step 1: Create a s3 bucket.

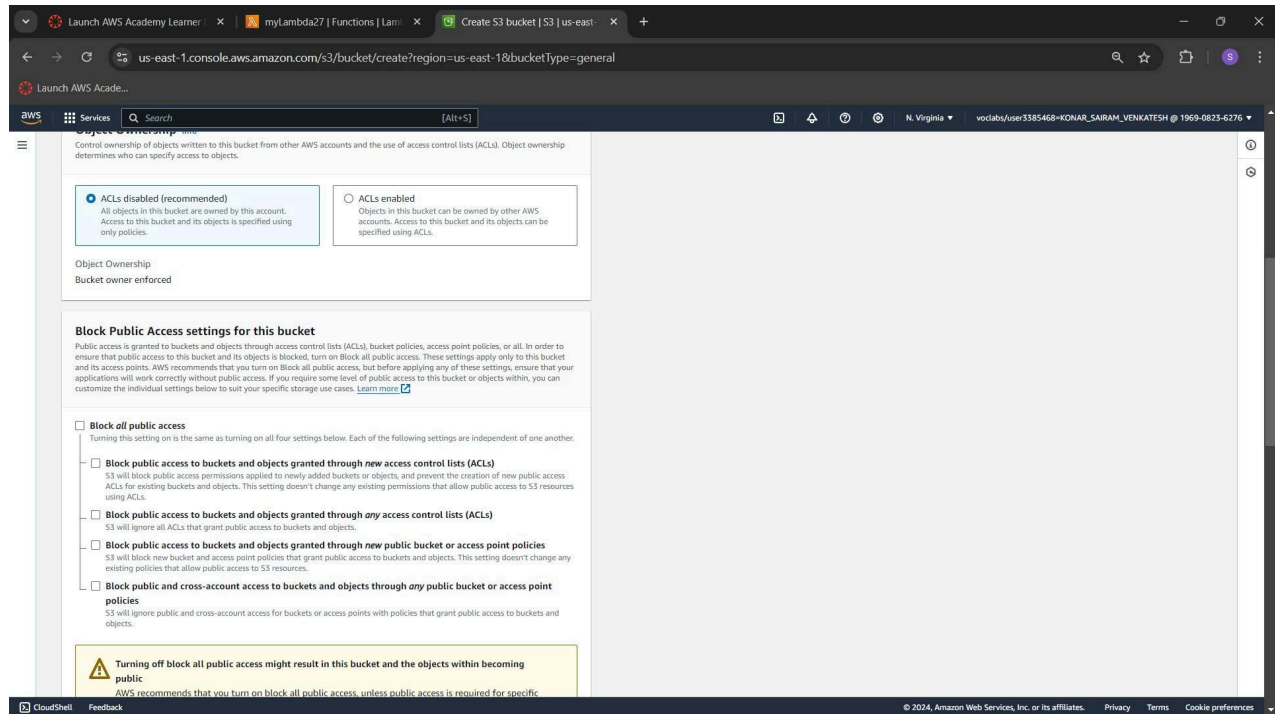
- 1) Search for S3 bucket in the services search. Then click on create bucket.



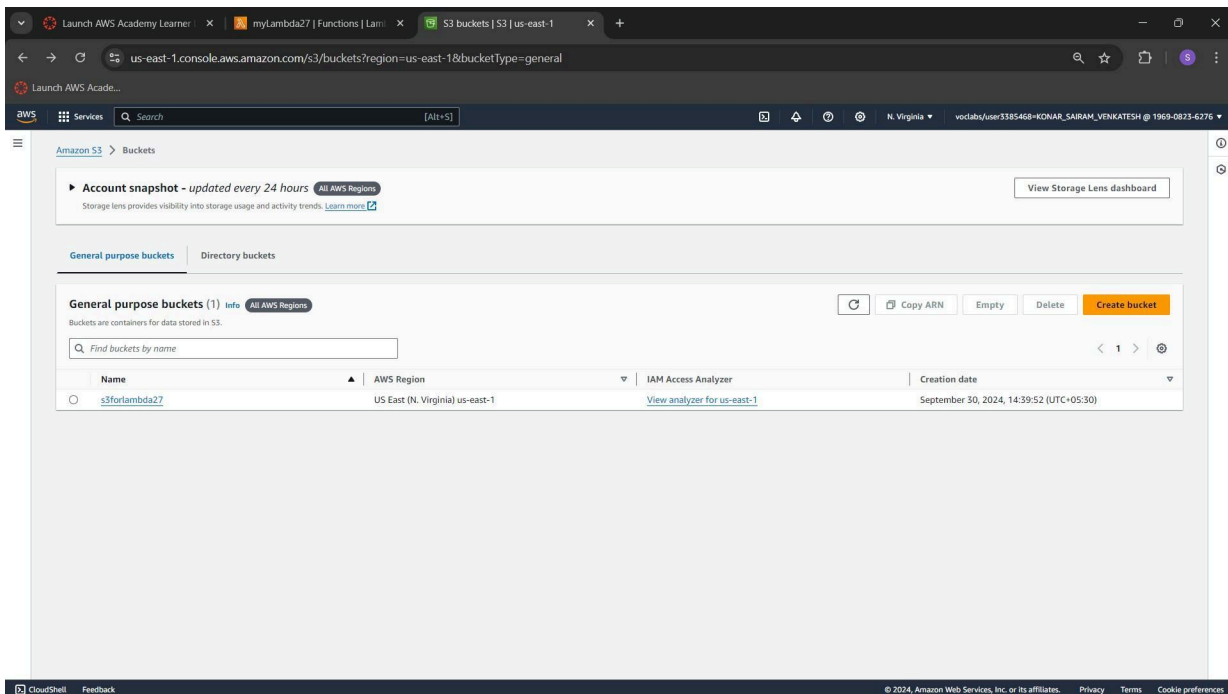
- 2) Keep the bucket as a general purpose bucket. Give a name to your bucket.



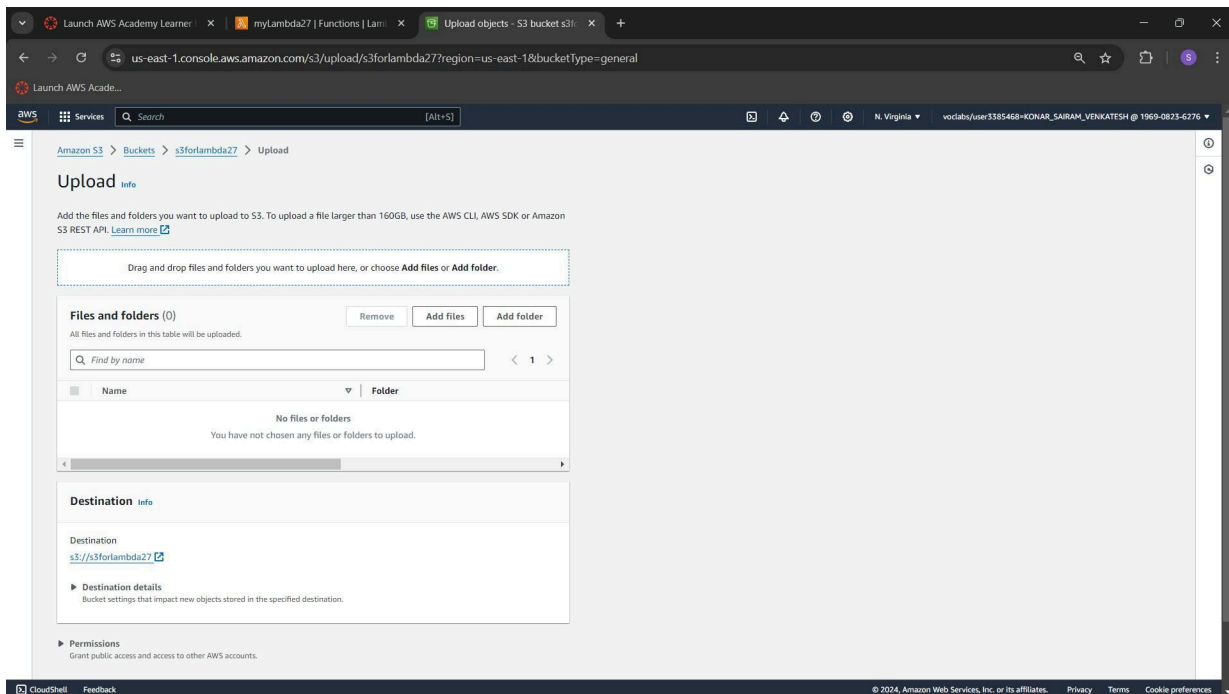
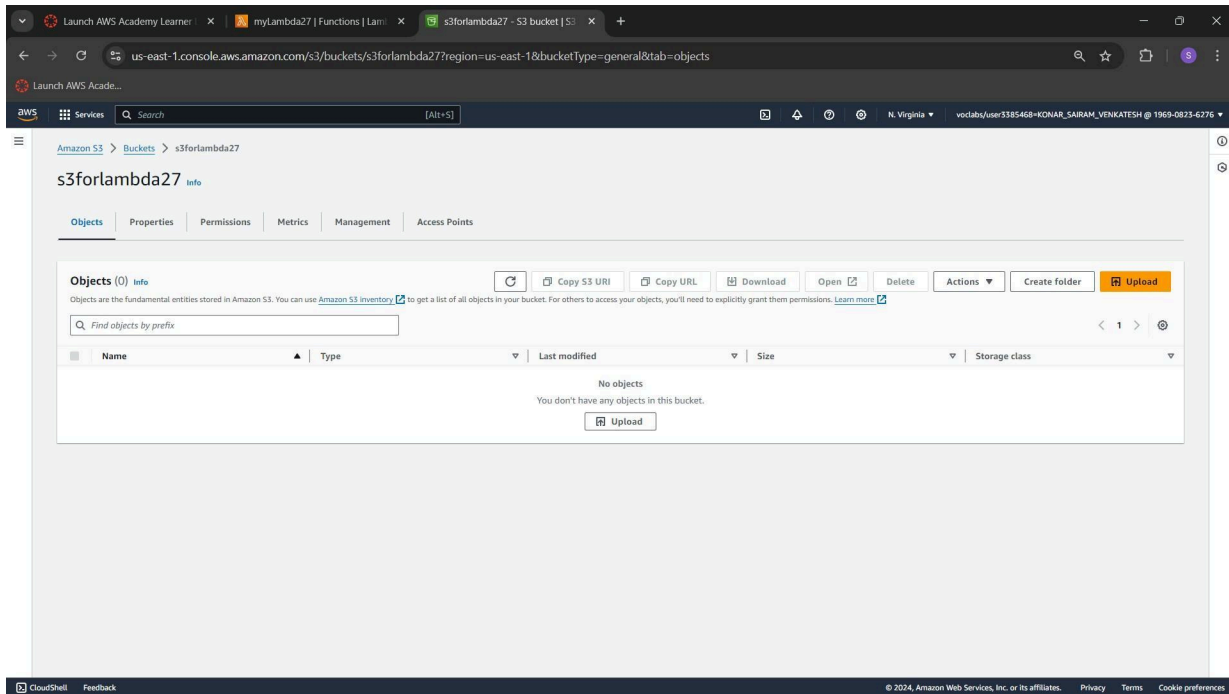
- 3) Uncheck block all public access.

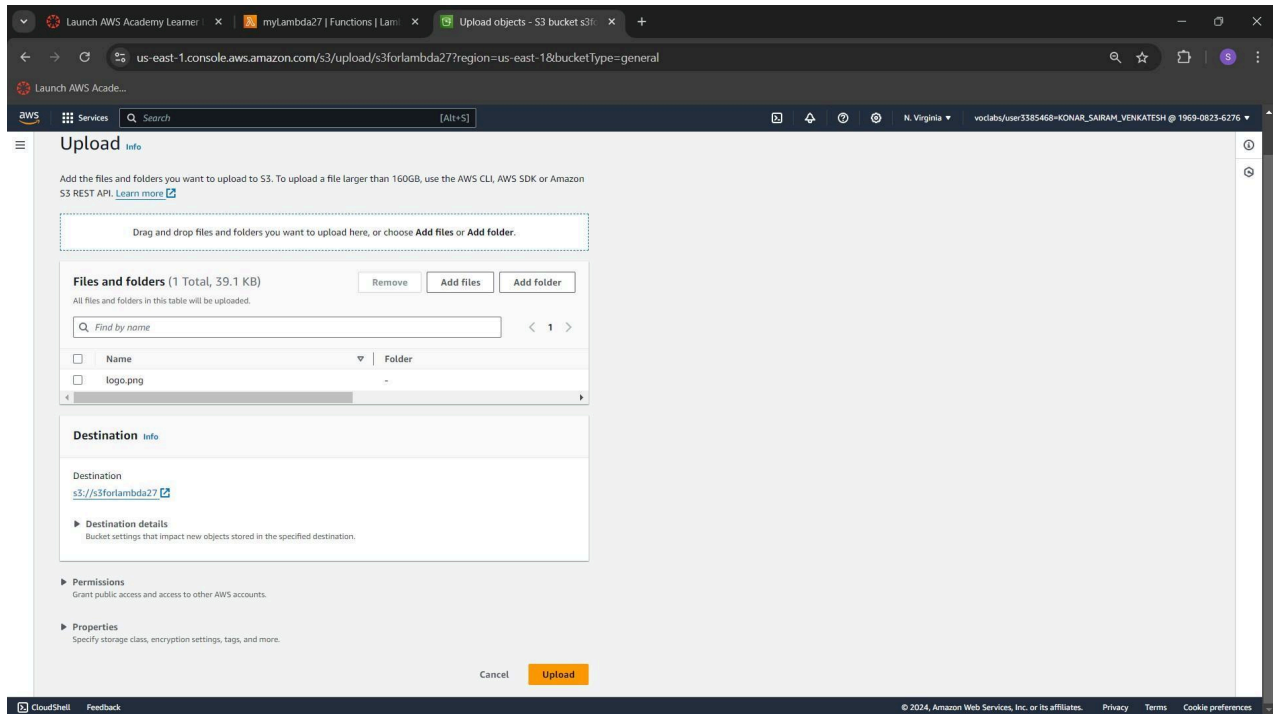


4) Keeping all other options same, click on create. This would create your bucket. Now click on the name of the bucket.

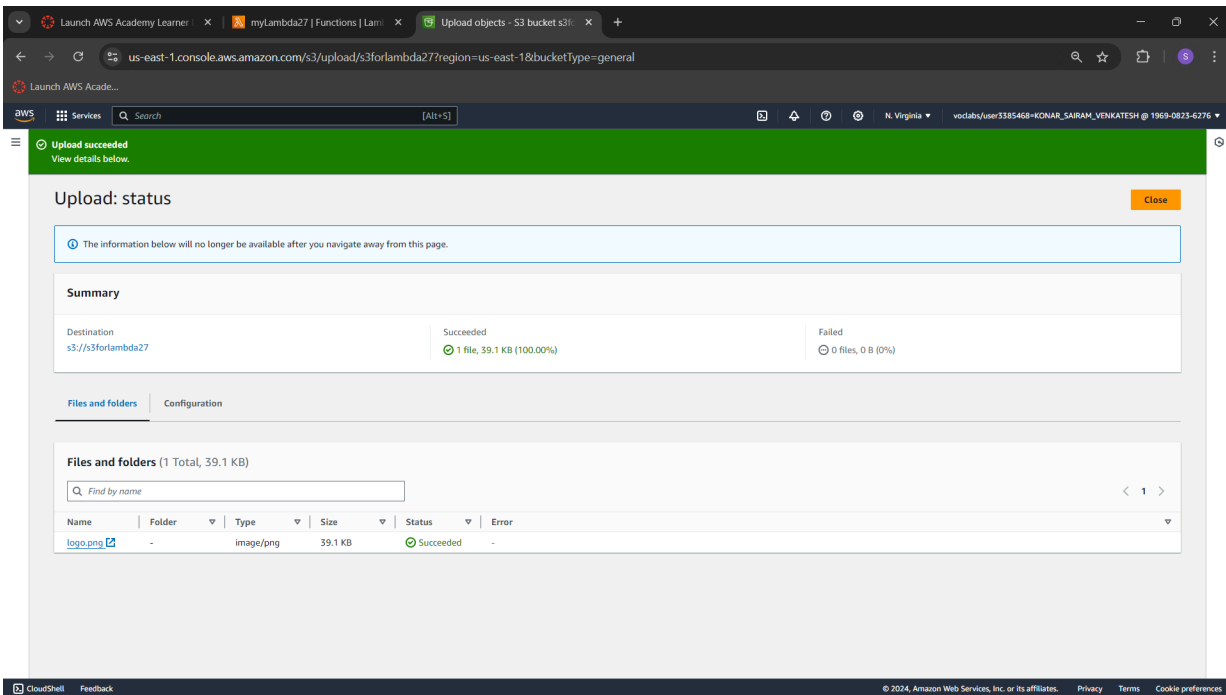


- 5) Here, click on upload, then add files. Select any image that you want to upload in the bucket and click on upload.



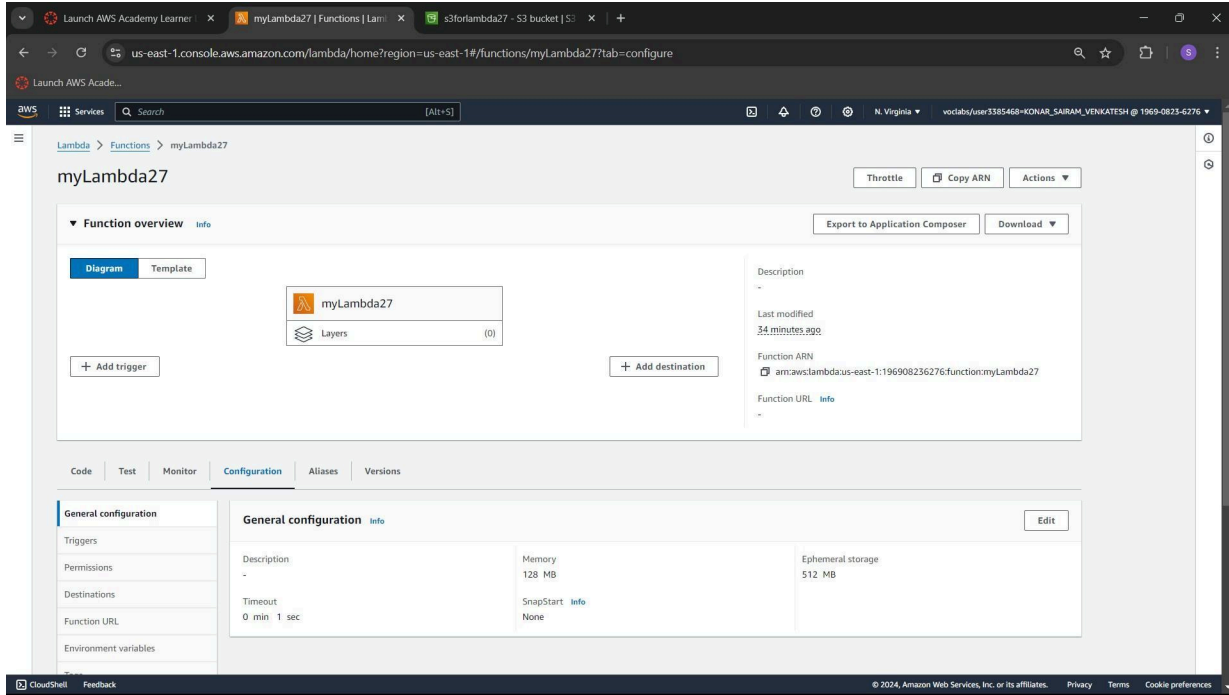


6) The image has been uploaded to the bucket.

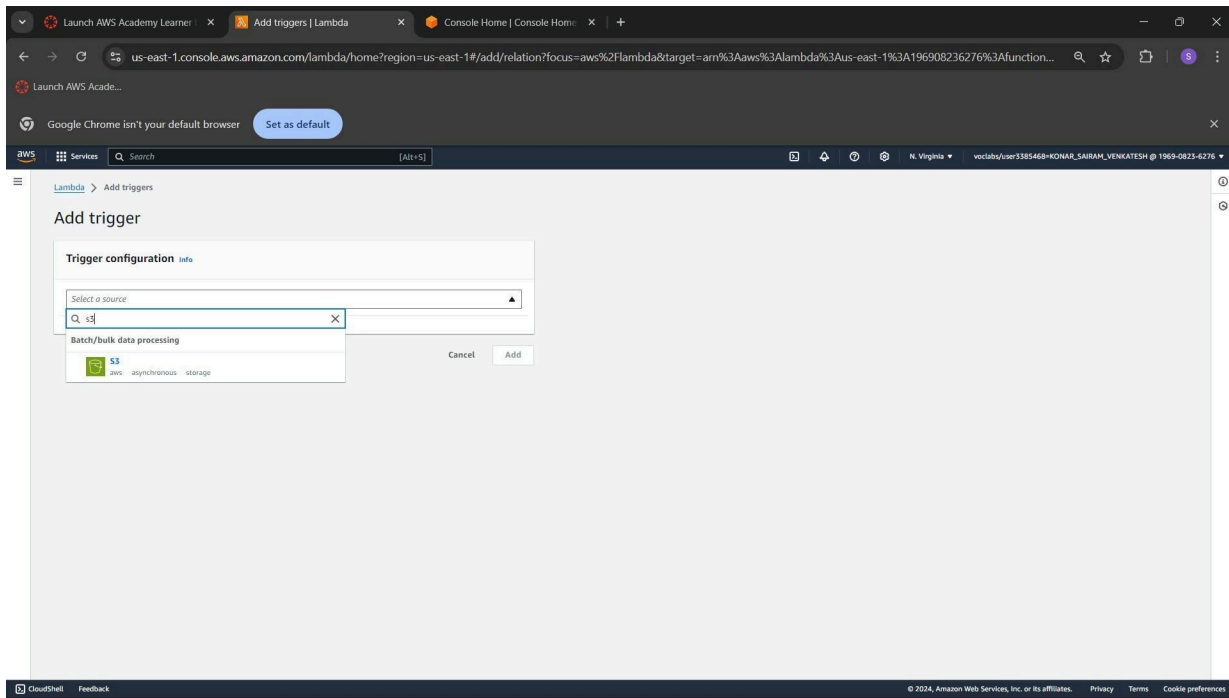


Step 2: Configure Lambda function

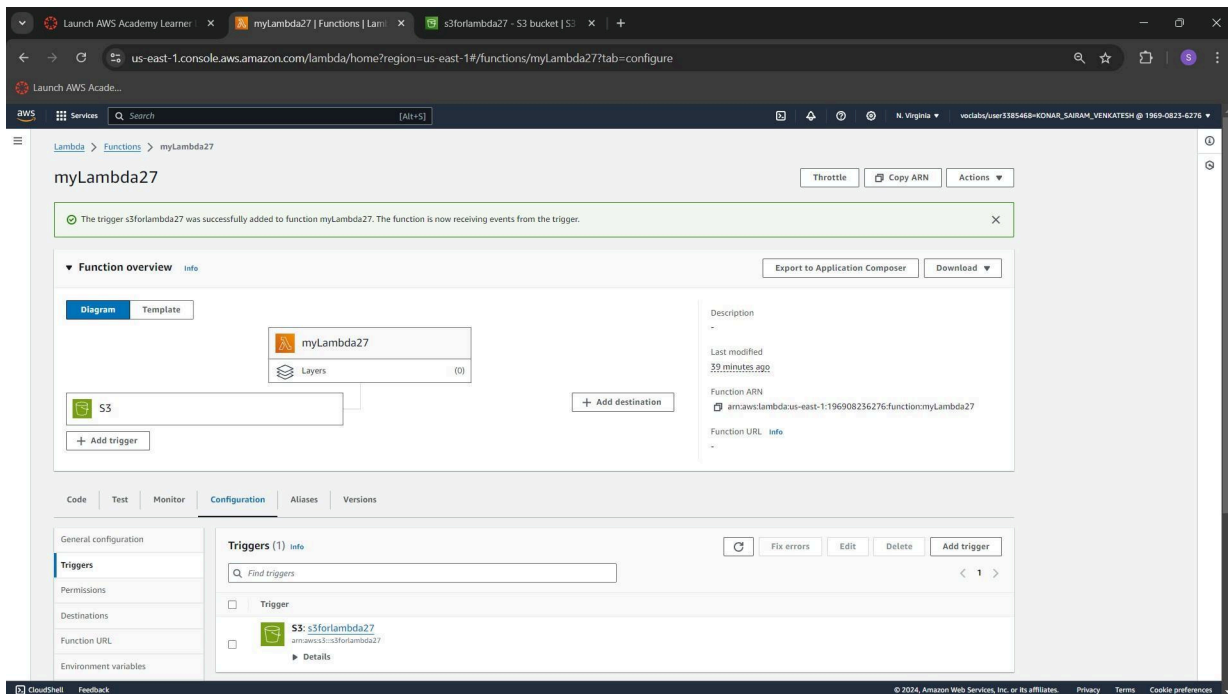
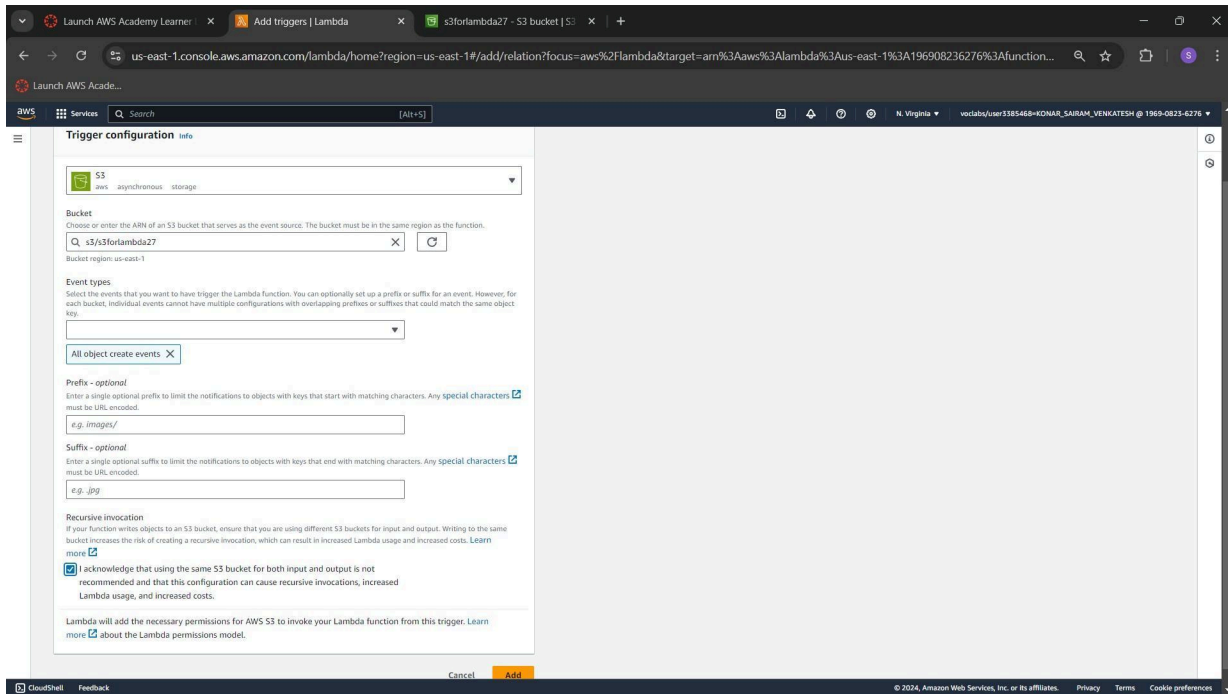
- 1) Go to the lambda function you had created berfor. (Services → Lambda → Click on name of function). Here, click on add trigger.



- 2) Under trigger configuration, search for S3 and select it.



- 3) Here, select the S3 bucket you created for this experiment. Acknowledge the condition given by AWS. then click on Add. This will add the S3 bucket trigger to your function.



- 4) Scroll down to the code section of the function. Add the following javascript code to the code area by replacing the existing code.

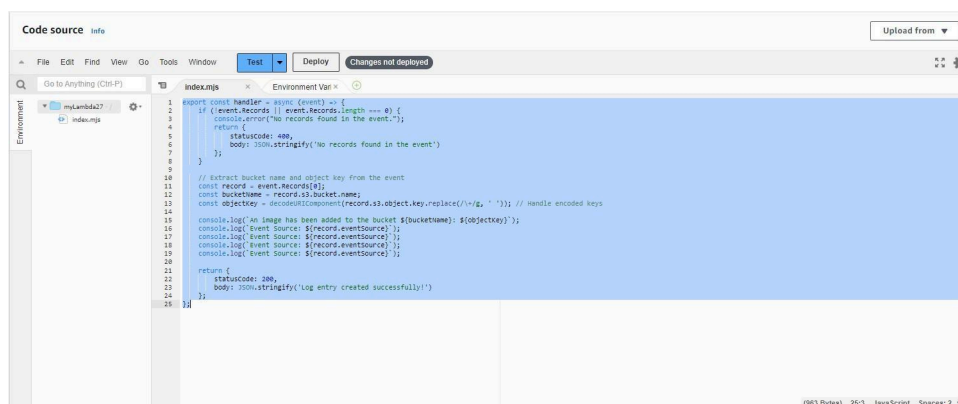
```
export const handler = async (event) => {
  if (!event.Records || event.Records.length === 0) {
    console.error("No records found in the event.");
    return {
      statusCode: 400,
      body: JSON.stringify('No records found in the event')
    };
  }

  // Extract bucket name and object key from the event
  const record = event.Records[0];
  const bucketName = record.s3.bucket.name;
  const objectKey = decodeURIComponent(record.s3.object.key.replace(/\+/g, ' ')); // Handle
  encoded keys

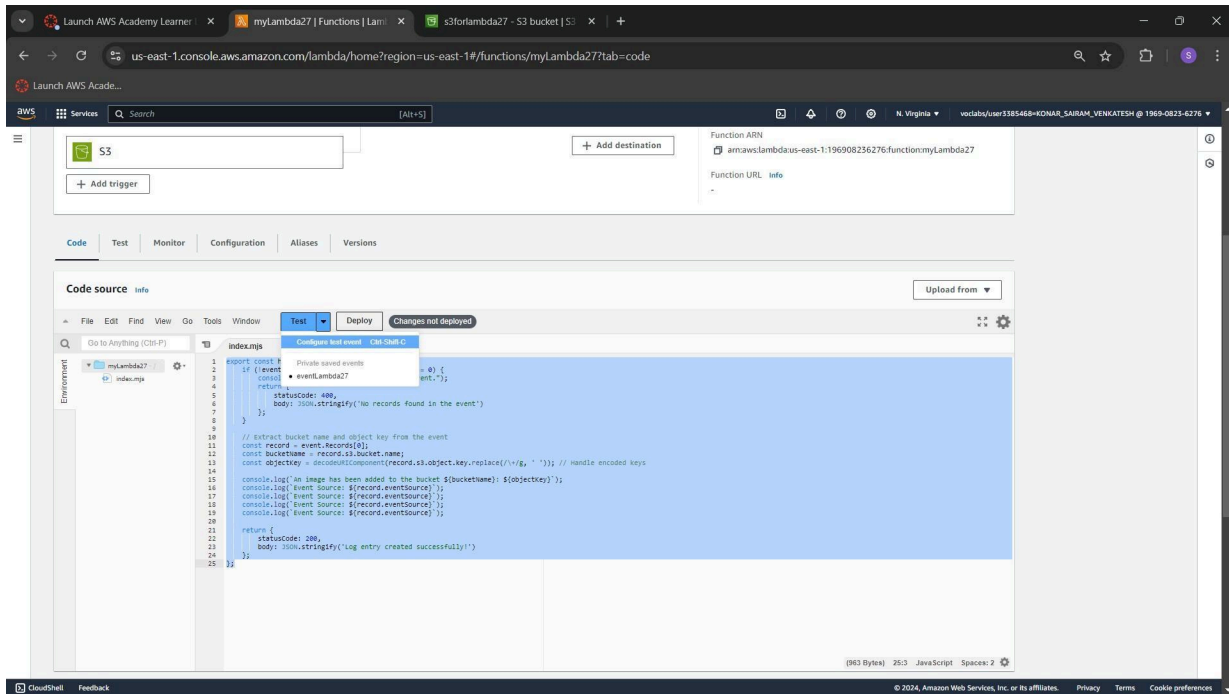
  console.log(`An image has been added to the bucket ${bucketName}: ${objectKey}`);
  console.log(`Event Source: ${record.eventSource}`);
  console.log(`Event Source: ${record.eventSource}`);
  console.log(`Event Source: ${record.eventSource}`);
  console.log(`Event Source: ${record.eventSource}`);

  return {
    statusCode: 200,
    body: JSON.stringify('Log entry created successfully!')
  };
};
```

This code checks for records in the event, extracts the bucket name and object key, logs the details, and returns a success message if an image is added to the bucket.



5) Now, click on the dropdown near test, then click on configure test event.



6) Here, select edit saved event. Select the event taht you had created before. Under Event JSON, paste the following code.

```
{
  "Records": [
    {
      "eventVersion": "2.0",
      "eventSource": "aws:s3",
      "awsRegion": "us-east-1",
      "eventTime": "1970-01-01T00:00:00.000Z",
      "eventName": "ObjectCreated:Put",
      "userIdentity": {
        "principalId": "EXAMPLE"
      },
      "requestParameters": {
        "sourceIPAddress": "127.0.0.1"
      },
      "responseElements": {
        "x-amz-request-id": "EXAMPLE123456789",
        "x-amz-id-2":
"EXAMPLE123/5678abcdefghijklmbdaisawesome/mnopqrstuvwxyzABCDEFGH"
      },
      "s3": {
        "s3SchemaVersion": "1.0",
        "configurationId": "testConfigRule",

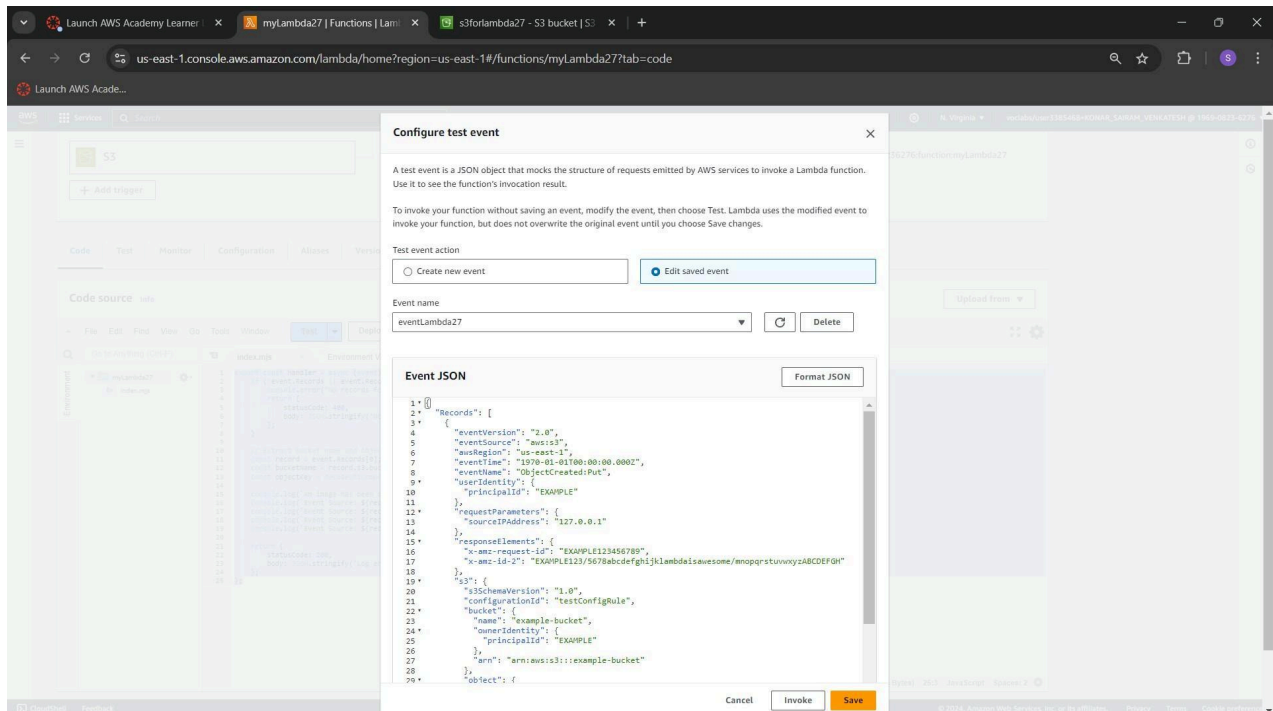
```

```

"bucket": {
  "name": "example-bucket",
  "ownerIdentity": {
    "principalId": "EXAMPLE"
  },
  "arn": "arn:aws:s3:::example-bucket"
},
"object": {
  "key": "test%2Fkey",
  "size": 1024,
  "eTag": "0123456789abcdef0123456789abcdef",
  "sequencer": "0A1B2C3D4E5F678901"
}
}
}
]
}

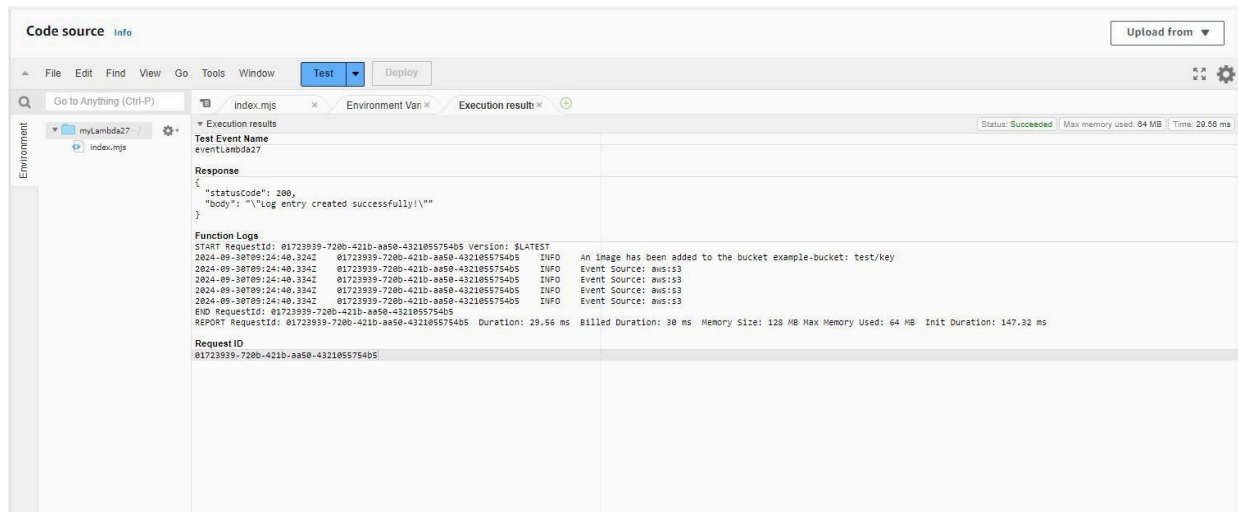
```

This JSON structure represents an S3 event notification triggered when an object is uploaded to an S3 bucket. It contains details about the event, including the bucket name (example-bucket), the object key (test/key), and metadata like the object's size, the event source (aws:s3), and the event time.



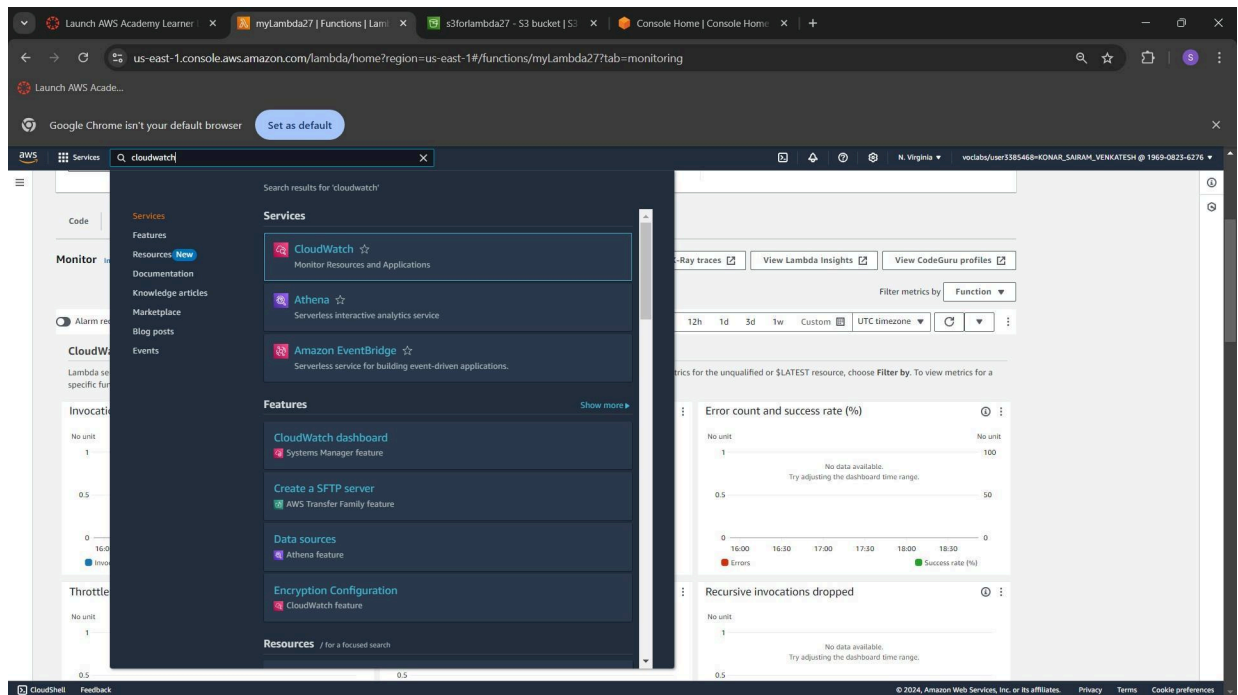
Save the changes. Then deploy the code changes by clicking on deploy.

- 7) After deploying, click on Test. The console output shows that 'an image has been added to the bucket'
The JSON response shows that the log entry was created successfully.

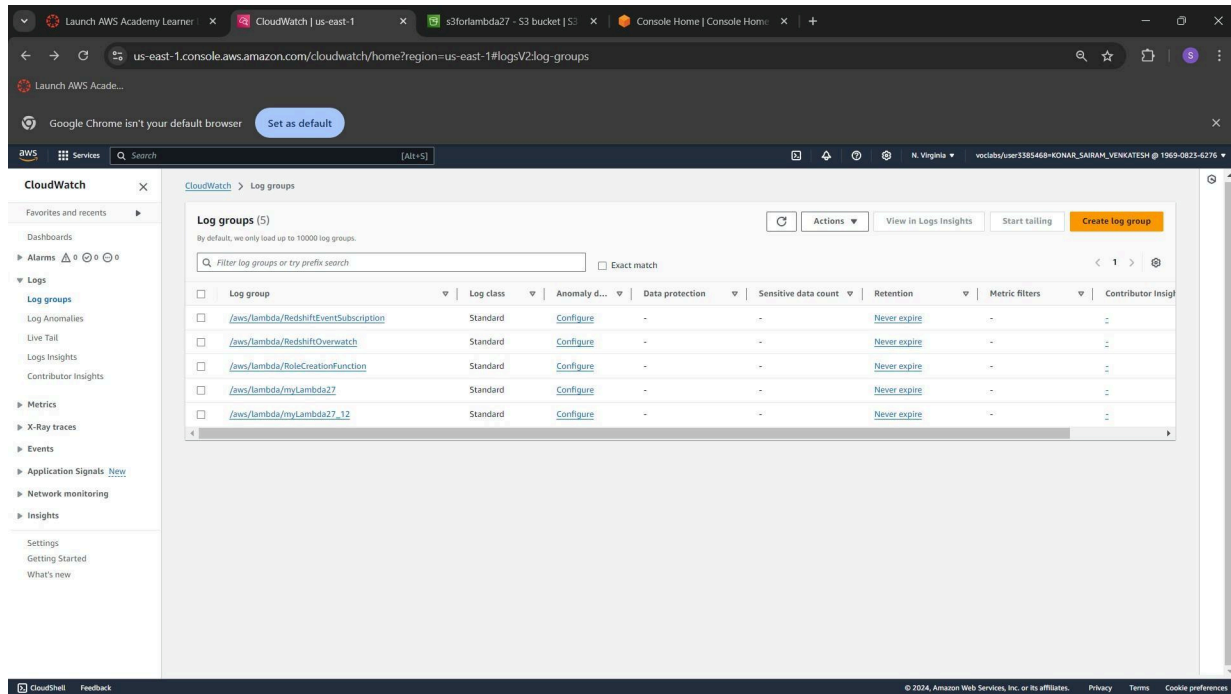


Step 3: Check the logs

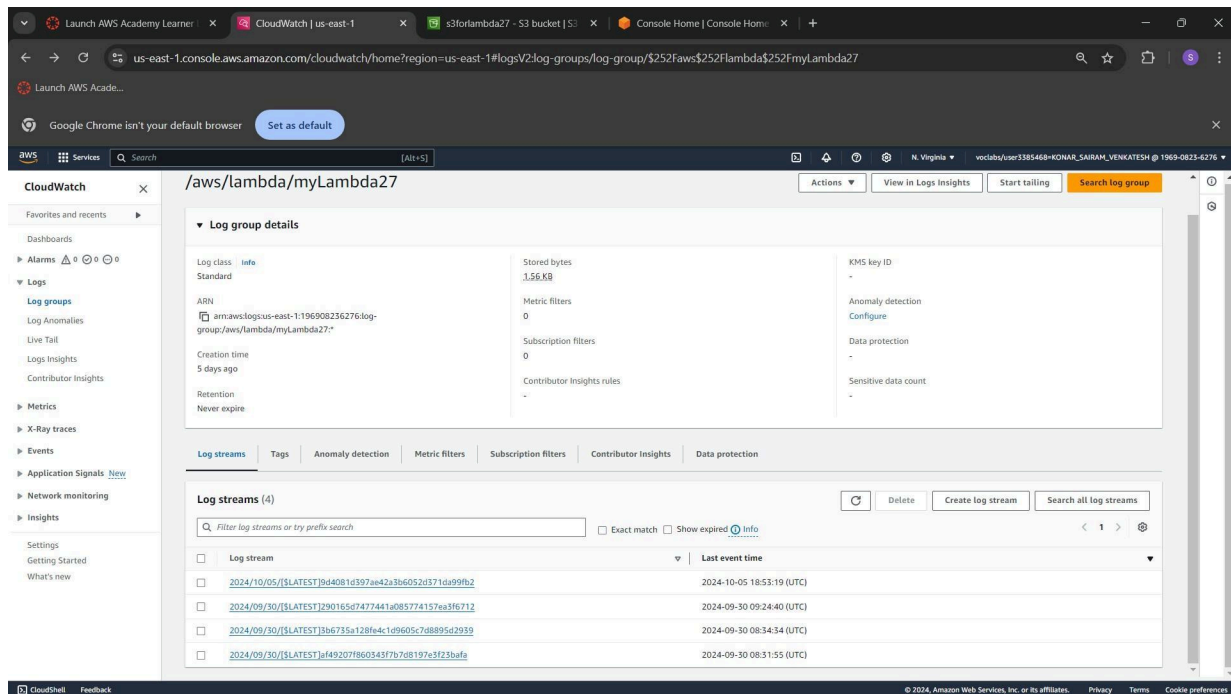
- 1) To check the logs explicitly, search for CloudWatch on services and open it in a new tab.



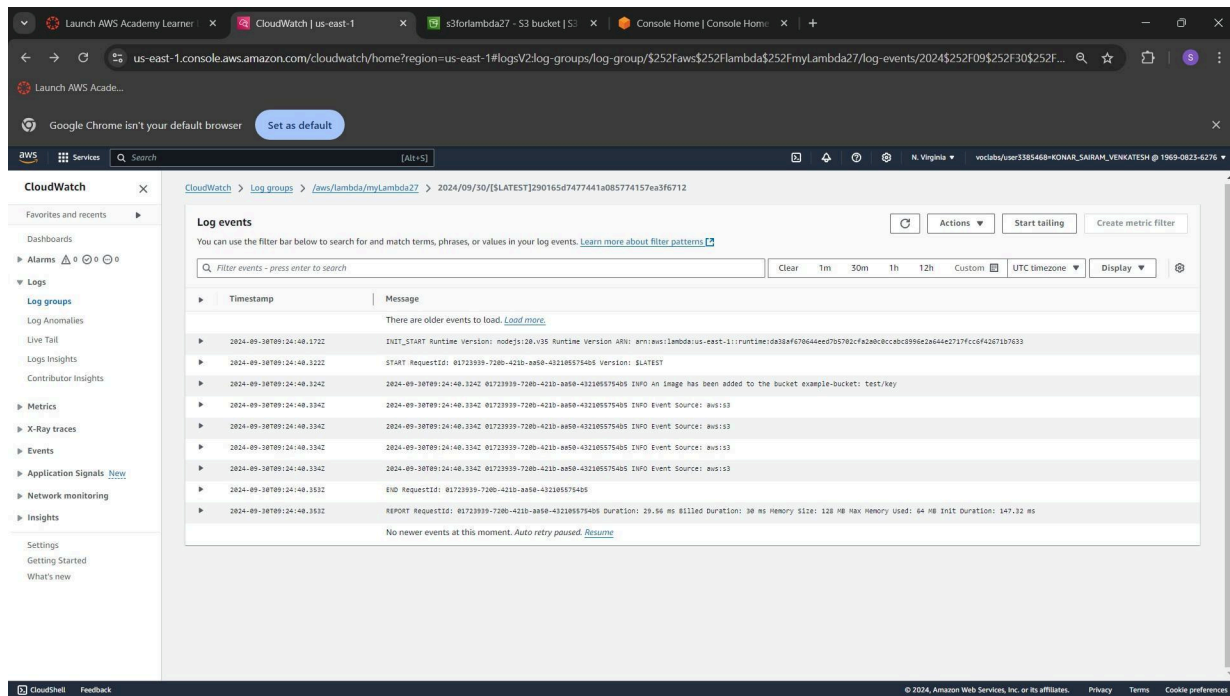
2) Here, Click on Logs → Log Groups. Select the log that has the lambda function name you just ran.



3) Here, under Log streams, select the log stream you want to check.



4) Here again, we can see that 'An image has been added to the bucket'.



Conclusion:

In this experiment, we successfully created a Lambda function that logs a message when an image is added to a specific S3 bucket. By configuring an S3 bucket trigger for the Lambda function, we demonstrated how AWS services can work together to automate tasks. The function logged important details about the event, including the bucket name and object key. After deploying the function and testing with a sample event, we verified the logs in CloudWatch, confirming that the Lambda function correctly detected and logged the addition of an image to the bucket. This experiment highlighted the seamless integration between AWS Lambda and S3 for event-driven processes.

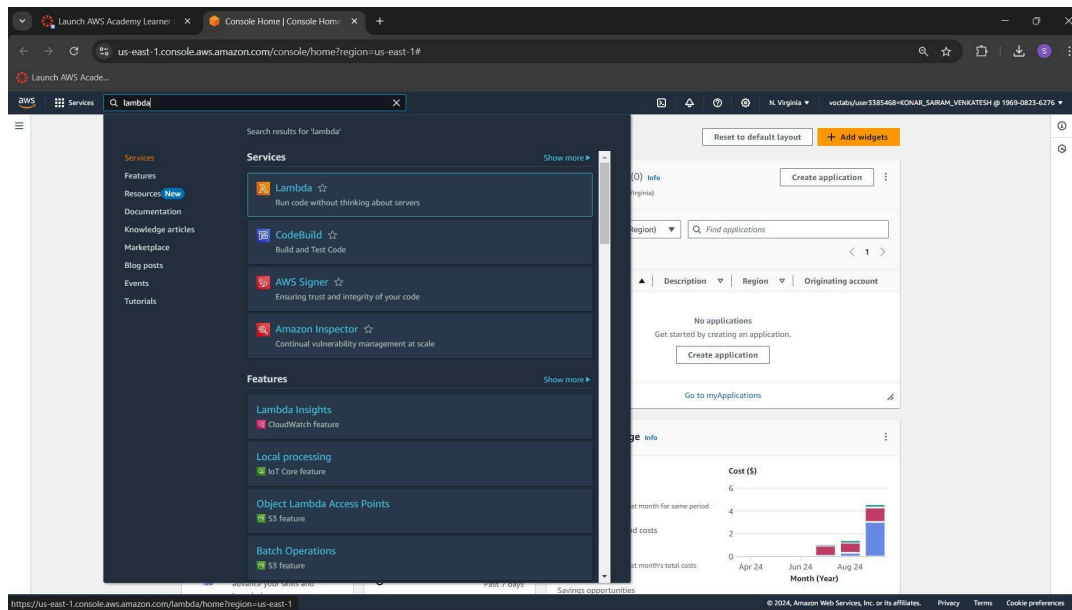
Aim: To understand AWS Lambda, its workflow, various functions and create your first Lambda functions using Python / Java / Nodejs.

Prerequisites:

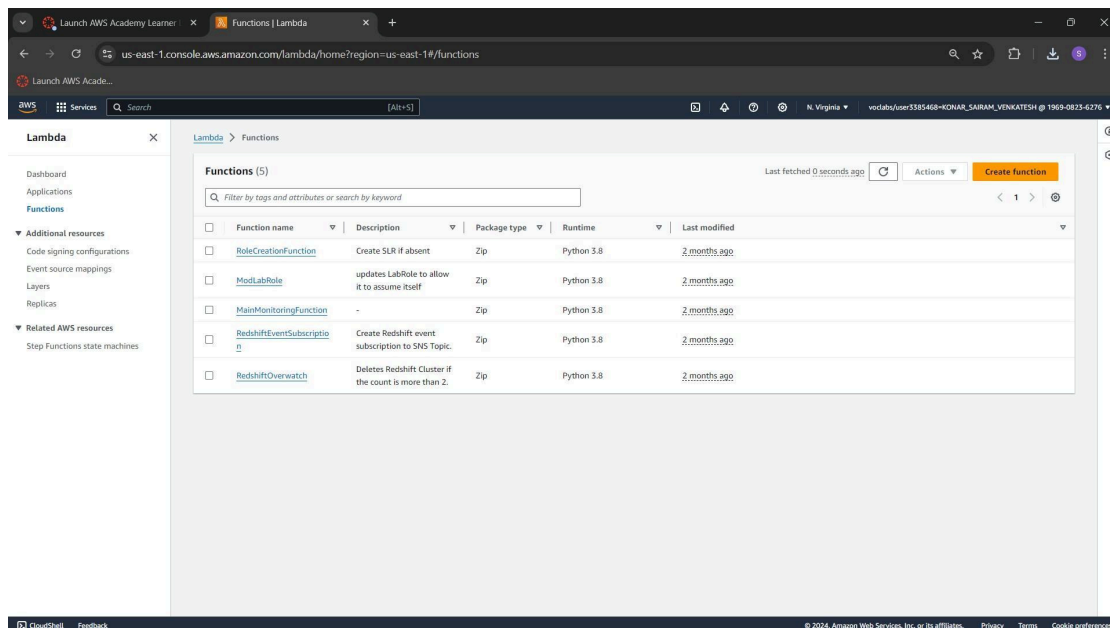
- 1) AWS account (academy recommended)

Step 1: Set up AWS Lambda Function

- 1) Search for Lambda in the services tab. Click on it once found.



- 2) Click on create functions.



- 3) Give a name to your Lambda function. Select the runtime as Node.js 20.x (You can also use python). Select the architecture as x86_64. Set the default execution role as LabRole if you are doing this on your academy account. (Use an existing role → LabRole)

The screenshot shows the AWS Lambda 'Create function' console page. The 'Basic information' section is visible, showing the function name 'myLambda27', runtime 'Node.js 20.x', architecture 'x86_64', and permissions 'LabRole'. The 'Execution role' section is also visible, showing the role 'LabRole'.

Basic information

Function name
Enter a name that describes the purpose of your function.
myLambda27
Use only letters, numbers, hyphens, or underscores with no spaces.

Runtime Info
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.
Node.js 20.x

Architecture Info
Choose the instruction set architecture you want for your function code.
☒ x86_64
☐ arm64

Permissions Info
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

▼ Change default execution role

Execution role
Choose a role that defines the permissions of your function. To create a custom role, go to the IAM console.

☐ Create a new role with basic Lambda permissions
☒ Use an existing role
☐ Create a new role from AWS policy templates

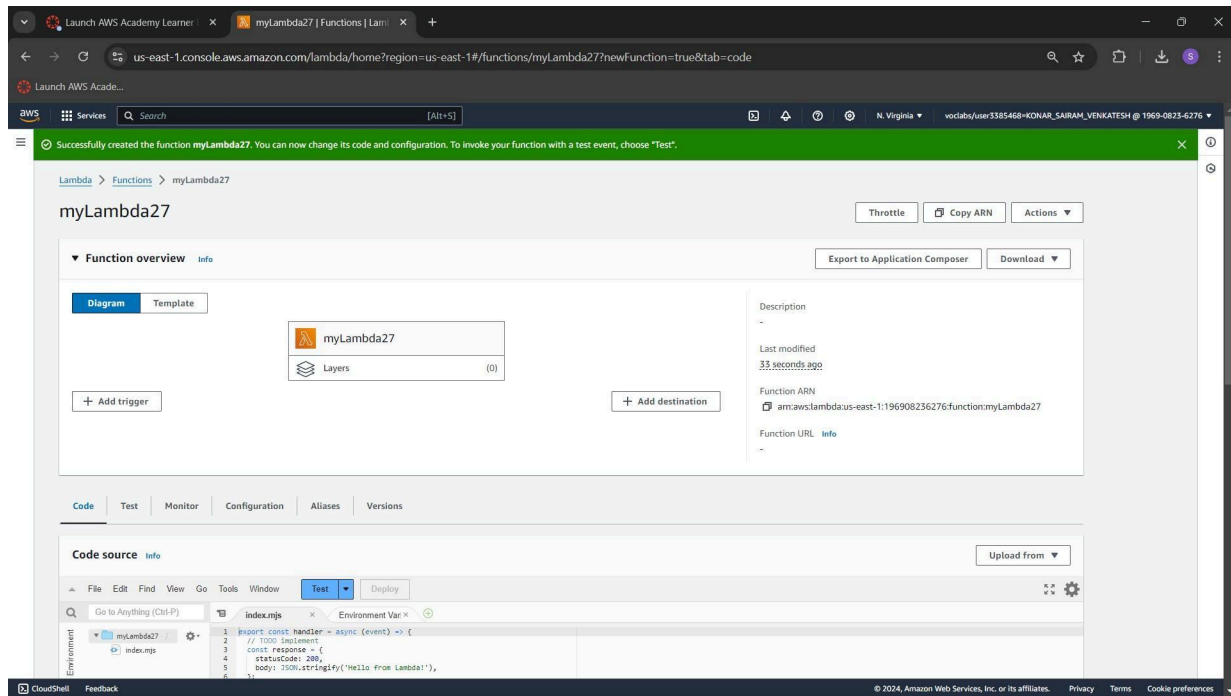
Existing role
Choose an existing role that you've created to be used with this Lambda function. The role must have permission to upload logs to Amazon CloudWatch Logs.
LabRole
View the LabRole role on the IAM console.

- 4) Once the function is created, click on the name of the function (myLambda27 in my case).

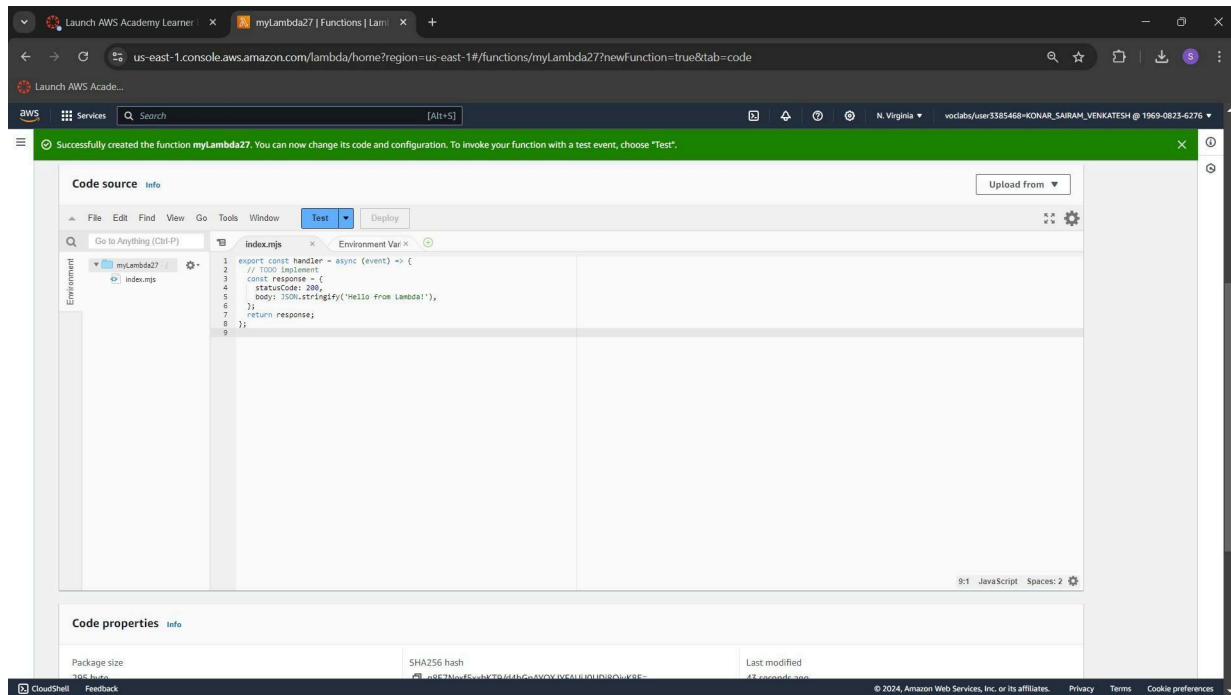
The screenshot shows the AWS Lambda 'Functions' console page. A table lists the functions, with 'myLambda27' highlighted. The table has columns for Function name, Description, Package type, Runtime, and Last modified.

Function name	Description	Package type	Runtime	Last modified
myLambda27	-	Zip	Node.js 20.x	5 days ago
RoleCreationFunction	Create SLR if absent	Zip	Python 3.8	2 months ago
ModLabRole	updates LabRole to allow it to assume itself	Zip	Python 3.8	2 months ago
myLambda27_12	-	Zip	Python 3.12	5 days ago
MainMonitoringFunction	-	Zip	Python 3.8	2 months ago
RedshiftEventSubscription	Create Redshift event subscription to SNS Topic.	Zip	Python 3.8	2 months ago
RedshiftOverwatch	Deletes Redshift Cluster if the count is more than 2.	Zip	Python 3.8	2 months ago

5) This is the dashboard of our lambda function.

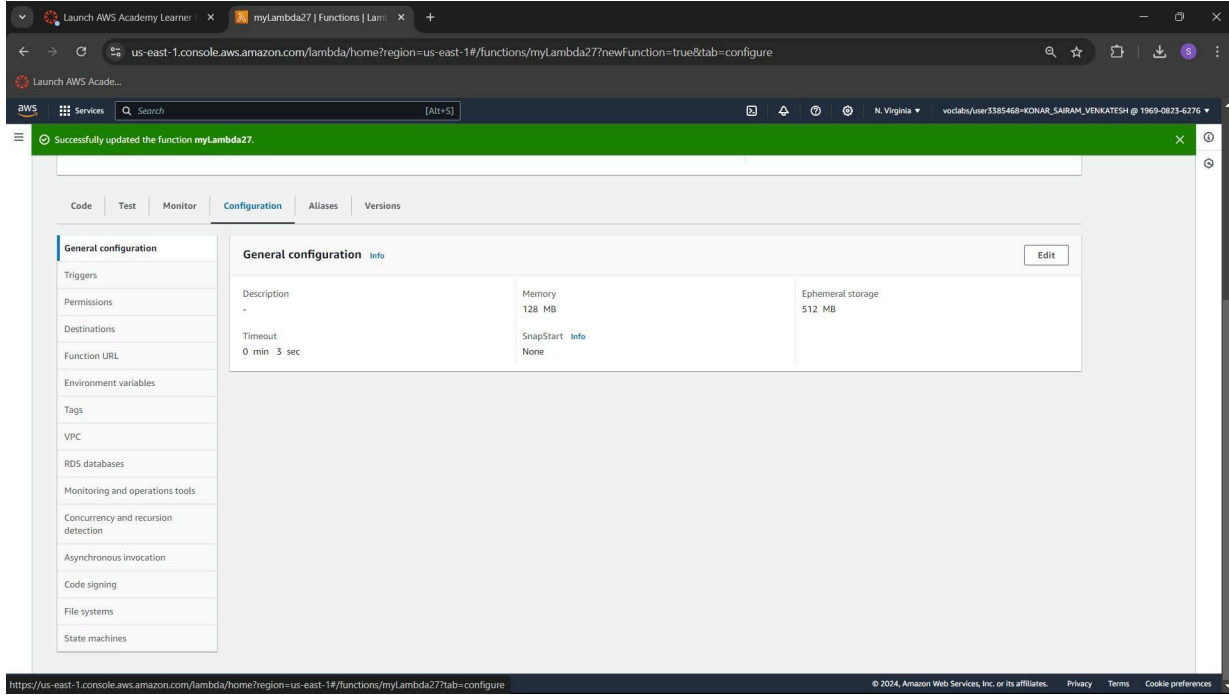


6) This function has the following default code, which is used to print "Hello form Lambda!".

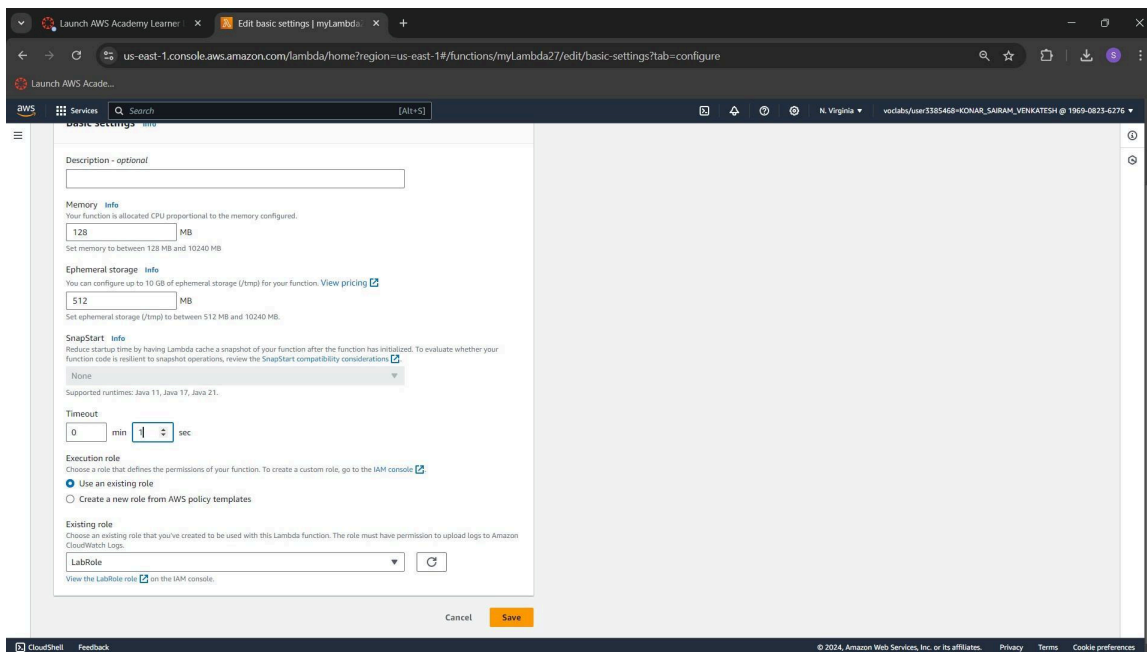


Step 3: Set up configurations and test events

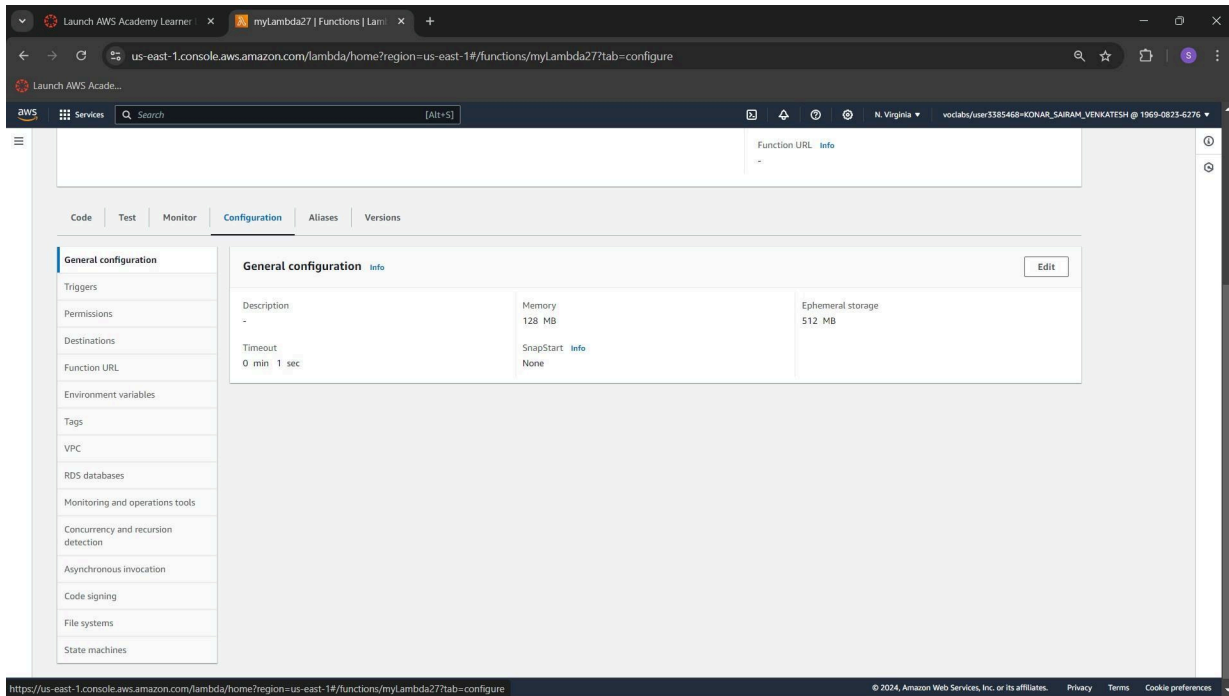
1) Just above the test code, you would find Configuration, click on it. Then click on Edit.



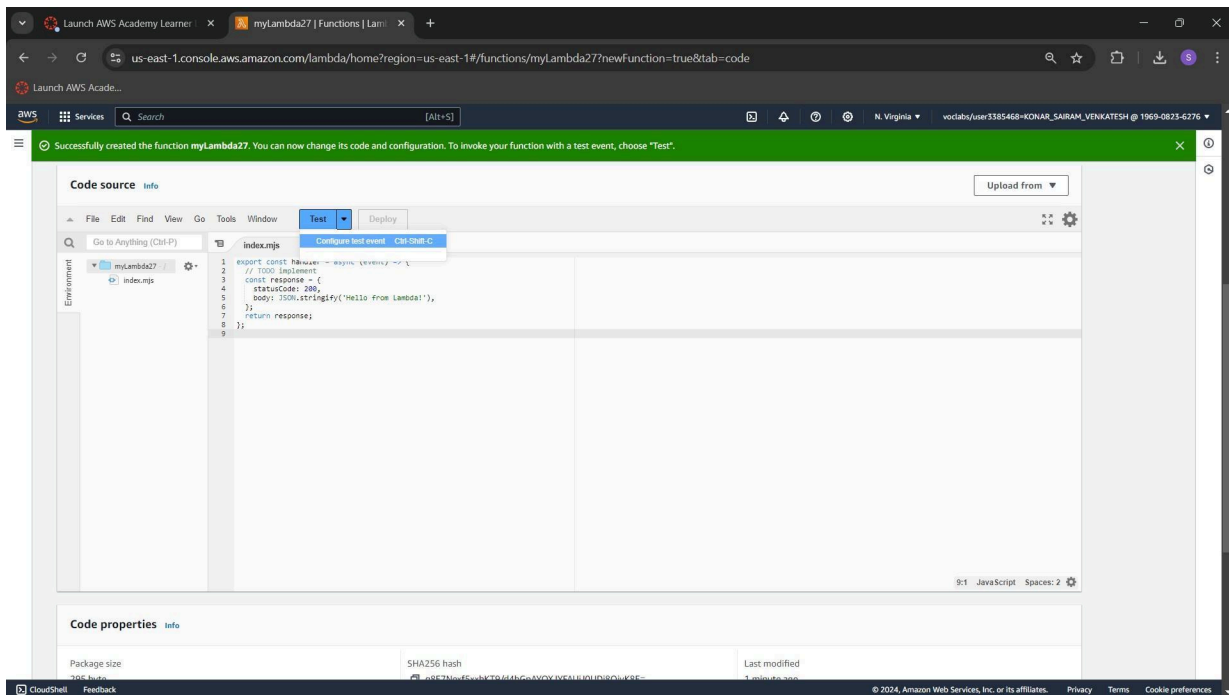
2) Here, change the Timeout to 1 sec. This is the time for which the function can be running before it is forcibly terminated.



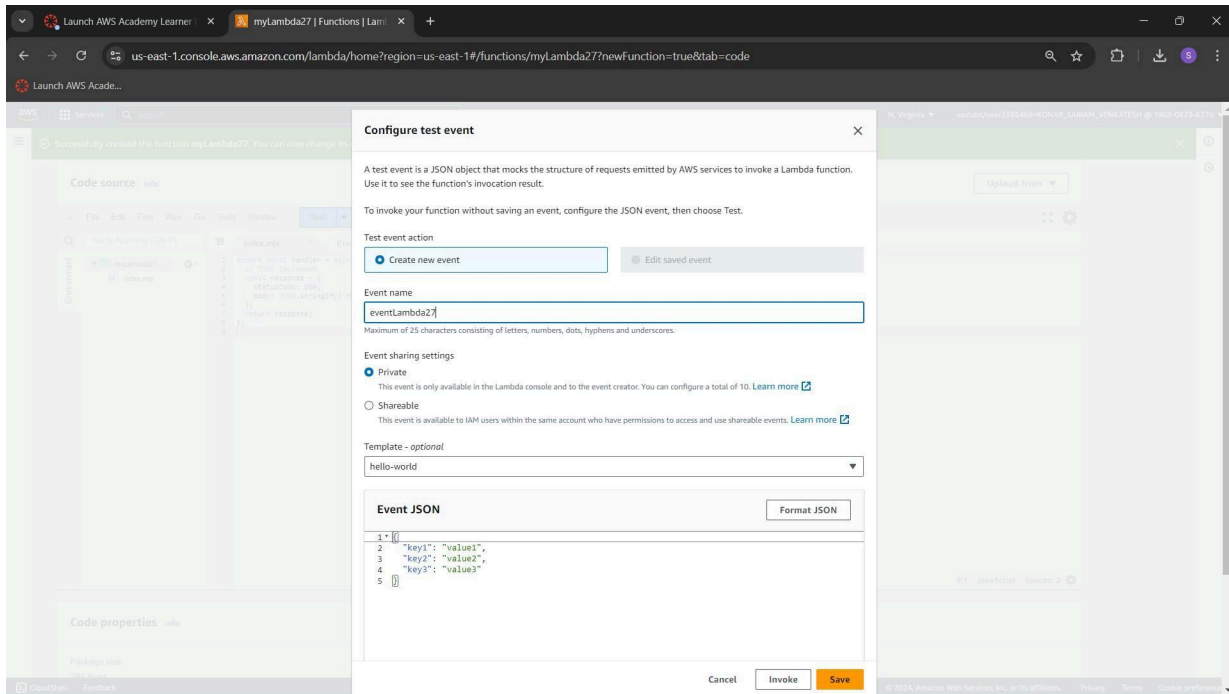
3) We can see the executed changes.



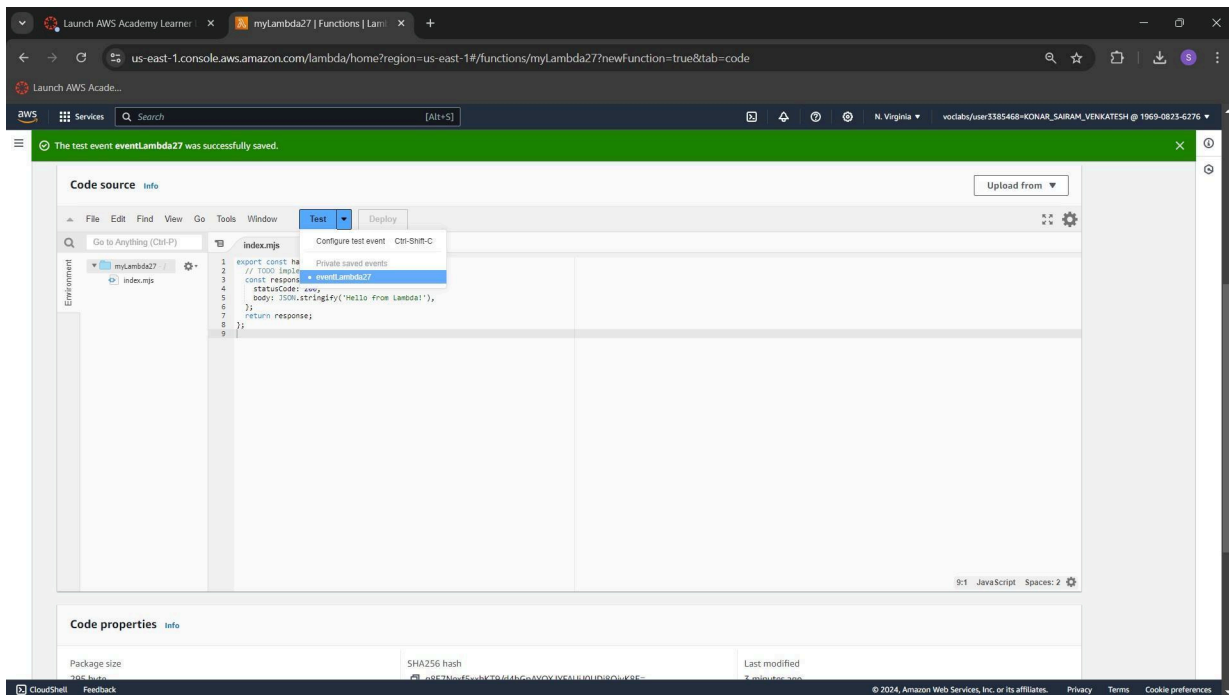
4) Switch back to the code tab. Click on the dropdown arrow near test. Then select configure test event.



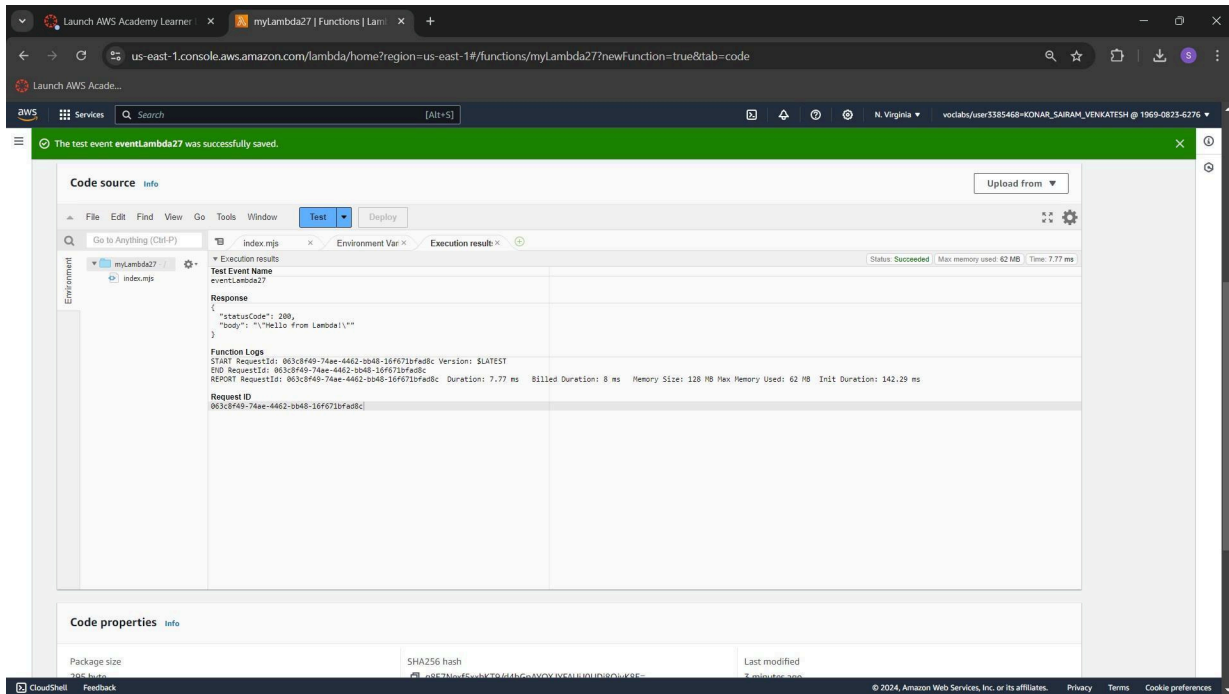
5) Here, create a new event, keep the other options default and save the event.



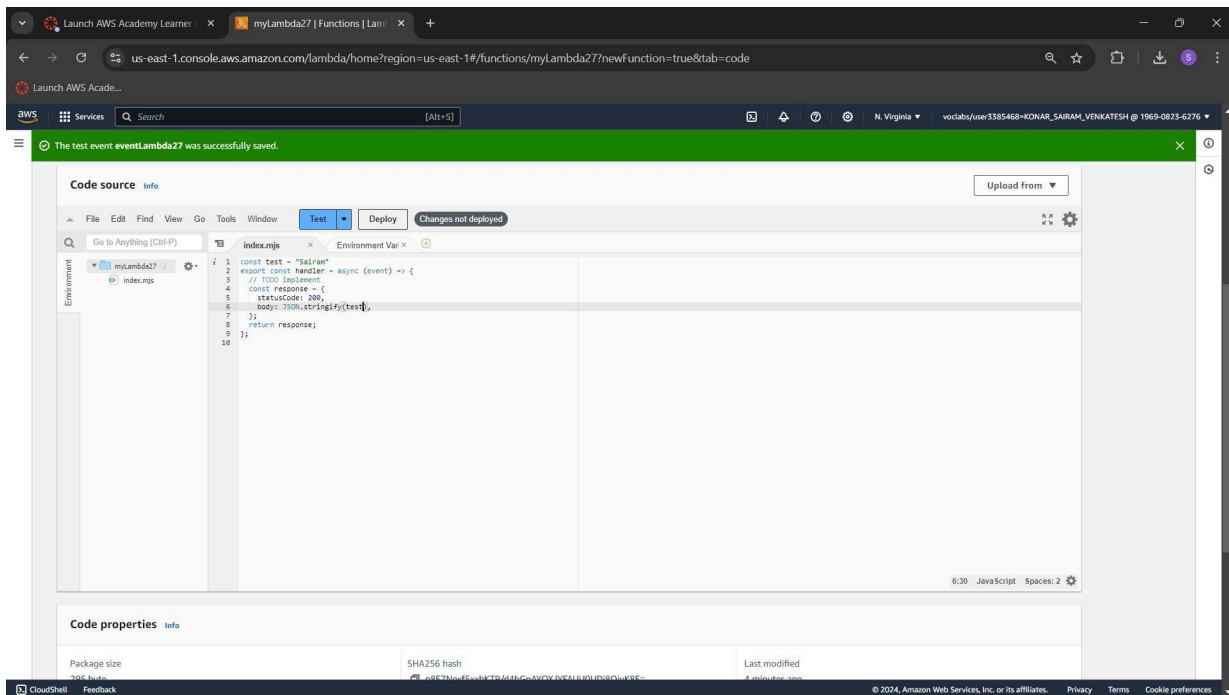
6) Now, again click on the dropdown. This time, select the event you have created. Then, click on TEST.



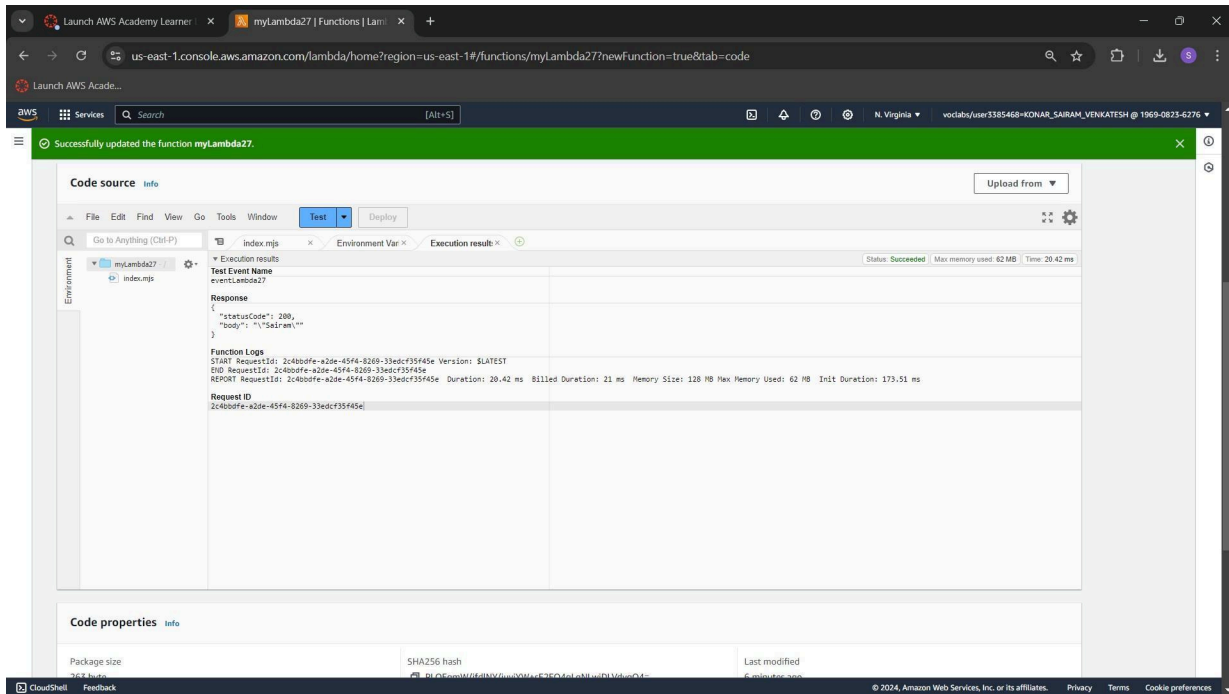
7) We can see the expected output for the sample code.



8) For a test, declare a string and call it in line 6. After making the changes click on deplopy.



9) Run the test. We can see that the string we declared has come in the output.



Conclusion:

In this experiment, we successfully explored the AWS Lambda service by creating and configuring Lambda functions using Python, Java, or Node.js. We learned how to set up a Lambda function, modify its configuration (such as adjusting the timeout), and test the function with custom events. Through this process, we observed how Lambda handles executions, including managing timeouts and returning expected outputs based on the code changes.

Aim: To perform Port, Service monitoring, Windows/Linux server monitoring using Nagios.

Prerequisites:

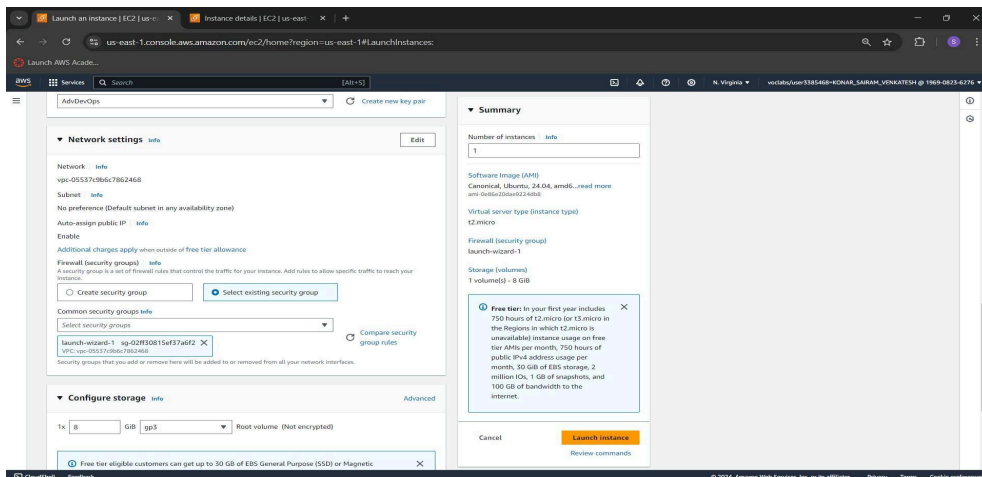
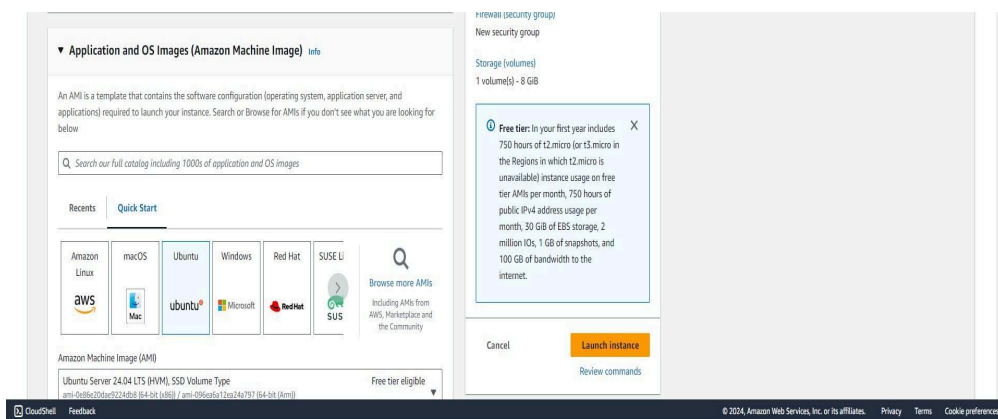
- 1) An Amazon Linux instance with nagios already set up.

Step 1: Set up ubuntu instance

- 1) Login to your AWS account. Search for EC2 on services. Open the interface and click on Create Instance.

Select The OS Image as Ubuntu.

- 2) Make sure to select the same private key that you created for the Amazon Linux instance. Also select the same security group as you created for the Linux instance.



- 3) Now come back to the instances screen. Click on the instance ID of your instance. Then click on Connect. Click on SSH client. Copy the example command. Now, we have to connect our local OS terminal to the instance using SSH. For this, open terminal where the private key file is located (.pem). Paste the copied SSH command and run it.

Step 2: Execute the following on Nagios Host machine (Linux)

- 1) We need to verify whether the nagios service is running or not. For that, run this command.

ps -ef | grep nagios

```
[ec2-user@ip-172-31-83-157 ~]$ ps -ef | grep nagios
nagios    66054      1    0 04:18 ?        00:00:00 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
nagios    66055    66054    0 04:18 ?        00:00:00 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
nagios    66056    66054    0 04:18 ?        00:00:00 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
nagios    66057    66054    0 04:18 ?        00:00:00 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
nagios    66058    66054    0 04:18 ?        00:00:00 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
nagios    66059    66054    0 04:18 ?        00:00:00 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
ec2-user  66758    66657    0 04:29 pts/0    00:00:00 grep --color=auto nagios
[ec2-user@ip-172-31-83-157 ~]$
```

- 2) Now, make yourself as the root user, and create a folder with the path '/usr/local/nagios/etc/objects/monitorhosts/linuxhosts'

sudo su

mkdir -p /usr/local/nagios/etc/objects/monitorhosts/linuxhosts

```
[ec2-user@ip-172-31-83-157 ~]$ sudo su
mkdir /usr/local/nagios/etc/objects/monitorhosts
mkdir /usr/local/nagios/etc/objects/monitorhosts/linuxhosts
[root@ip-172-31-83-157 ec2-user]#
```

- 3) We need to create a config file in this folder. So, copy the contents of the existing localhost config to the new file 'linuxserver.cfg'.

cp /usr/local/nagios/etc/objects/localhost.cfg

/usr/local/nagios/etc/objects/monitorhosts/linuxhosts/linuxserver.cfg

- 4) We need to make some changes in this config file. Open it using nano editor. nano

/usr/local/nagios/etc/objects/monitorhosts/linuxhosts/linuxserver.cfg

Change **hostname** and **alias** to **linuxserver**

Change address to **public ip address of client instance** (Ubuntu instance)

```
# Define a host for the local machine

define host{
    use                linux-server      ; Name of host template
                                ; This host definition
                                ; inherits from the template
    host_name          linuxserver
    alias              linuxserver
    address            32.226.136.73
}
```

Change `hostgroup_name` to **linux-servers1**

```
define hostgroup{
    hostgroup_name    linux-servers1 ; The name of the hostgroup
    alias            Linux Servers ; Long name of the group
    members          localhost      ; Comma separated list of hosts that
}
```

Change the **occurrences of hostname** further in the document from **localhost** to **linuxserver**

5) Now, we need to edit the nagios configuration file to add this directory.

nano /usr/local/nagios/etc/nagios.cfg

Run this command and add the following line

cfg_dir=/usr/local/nagios/etc/objects/monitorhosts/

```
root@ip-172-31-83-159:/home  x  +  v
GNU nano 5.8 /usr/local/nagios/etc/nagios.cfg

# Definitions for monitoring the local (Linux) host
cfg_file=/usr/local/nagios/etc/objects/localhost.cfg

# Definitions for monitoring a Windows machine
#cfg_file=/usr/local/nagios/etc/objects/windows.cfg

# Definitions for monitoring a router/switch
#cfg_file=/usr/local/nagios/etc/objects/switch.cfg

# Definitions for monitoring a network printer
#cfg_file=/usr/local/nagios/etc/objects/printer.cfg

# You can also tell Nagios to process all config files (with a .cfg
# extension) in a particular directory by using the cfg_dir
# directive as shown below:

#cfg_dir=/usr/local/nagios/etc/servers
#cfg_dir=/usr/local/nagios/etc/printers
#cfg_dir=/usr/local/nagios/etc/switches
#cfg_dir=/usr/local/nagios/etc/routers
cfg_dir=/usr/local/nagios/etc/objects/monitorhosts/

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File ^\ Replace   ^J Paste     ^_ Justify  ^/ Go To Line M-E Redo     M-C Copy
```

6) Now we verify the configuration files.

/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg

```
[root@ip-172-31-83-157 ec2-user]# /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg

Nagios Core 4.5.5
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2024-09-17
License: GPL

Website: https://www.nagios.org
Reading configuration data...
  Read main config file okay...
  Read object config files okay...

Running pre-flight check on configuration data...

Checking objects...
  Checked 16 services.
  Checked 2 hosts.
  Checked 2 host groups.
  Checked 0 service groups.
  Checked 1 contacts.
  Checked 1 contact groups.
  Checked 24 commands.
  Checked 5 time periods.
  Checked 0 host escalations.
  Checked 0 service escalations.
Checking for circular paths...
  Checked 2 hosts
  Checked 0 service dependencies
  Checked 0 host dependencies
  Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

Total Warnings: 0
Total Errors: 0

Things look okay - No serious problems were detected during the pre-flight check
[root@ip-172-31-83-157 ec2-user]#
```

7) Once the files are verified, we need to restart the server.

service nagios restart

```
[root@ip-172-31-83-159 nagios-plugins-2.0.3]# service nagios restart
Restarting nagios (via systemctl): [ OK ]
[root@ip-172-31-83-159 nagios-plugins-2.0.3]#
```

Step 3: Execute the following on Nagios Client machine (Ubuntu)

- 1) First, we check for any new updates, then we install gcc, nagios nrpe server and nagios plugins.

sudo apt update -y

sudo apt install gcc -y

sudo apt install -y nagios-nrpe-server nagios-plugins

```
ubuntu@ip-172-31-81-89:~$ sudo apt update -y
sudo apt install gcc -y
sudo apt install -y nagios-nrpe-server nagios-plugins
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [380 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
```

Running kernel seems to be up-to-date.

Restarting services...

```
Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service
```

No containers need to be restarted.

```
User sessions running outdated binaries:
ubuntu @ session #4: sshd[1495,1569]
ubuntu @ user manager service: systemd[1500]
```

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```
ubuntu@ip-172-31-81-89:~$ |
```

- 2) We need to add the public IP address of our host Nagios machine (Linux) to the nrpe configuration file.

sudo nano /etc/nagios/nrpe.cfg

Under `allowed_hosts`, add the nagios host ip address (public)

```
GNU nano 7.2 /etc/nagios/nrpe.cfg *
# You can either supply a username or a UID.
#
# NOTE: This option is ignored if NRPE is running under either inetd or xinetd

nrpe_user=nagios

# NRPE GROUP
# This determines the effective group that the NRPE daemon should run as.
# You can either supply a group name or a GID.
#
# NOTE: This option is ignored if NRPE is running under either inetd or xinetd

nrpe_group=nagios

# ALLOWED HOST ADDRESSES
# This is an optional comma-delimited list of IP address or hostnames
# that are allowed to talk to the NRPE daemon. Network addresses with a bit mask
# (i.e. 192.168.1.0/24) are also supported. Hostname wildcards are not currently
# supported.
#
# Note: The daemon only does rudimentary checking of the client's IP
# address. I would highly recommend adding entries in your /etc/hosts.allow
# file to allow only the specified host to connect to the port
# you are running this daemon on.
#
# NOTE: This option is ignored if NRPE is running under either inetd or xinetd

allowed_hosts=127.0.0.1,54.210.81.106

# COMMAND ARGUMENT PROCESSING
# This option determines whether or not the NRPE daemon will allow clients
# to specify arguments to commands that are executed. This option only works
# if the daemon was configured with the --enable-command-args configure script
# option.

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^J Execute   ^C Location  ^M-U Undo     ^M-A Set Mark ^M-J To Bracket
^X Exit      ^R Read File  ^L Replace    ^U Paste      ^_ Justify   ^_/ Go To Line ^M-E Redo     ^M-G Copy     ^M-Q Where Was
```

Step 4: Check the Nagios Dashboard

- 1) Go to Nagios dashboard, click on hosts.

Here, we can see that the linuxserver is also added as a host.

Nagios
Current Network Status
Last Updated: Sat Sep 28 04:42:59 UTC 2024
Updated every 30 seconds
Nagios Core™ 4.5.5 - www.nagios.org
Logged in as nagiosadmin

Host Status Totals
Up: 2, Down: 0, Unreachable: 0, Pending: 0
All Problems: 0, All Types: 2

Service Status Totals
Ok: 8, Warning: 1, Unknown: 0, Critical: 2, Pending: 5
All Problems: 3, All Types: 16

Host Status Details For All Host Groups

Host	Status	Last Check	Duration	Status Information
linuxserver	UP	09-28-2024 04:42:16	0d 0h 2m 35s	PING OK - Packet loss = 0%, RTA = 1.15 ms
localhost	UP	09-28-2024 04:38:21	0d 0h 24m 0s	PING OK - Packet loss = 0%, RTA = 0.03 ms

Results 1 - 2 of 2 Matching Hosts

Page View

2) Click on linuxserver. Here, we can check all the information about linuxserver host.

The screenshot displays the Nagios web interface for the host 'linuxserver (linuxserver)'. The interface includes a sidebar with navigation links such as General, Current Status, Tactical Overview, Hosts, Services, Host Groups, Service Groups, Problems, Reports, and System. The main content area shows the following information:

- Host Information:** Last updated: Sat Sep 28 04:37:37 UTC 2024. Nagios Core 4.5.5 - www.nagios.org. Logged in as nagiosadmin.
- Host State Information:**
 - Host Status: UP (for 0d 0h 3m 13s)
 - Status Information: PING OK - Packet loss = 0%, RTA = 1.15 ms
 - Performance Data: rta=1.151000ms;3000.000000;5000.000000 pl=0%;80;100;0
 - Current Attempt: 1/10 (HARD state)
 - Last Check Time: 09-28-2024 04:42:16
 - Check Type: ACTIVE
 - Check Latency / Duration: 0.000 / 4.033 seconds
 - Next Scheduled Active Check: 09-28-2024 04:47:16
 - Last State Change: 09-28-2024 04:40:24
 - Last Notification: N/A (notification 0)
 - Is This Host Flapping? NO (0.00% state change)
 - In Scheduled Downtime? NO
 - Last Update: 09-28-2024 04:43:33 (0d 0h 0m 4s ago)
- Host Commands:** A list of commands with checkboxes, including 'Locate host on map', 'Disable active checks of this host', 'Re-schedule the next check of this host', 'Submit passive check result for this host', 'Stop accepting passive checks for this host', 'Stop obsessing over this host', 'Disable notifications for this host', 'Send custom host notification', 'Schedule downtime for this host', 'Schedule downtime for all services on this host', 'Enable notifications for all services on this host', 'Schedule a check of all services on this host', 'Disable checks of all services on this host', 'Enable checks of all services on this host', 'Disable event handler for this host', 'Disable flap detection for this host', and 'Clear flapping state for this host'.
- Host Comments:** A section for adding or deleting comments. It shows 'This host has no comments associated with it'.

3) Click on services. Here we can see all the services that are being monitored by linuxserver.

The screenshot displays the Nagios web interface for the 'Services' section. The interface includes a sidebar with navigation links such as General, Current Status, Tactical Overview, Hosts, Services, Host Groups, Service Groups, Problems, Reports, and System. The main content area shows the following information:

- Current Network Status:** Last updated: Sat Sep 28 04:44:10 UTC 2024. Nagios Core 4.5.5 - www.nagios.org. Logged in as nagiosadmin.
- Host Status Totals:** Up: 2, Down: 0, Unreachable: 0, Pending: 0.
- Service Status Totals:** OK: 16, Warning: 1, Unknown: 0, Critical: 2, Pending: 3.
- Service Status Details For All Hosts:** A table showing the status of services for all hosts. The table has columns for Host, Service, Status, Last Check, Duration, Attempt, and Status Information.

Host	Service	Status	Last Check	Duration	Attempt	Status Information
linuxserver	Current Load	OK	09-28-2024 04:41:01	0d 0h 3m 46s+	1/4	OK - load average: 0.00, 0.01, 0.00
linuxserver	Current Users	OK	09-28-2024 04:41:39	0d 0h 3m 46s+	1/4	USERS OK - 2 users currently logged in
linuxserver	HTTP	CRITICAL	09-28-2024 04:43:16	0d 0h 1m 54s	2/4	connect to address 107.22.153.120 and port 80: Connection refused
linuxserver	PING	OK	09-28-2024 04:42:54	0d 0h 3m 46s+	1/4	PING OK - Packet loss = 0%, RTA = 1.11 ms
linuxserver	Root Partition	OK	09-28-2024 04:43:31	0d 0h 3m 46s+	1/4	DISK OK - free space: / 6116 MiB (75.36% inodes=98%)
linuxserver	SSH	PENDING	N/A	0d 0h 3m 46s+	1/4	Service check scheduled for Sat Sep 28 04:44:05 UTC 2024
linuxserver	Swap Usage	PENDING	N/A	0d 0h 3m 46s+	1/4	Service check scheduled for Sat Sep 28 04:44:46 UTC 2024
linuxserver	Total Processes	PENDING	N/A	0d 0h 3m 46s+	1/4	Service check scheduled for Sat Sep 28 04:45:24 UTC 2024
localhost	Current Load	OK	09-28-2024 04:39:36	0d 0h 24m 34s	1/4	OK - load average: 0.00, 0.02, 0.00
localhost	Current Users	OK	09-28-2024 04:40:14	0d 0h 23m 56s	1/4	USERS OK - 2 users currently logged in
localhost	HTTP	WARNING	09-28-2024 04:43:51	0d 0h 20m 19s	4/4	HTTP WARNING: HTTP/1.1 403 Forbidden - 319 bytes in 0.000 second response time
localhost	PING	OK	09-28-2024 04:41:29	0d 0h 22m 41s	1/4	PING OK - Packet loss = 0%, RTA = 0.03 ms
localhost	Root Partition	OK	09-28-2024 04:42:06	0d 0h 22m 4s	1/4	DISK OK - free space: / 6116 MiB (75.36% inodes=98%)
localhost	SSH	OK	09-28-2024 04:42:44	0d 0h 21m 26s	1/4	SSH OK - OpenSSH_8.7 (protocol 2.0)
localhost	Swap Usage	CRITICAL	09-28-2024 04:41:21	0d 0h 17m 49s	4/4	SWAP CRITICAL - 0% free (0 MB out of 0 MB) - Swap is either disabled, not present, or of zero size.
localhost	Total Processes	OK	09-28-2024 04:43:59	0d 0h 20m 11s	1/4	PROCS OK: 37 processes with STATE = RSZDT

In this case, we have monitored -

Servers: 1 linux server

Services: swap

Ports: 22, 80 (ssh, http)

Processes: User status, Current load, total processes, root partition, etc.

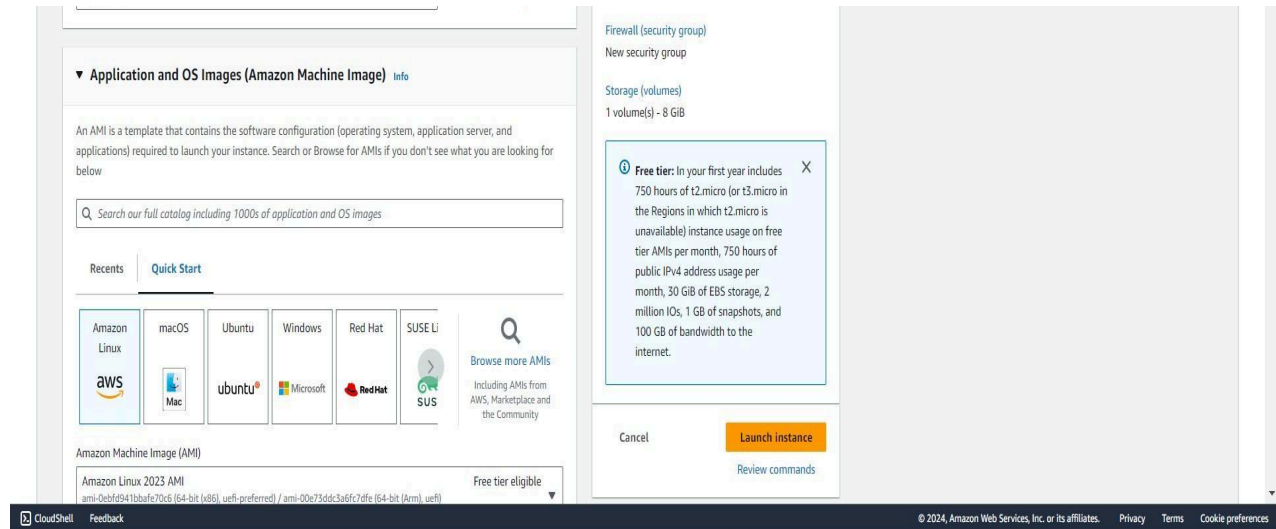
Conclusion:

In this experiment, we learned to perform port service monitoring and server monitoring using Nagios. For this, we need the Linux instance used to host the Nagios dashboard and server.

Also, we would need an Ubuntu instance which would be linked to a second host. We need to set up some configurations on the Linux instance and add the IP address of the Ubuntu instance. After that, we need to make the same initial setup on the ubuntu instance as the linux instance. Add the IP address of linux instance in allowed hosts. After restarting the NRPE server, we can see the 'linuxserver' host added.

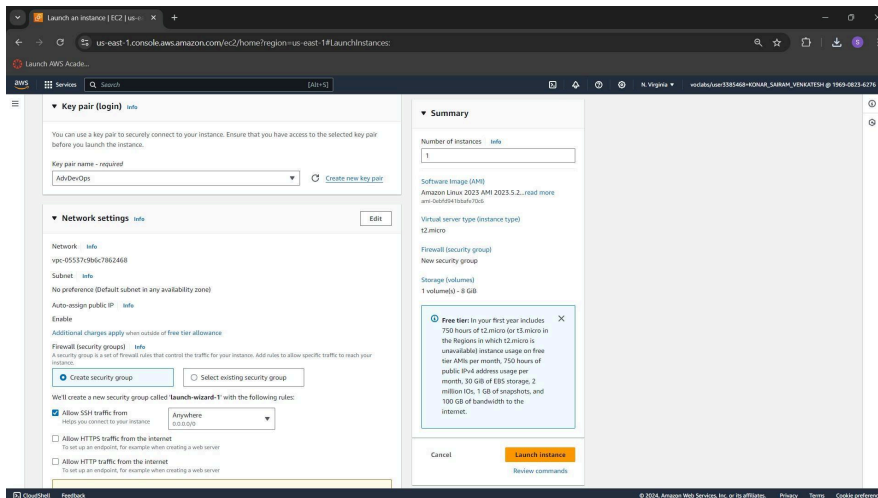
Aim: To Understand Continuous monitoring and Installation and configuration of Nagios Core, Nagios Plugins and NRPE (Nagios Remote Plugin Executor) on Linux Machine.

Step 1: Login to your AWS account. Search for EC2 on services. Open the interface and click on Create Instance.



Select The OS Image as Amazon Linux.

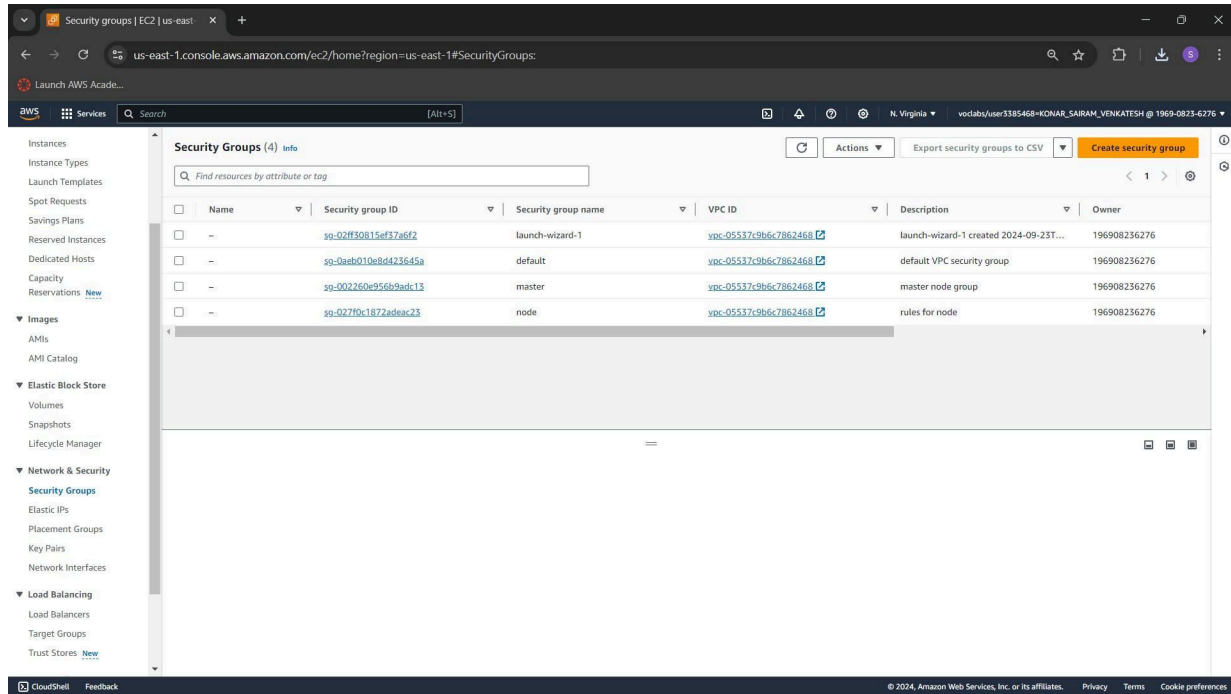
Step 2: If you do not have a private key created or a .pem file created, click on create a key pair. Else select the key pair that you had created before. (Make sure you know where the .pem file for that key is present on your system)



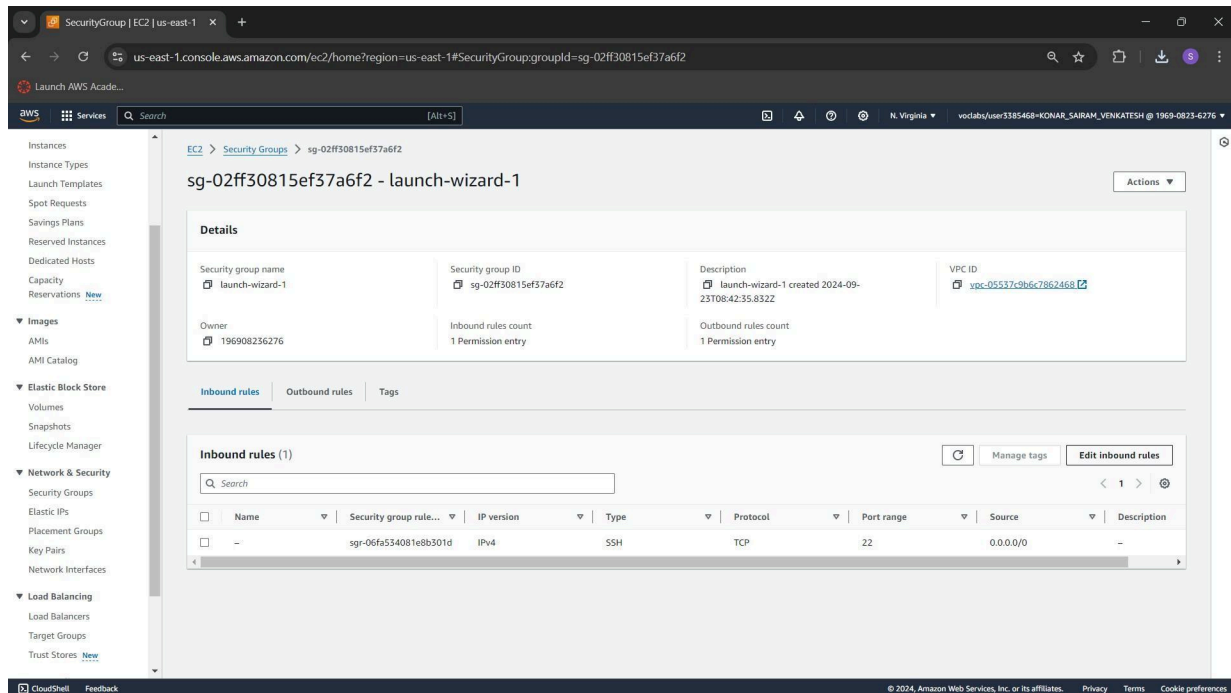
AWS will create a security group for this instance. Keep the name of that instance saved.

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Step 3: After creating the instance, click on Security Groups from the left side pane. Find the security group that was created for your instance. Click on the instance ID for that group.



Here, click on Edit Inbound Rules.



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Now, click on add rules, and add the rules for the following protocols:
HTTP, All ICMP - IPv6, HTTPS, All traffic, Custom TCP (Port 5666), All ICMP - IPv4

The screenshot shows the 'Inbound rules' section of the AWS Management Console. The rules are being added one by one. The rules are:

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-06fa534081e8b301d	SSH	TCP	22	Custom	0.0.0.0/0
-	HTTP	TCP	80	Anywhere-I...	:::0
-	All ICMP - IPv6	IPv6 ICMP	All	Anywhere-I...	:::0
-	HTTPS	TCP	443	Anywhere-I...	0.0.0.0/0
-	All traffic	All	All	Anywhere-I...	0.0.0.0/0
-	Custom TCP	TCP	5666	Anywhere-I...	0.0.0.0/0
-	All ICMP - IPv4	ICMP	All	Anywhere-I...	0.0.0.0/0

Click on save. This will add all the inbound rules to the security group.

The screenshot shows the 'Details' section of the AWS Management Console. The 'Inbound rules' tab is selected, showing all the rules added. The rules are:

Name	Security group rule...	IP version	Type	Protocol	Port range	Source	Description
-	sgr-0c2dac6846adfe9	IPv4	Custom TCP	TCP	5666	0.0.0.0/0	-
-	sgr-0fb3cb0919d7daaea	IPv4	All ICMP - IPv4	ICMP	All	0.0.0.0/0	-
-	sgr-04f91c013445aeb11	IPv4	All traffic	All	All	0.0.0.0/0	-
-	sgr-026fb1f12ee815f82	IPv6	HTTP	TCP	80	:::0	-
-	sgr-06fa534081e8b301d	IPv4	SSH	TCP	22	0.0.0.0/0	-
-	sgr-028394abd850b1f...	IPv6	All ICMP - IPv6	IPv6 ICMP	All	:::0	-
-	sgr-041e8aff5ec14d7af	IPv4	HTTPS	TCP	443	0.0.0.0/0	-

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Step 4: Now come back to the instances screen. Click on the instance ID of your instance. Then click on Connect.

Click on SSH client. Copy the example command.

The screenshot shows the AWS Management Console interface for connecting to an EC2 instance. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#ConnectToInstance:instanceId=i-0e1c706c2051d5121`. The console header includes the AWS logo, a search bar, and the user's profile information: `veclab4/user3385468-KONAR_SAIRAM_VENKATESH @ 1969-0823-6276`.

The main content area is titled "Connect to instance" with a sub-header "Connect to your instance i-0e1c706c2051d5121 (nagios-host-27) using any of these options". Below this, there are four tabs: "EC2 Instance Connect", "Session Manager", "SSH client" (which is selected), and "EC2 serial console".

Under the "SSH client" tab, the "Instance ID" is listed as `i-0e1c706c2051d5121 (nagios-host-27)`. A list of steps is provided:

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is `AdvDevOps.pem`.
3. Run this command, if necessary, to ensure your key is not publicly viewable.
`chmod 400 "AdvDevOps.pem"`
4. Connect to your instance using its Public DNS:
`ec2-3-89-111-169.compute-1.amazonaws.com`

An "Example:" section shows the command: `ssh -i "AdvDevOps.pem" ec2-user@ec2-3-89-111-169.compute-1.amazonaws.com`. Below this, a note states: "Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username."

A "Cancel" button is located at the bottom right of the content area.

The footer of the console shows "cloudShell" and "Feedback" on the left, and copyright information "© 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences" on the right.

Step 5: Now, we have to connect our local OS terminal to the instance using SSH. For this, Open terminal where the private key file is located (.pem)
Paste the copied SSH command and run it.

```
C:\Users\saira\OneDrive\Desktop\AdvDevOps\lab 9>ssh -i "AdvDevOps.pem" ec2-user@ec2-3-89-111-169.compute-1.amazonaws.com

      #_
     ##
  ##  ###
 ##  #####
##  #####|
##  \###|
     \#/
     V~'  ---
      /~'  '--->

      Amazon Linux 2023

      https://aws.amazon.com/linux/amazon-linux-2023

      _-_-
     _-_-
    _-_-
   _-_-
  _-_-
 _-_-
_-_-
_-m/'

[ec2-user@ip-172-31-90-224 ~]$
```

Step 6: Now we start working on this terminal. First run the command `sudo yum update`

This command will check for any updates for the YUM library.

```
[ec2-user@ip-172-31-83-157 ~]$ sudo yum update
Last metadata expiration check: 0:04:50 ago on Sat Sep 28 03:46:46 2024.
Dependencies resolved.
Nothing to do.
Complete!
```

Step 7: We are going to install an Apache server and a PHP on this instance. For that, run this command.

```
sudo yum install httpd php
```

```
[ec2-user@ip-172-31-83-157 ~]$ sudo yum install httpd php
Last metadata expiration check: 0:04:58 ago on Sat Sep 28 03:46:46 2024.
Dependencies resolved.
=====
Package                               Architecture      Version           Size
=====
Installing:
httpd                                 x86_64            2.4.62-1.amzn2023      48 k
php8.3                               x86_64            8.3.10-1.amzn2023.0.1  10 k
Installing dependencies:
apr                                   x86_64            1.7.2-2.amzn2023.0.2   129 k
apr-util                             x86_64            1.6.3-1.amzn2023.0.1   98 k
generic-logos-httpd                 noarch            18.0-0-12.amzn2023.0.3  19 k
httpd-core                           x86_64            2.4.62-1.amzn2023      1.4 M
httpd-filesystem                     noarch            2.4.62-1.amzn2023      14 k
httpd-tools                          x86_64            2.4.62-1.amzn2023      81 k
libbrotli                             x86_64            1.0.9-4.amzn2023.0.2   315 k
libsodium                             x86_64            1.0.19-4.amzn2023      176 k
libxslt                              x86_64            1.1.34-5.amzn2023.0.2  241 k
mailcap                              noarch            2.1.49-3.amzn2023.0.3   33 k
nginx-filesystem                     noarch            1.1.24.0-1.amzn2023.0.4  9.8 k
php8.3-cli                           x86_64            8.3.10-1.amzn2023.0.1   3.7 M
php8.3-common                         x86_64            8.3.10-1.amzn2023.0.1   737 k
=====

Installed:
apr-1.7.2-2.amzn2023.0.2.x86_64
generic-logos-httpd-18.0-0-12.amzn2023.0.3.noarch
httpd-filesystem-2.4.62-1.amzn2023.noarch
libsodium-1.0.19-4.amzn2023.x86_64
mod_http2-2.0.27-1.amzn2023.0.3.x86_64
php8.3-8.3.10-1.amzn2023.0.1.x86_64
php8.3-fpm-8.3.10-1.amzn2023.0.1.x86_64
php8.3-pdo-8.3.10-1.amzn2023.0.1.x86_64
php8.3-xml-8.3.10-1.amzn2023.0.1.x86_64
apr-util-1.6.3-1.amzn2023.0.1.x86_64
httpd-2.4.62-1.amzn2023.x86_64
httpd-tools-2.4.62-1.amzn2023.x86_64
libxslt-1.1.34-5.amzn2023.0.2.x86_64
mod_lua-2.4.62-1.amzn2023.x86_64
php8.3-cli-8.3.10-1.amzn2023.0.1.x86_64
php8.3-mbstring-8.3.10-1.amzn2023.0.1.x86_64
php8.3-process-8.3.10-1.amzn2023.0.1.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64
httpd-core-2.4.62-1.amzn2023.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch
nginx-filesystem-1.1.24.0-1.amzn2023.0.4.noarch
php8.3-common-8.3.10-1.amzn2023.0.1.x86_64
php8.3-opcache-8.3.10-1.amzn2023.0.1.x86_64
php8.3-sodium-8.3.10-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-83-157 ~]$ sudo yum install gcc glibc glibc-common
```

Step 8: Next we install C/C++ compiler (GCC) along with the necessary C libraries required for compiling and running C programs. Use the following command.

sudo yum install gcc glibc glibc-common

```
[ec2-user@ip-172-31-83-157 ~]$ sudo yum install gcc glibc glibc-common
Last metadata expiration check: 0:05:58 ago on Sat Sep 28 03:46:46 2024.
Package glibc-2.34-52.amzn2023.0.11.x86_64 is already installed.
Package glibc-common-2.34-52.amzn2023.0.11.x86_64 is already installed.
Dependencies resolved.
=====
Package                               Architecture      Version                               Repository      Size
=====
Installing:
gcc                                   x86_64            11.4.1-2.amzn2023.0.2                amazonlinux     32 M
Installing dependencies:
annobin-docs                         noarch            10.93-1.amzn2023.0.1                amazonlinux     92 k
annobin-plugin-gcc                   x86_64            10.93-1.amzn2023.0.1                amazonlinux     887 k
cpp                                   x86_64            11.4.1-2.amzn2023.0.2                amazonlinux     10 M
gc                                    x86_64            8.0.4-5.amzn2023.0.2                amazonlinux     105 k
glibc-devel                           x86_64            2.34-52.amzn2023.0.11               amazonlinux     27 k
glibc-headers-x86_64                 noarch            2.34-52.amzn2023.0.11               amazonlinux     427 k
guile22                              x86_64            2.2.7-2.amzn2023.0.3                amazonlinux     6.4 M
kernel-headers                       x86_64            6.1.109-118.189.amzn2023           amazonlinux     1.4 M
libmpc                               x86_64            1.2.1-2.amzn2023.0.2                amazonlinux     62 k
libtool-ltdl                         x86_64            2.4.7-1.amzn2023.0.3                amazonlinux     38 k
libxcrypt-devel                      x86_64            4.4.33-7.amzn2023                   amazonlinux     32 k
=====

Installed:
annobin-docs-10.93-1.amzn2023.0.1.noarch      annobin-plugin-gcc-10.93-1.amzn2023.0.1.x86_64      cpp-11.4.1-2.amzn2023.0.2.x86_64
gcc-8.0.4-5.amzn2023.0.2.x86_64              gcc-11.4.1-2.amzn2023.0.2.x86_64                  glibc-devel-2.34-52.amzn2023.0.11.x86_64
glibc-headers-x86-2.34-52.amzn2023.0.11.noarch  guile22-2.2.7-2.amzn2023.0.3.x86_64              kernel-headers-6.1.109-118.189.amzn2023.x86_64
libmpc-1.2.1-2.amzn2023.0.2.x86_64           libtool-ltdl-2.4.7-1.amzn2023.0.3.x86_64          libxcrypt-devel-4.4.33-7.amzn2023.x86_64
make-1:4.3-5.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-83-157 ~]$
```

Step 9: We would also need GD library and its development tools. For that, run this command

sudo yum install gd gd-devel

```
[ec2-user@ip-172-31-83-157 ~]$ sudo yum install gd gd-devel
Last metadata expiration check: 0:11:36 ago on Sat Sep 28 03:46:46 2024.
Dependencies resolved.
=====
Package                               Architecture      Version                               Repository      Size
=====
Installing:
gd                                    x86_64            2.3.3-5.amzn2023.0.3                amazonlinux     139 k
gd-devel                             x86_64            2.3.3-5.amzn2023.0.3                amazonlinux     38 k
Installing dependencies:
brotli                               x86_64            1.0.9-4.amzn2023.0.2                amazonlinux     314 k
brotli-devel                         x86_64            1.0.9-4.amzn2023.0.2                amazonlinux     31 k
bzip2-devel                          x86_64            1.0.8-6.amzn2023.0.2                amazonlinux     214 k
cairo                                 x86_64            1.17.6-2.amzn2023.0.1               amazonlinux     684 k
cmake-filesystem                     x86_64            3.22.2-1.amzn2023.0.4               amazonlinux     16 k
fontconfig                           x86_64            2.13.94-2.amzn2023.0.2              amazonlinux     273 k
fontconfig-devel                     x86_64            2.13.94-2.amzn2023.0.2              amazonlinux     128 k
fonts-filessystem                     noarch            1:2.0.5-12.amzn2023.0.2             amazonlinux     9.5 k
freetype                             x86_64            2.13.2-5.amzn2023.0.1               amazonlinux     423 k
freetype-devel                       x86_64            2.13.2-5.amzn2023.0.1               amazonlinux     912 k
glib2-devel                           x86_64            2.74.7-689.amzn2023.0.2            amazonlinux     486 k
=====

Installed:
brotli-1.0.9-4.amzn2023.0.2.x86_64          brotli-devel-1.0.9-4.amzn2023.0.2.x86_64
bzip2-devel-1.0.8-6.amzn2023.0.2.x86_64    cairo-1.17.6-2.amzn2023.0.1.x86_64
cmake-filesystem-3.22.2-1.amzn2023.0.4.x86_64  fontconfig-2.13.94-2.amzn2023.0.2.x86_64
fontconfig-devel-2.13.94-2.amzn2023.0.2.x86_64  fonts-filessystem-1:2.0.5-12.amzn2023.0.2.noarch
freetype-2.13.2-5.amzn2023.0.1.x86_64        freetype-devel-2.13.2-5.amzn2023.0.1.x86_64
gd-2.3.3-5.amzn2023.0.3.x86_64              gd-devel-2.3.3-5.amzn2023.0.3.x86_64
glib2-devel-2.74.7-689.amzn2023.0.2.x86_64  google-noto-sans-vf-fonts-20201206-2.amzn2023.0.2.noarch
google-noto-sans-vf-fonts-20201206-2.amzn2023.0.2.noarch  graphite2-1.3.14-7.amzn2023.0.2.x86_64
graphite2-devel-1.3.14-7.amzn2023.0.2.x86_64  harfbuzz-7.0.0-2.amzn2023.0.1.x86_64
harfbuzz-devel-7.0.0-2.amzn2023.0.1.x86_64  harfbuzz-icu-7.0.0-2.amzn2023.0.1.x86_64
jbigkit-libs-2.1-21.amzn2023.0.2.x86_64      langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch
libICE-1.0.10-6.amzn2023.0.2.x86_64         libSM-1.2.3-8.amzn2023.0.2.x86_64
libX11-1.7.2-3.amzn2023.0.4.x86_64          libX11-common-1.7.2-3.amzn2023.0.4.noarch
libX11-devel-1.7.2-3.amzn2023.0.4.x86_64    libX11-xcb-1.7.2-3.amzn2023.0.4.x86_64
libXau-1.0.9-6.amzn2023.0.2.x86_64          libXau-devel-1.0.9-6.amzn2023.0.2.x86_64
libXext-1.3.4-6.amzn2023.0.2.x86_64         libXpm-3.5.15-2.amzn2023.0.3.x86_64
libXpm-devel-3.5.15-2.amzn2023.0.3.x86_64   libXrender-0.9.10-14.amzn2023.0.2.x86_64
libXt-1.2.0-4.amzn2023.0.2.x86_64           libblkid-devel-2.37.4-1.amzn2023.0.4.x86_64
libffi-devel-3.4.4-1.amzn2023.0.1.x86_64    libicu-67.1-7.amzn2023.0.3.x86_64
libicu-devel-67.1-7.amzn2023.0.3.x86_64    libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
libjpeg-turbo-devel-2.1.4-2.amzn2023.0.5.x86_64  libmount-devel-2.37.4-1.amzn2023.0.4.x86_64
libpng-2.1.6.37-10.amzn2023.0.6.x86_64     libpng-devel-2.1.6.37-10.amzn2023.0.6.x86_64
libselinux-devel-3.4-5.amzn2023.0.2.x86_64  libsepol-devel-3.4-3.amzn2023.0.3.x86_64
libtiff-4.0.0-4.amzn2023.0.18.x86_64       libtiff-devel-4.0.0-4.amzn2023.0.18.x86_64
libwebp-1.2.4-1.amzn2023.0.6.x86_64        libwebp-devel-1.2.4-1.amzn2023.0.6.x86_64
libxcb-1.13.1-7.amzn2023.0.2.x86_64        libxcb-devel-1.13.1-7.amzn2023.0.2.x86_64
libxml2-devel-2.10.4-1.amzn2023.0.6.x86_64  pcre2-devel-10.40-1.amzn2023.0.3.x86_64
pcre2-utf16-10.40-1.amzn2023.0.3.x86_64    pcre2-utf32-10.40-1.amzn2023.0.3.x86_64
pixman-0.40.0-3.amzn2023.0.3.x86_64        sysprof-capture-devel-3.40.1-2.amzn2023.0.2.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch     xorg-x11-proto-devel-2021.4-1.amzn2023.0.2.noarch
xz-devel-5.2.5-9.amzn2023.0.2.x86_64       zlib-devel-1.2.11-33.amzn2023.0.5.x86_64

Complete!
[ec2-user@ip-172-31-83-157 ~]$
```


Step 10: Now, we create a user called as 'nagios' and make sure that it has a home directory, and set up a password for it.

```
sudo adduser -m
```

```
nagios sudo passwd
```

```
[ec2-user@ip-172-31-83-157 ~]$ sudo adduser -m nagios
sudo passwd nagios
Changing password for user nagios.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
```

nagios

Step 11: Create a user group called as 'nagcmd' to execute nagios commands.

```
sudo groupadd nagcmd
```

```
[ec2-user@ip-172-31-83-157 ~]$ sudo groupadd nagcmd
[ec2-user@ip-172-31-83-157 ~]$ |
```

Step 12: Add users apache and nagios to this user group.

```
sudo usermod -a -G nagcmd nagios
```

```
sudo usermod -a -G nagcmd apache
```

```
[ec2-user@ip-172-31-83-157 ~]$ sudo usermod -a -G nagcmd nagios
sudo usermod -a -G nagcmd apache
[ec2-user@ip-172-31-83-157 ~]$
```

Step 13: We create a directory downloads, to store the files of nagios server that are downloaded.

```
mkdir ~/downloads
```

```
cd ~/downloads
```

```
[ec2-user@ip-172-31-83-157 ~]$ mkdir ~/downloads
cd ~/downloads
[ec2-user@ip-172-31-83-157 downloads]$ |
```

Step 14: Now we need to install the latest versions of nagios-core and nagios-plugins. Go to the respective websites and check whether a better version is available. If newer versions are available, then right click on the download button → Copy link address.

Paste this link address in place of the current link in command.

If not run these commands.

wget <https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.5.tar.gz>

```
[ec2-user@ip-172-31-83-157 downloads]$ wget https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.5.tar.gz
--2024-09-28 04:04:23-- https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.5.tar.gz
Resolving assets.nagios.com (assets.nagios.com)... 45.79.49.120, 2600:3c00:f03c:92ff:fe7:45ce
Connecting to assets.nagios.com (assets.nagios.com)|45.79.49.120|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2065473 (2.0M) [application/x-gzip]
Saving to: 'nagios-4.5.5.tar.gz'

nagios-4.5.5.tar.gz          100%[=====] 1.97M  5.36MB/s  in 0.4s
2024-09-28 04:04:24 (5.36 MB/s) - 'nagios-4.5.5.tar.gz' saved [2065473/2065473]

[ec2-user@ip-172-31-83-157 downloads]$
```

wget <https://nagios-plugins.org/download/nagios-plugins-2.4.11.tar.gz>

```
[ec2-user@ip-172-31-83-157 downloads]$ wget https://nagios-plugins.org/download/nagios-plugins-2.4.11.tar.gz
--2024-09-28 04:06:15-- https://nagios-plugins.org/download/nagios-plugins-2.4.11.tar.gz
Resolving nagios-plugins.org (nagios-plugins.org)... 45.56.123.251
Connecting to nagios-plugins.org (nagios-plugins.org)|45.56.123.251|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2753049 (2.6M) [application/x-gzip]
Saving to: 'nagios-plugins-2.4.11.tar.gz'

nagios-plugins-2.4.11.tar.gz 100%[=====] 2.62M  5.22MB/s  in 0.5s
2024-09-28 04:06:16 (5.22 MB/s) - 'nagios-plugins-2.4.11.tar.gz' saved [2753049/2753049]

[ec2-user@ip-172-31-83-157 downloads]$
```

Step 15: Now, we need to extract nagios-core file into the same directory. For that, we will use tar command.

tar zxvf nagios-4.5.5.tar.gz

```
[ec2-user@ip-172-31-83-157 downloads]$ tar zxvf nagios-4.5.5.tar.gz
nagios-4.5.5/
nagios-4.5.5/.github/
nagios-4.5.5/.github/workflows/
nagios-4.5.5/.github/workflows/test.yml
nagios-4.5.5/.gitignore
nagios-4.5.5/CONTRIBUTING.md
nagios-4.5.5/Changelog
nagios-4.5.5/INSTALLING
nagios-4.5.5/LLEGAL
nagios-4.5.5/LICENSE
nagios-4.5.5/Makefile.in
nagios-4.5.5/README.md
nagios-4.5.5/THANKS
nagios-4.5.5/UPGRADING
nagios-4.5.5/aclocal.m4
nagios-4.5.5/autoconf-macros/
nagios-4.5.5/autoconf-macros/.gitignore
nagios-4.5.5/autoconf-macros/CHANGELOG.md
nagios-4.5.5/autoconf-macros/LICENSE
nagios-4.5.5/autoconf-macros/LICENSE.md
```



```
nagios-4.5.5/xdata/.gitignore
nagios-4.5.5/xdata/Makefile.in
nagios-4.5.5/xdata/xcddefault.c
nagios-4.5.5/xdata/xcddefault.h
nagios-4.5.5/xdata/xodtemplate.c
nagios-4.5.5/xdata/xodtemplate.h
nagios-4.5.5/xdata/xpddefault.c
nagios-4.5.5/xdata/xpddefault.h
nagios-4.5.5/xdata/xrddefault.c
nagios-4.5.5/xdata/xrddefault.h
nagios-4.5.5/xdata/xsddefault.c
nagios-4.5.5/xdata/xsddefault.h
[ec2-user@ip-172-31-83-157 downloads]$
```

Step 16: We need to ensure that Nagios uses a specific group (in this case, `nagcmd`) for executing external commands.

`./configure --with-command-group=nagcmd`

An **error** could be encountered here: `./configure: no such path or directory`

Solution: Navigate to the nagios-4.5.5 folder in downloads. (version could vary)

Steps: ls

- `cd nagios-4.5.5` (use the version shown by your ls command)
- `./configure --with-command-group=nagcmd`

```
[ec2-user@ip-172-31-83-157 downloads]$ ls
nagios-4.5.5  nagios-4.5.5.tar.gz  nagios-plugins-2.4.11.tar.gz
```

Another **error** could be **Cannot find SSL headers**.

To solve this, we need to install OpenSSL Dev Library

Steps:

`sudo yum install openssl-devel`

```
checking for a BSD-compatible install... /usr/bin/install -c
[ec2-user@ip-172-31-83-157 nagios-4.5.5]$ sudo yum install openssl-devel
Last metadata expiration check: 0:21:59 ago on Sat Sep 28 03:46:46 2024.
Dependencies resolved.
=====
Package                                Repository                               Architecture      Size              Version
=====
Installing:
openssl-devel                          amazonlinux                               x86_64             3.0 M             1:3.0.8-1.amzn2023.0.14
=====
Transaction Summary
=====
Install 1 Package
Total download size: 3.0 M
Installed size: 4.7 M
Is this ok [y/N]: y

```

`./configure --with-command-group=nagcmd`

```
[ec2-user@ip-172-31-83-157 nagios-4.5.5]$ ./configure --with-command-group=nagcmd
checking for a BSD-compatible install... /usr/bin/install -c
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether gcc accepts -g... yes
checking for gcc option to enable C11 features... none needed
checking whether make sets $(MAKE)... yes
checking whether ln -s works... yes
checking for strip... /usr/bin/strip
checking for sys/wait.h that is POSIX.1 compatible... yes
checking for stdio.h... yes
checking for stdlib.h... yes
checking for string.h... yes
checking for inttypes.h... yes
checking for stdint.h... yes
checking for strings.h... yes
checking for sys/stat.h... yes

```

```
*** Configuration summary for nagios 4.5.5 2024-09-17 ***:

General Options:
-----
Nagios executable: nagios
Nagios user/group: nagios,nagios
Command user/group: nagios,nagcmd
Event Broker: yes
Install ${prefix}: /usr/local/nagios
Install ${includedir}: /usr/local/nagios/include/nagios
Lock file: /run/nagios.lock
Check result directory: /usr/local/nagios/var/spool/checkresults
Init directory: /lib/systemd/system
Apache conf.d directory: /etc/httpd/conf.d
Mail program: /bin/mail
Host OS: linux-gnu
IOBroker Method: epoll

Web Interface Options:
-----
HTML URL: http://localhost/nagios/
CGI URL: http://localhost/nagios/cgi-bin/
Traceroute (used by WAP): /usr/bin/traceroute

Review the options above for accuracy. If they look okay,
type 'make all' to compile the main program and CGIs.

[ec2-user@ip-172-31-83-157 nagios-4.5.5]$
```

Step 17: We need to compile all components of this software according to the instruction in the Makefile. For that, use this command:
make all

Then,

sudo make install

sudo make install-init

sudo make install-config

sudo make install-commandmode

```
[ec2-user@ip-172-31-83-157 nagios-4.5.5]$ sudo make install
sudo make install-init
sudo make install-config
sudo make install-commandmode
cd ./base && make install
make[1]: Entering directory '/home/ec2-user/downloads/nagios-4.5.5/base'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/bin
/usr/bin/install -c -s -m 774 -o nagios -g nagios nagios /usr/local/nagios/bin
/usr/bin/install -c -s -m 774 -o nagios -g nagios nagiosstats /usr/local/nagios/bin
make[1]: Leaving directory '/home/ec2-user/downloads/nagios-4.5.5/base'
cd ./cgi && make install
make[1]: Entering directory '/home/ec2-user/downloads/nagios-4.5.5/cgi'
make install-basic
make[2]: Entering directory '/home/ec2-user/downloads/nagios-4.5.5/cgi'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/sbin
```

```
make[1]: Leaving directory '/home/ec2-user/downloads/nagios-4.5.5'
/usr/bin/install -c -m 755 -d -o root -g root /lib/systemd/system
/usr/bin/install -c -m 755 -o root -g root startup/default-service /lib/systemd/system/nagios.service
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc/objects
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/nagios.cfg /usr/local/nagios/etc/nagios.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/cgi.cfg /usr/local/nagios/etc/cgi.cfg
/usr/bin/install -c -b -m 660 -o nagios -g nagios sample-config/resource.cfg /usr/local/nagios/etc/resource.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/templates.cfg /usr/local/nagios/etc/objects/templates.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/commands.cfg /usr/local/nagios/etc/objects/commands.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/contacts.cfg /usr/local/nagios/etc/objects/contacts.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/timeperiods.cfg /usr/local/nagios/etc/objects/timeperiods.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/localhost.cfg /usr/local/nagios/etc/objects/localhost.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/windows.cfg /usr/local/nagios/etc/objects/windows.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/printer.cfg /usr/local/nagios/etc/objects/printer.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/switch.cfg /usr/local/nagios/etc/objects/switch.cfg

*** Config files installed ***

Remember, these are *SAMPLE* config files. You'll need to read
the documentation for more information on how to actually define
services, hosts, etc. to fit your particular needs.

/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/var/rw
chmod g+s /usr/local/nagios/var/rw

*** External command directory configured ***
```

Step 18: We need to update the email linked with this server to our email for it to send notifications (if any needed).

`sudo nano /usr/local/nagios/etc/objects/contacts.cfg`

```
GNU nano 5.8 /usr/local/nagios/etc/objects/contacts.cfg Modified
# CONTACTS.CFG - SAMPLE CONTACT/CONTACTGROUP DEFINITIONS
#
# NOTES: This config file provides you with some example contact and contact
# group definitions that you can reference in host and service
# definitions.
#
# You don't need to keep these definitions in a separate file from your
# other object definitions. This has been done just to make things
# easier to understand.
#
#####

# CONTACTS
#
#####

# Just one contact defined by default - the Nagios admin (that's you)
# This contact definition inherits a lot of default values from the
# 'generic-contact' template which is defined elsewhere.
define contact {
    contact_name      nagiosadmin          ; Short name of user
    use                generic-contact      ; Inherit default values from generic-contact template (defined above)
    alias              Nagios Admin         ; Full name of user
    email              2022.salram.konar@ves.ac.in ; <***** CHANGE THIS TO YOUR EMAIL ADDRESS *****>
}

#####

# CONTACT GROUPS
#
#####

# We only have one contact in this simple configuration file, so there is
# no need to create more than one contact group.
define contactgroup {
    contactgroup_name admins
    alias              Nagios Administrators
    members             nagiosadmin
}
```

Here, change the email under 'define contact{' to your email address.

To save this use the following shortcut sequence CTRL+O→Enter→CTRL+X.

CTRL+O: Overwrite the existing file with edited file

CTRL+X: Exit nano editor.

Step 19: We need to install the necessary configuration files for the Nagios web interface. Run the following command.

`sudo make install-webconf`

```
[ec2-user@ip-172-31-83-157 nagios-4.5.5]$ sudo make install-webconf
/usr/bin/install -c -m 644 sample-config/httpd.conf /etc/httpd/conf.d/nagios.conf
if [ 0 -eq 1 ]; then \
    ln -s /etc/httpd/conf.d/nagios.conf /etc/apache2/sites-enabled/nagios.conf; \
fi

*** Nagios/Apache conf file installed ***

[ec2-user@ip-172-31-83-157 nagios-4.5.5]$ |
```

Step 20: Now we need to setup a user to access this nagios web interface. So we run this command to create a user called 'nagiosadmin'.

Keep this username and password saved as it is needed to login to the web interface.

`sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin`

```
[ec2-user@ip-172-31-83-157 nagios-4.5.5]$ sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
```

Step 21: Restart the apache server to apply all the recent configurations.

`sudo service httpd restart`

```
Adding password for user nagiosadmin
[ec2-user@ip-172-31-83-157 nagios-4.5.5]$ sudo service httpd restart
Redirecting to /bin/systemctl restart httpd.service
```

Step 22: Now we go back to the downloads folder and extract the files of nagios plugin.

`cd ~/downloads`

`tar zxvf nagios-plugins-2.4.11.tar.gz (Version may vary)`

```
[ec2-user@ip-172-31-83-157 downloads]$ tar zxvf nagios-plugins-2.4.11.tar.gz
nagios-plugins-2.4.11/
nagios-plugins-2.4.11/build-aux/
nagios-plugins-2.4.11/build-aux/compile
nagios-plugins-2.4.11/build-aux/config.guess
nagios-plugins-2.4.11/build-aux/config.rpath
nagios-plugins-2.4.11/build-aux/config.sub
nagios-plugins-2.4.11/build-aux/install-sh
nagios-plugins-2.4.11/build-aux/ltmain.sh
nagios-plugins-2.4.11/build-aux/missing
nagios-plugins-2.4.11/build-aux/mkinstalldirs
nagios-plugins-2.4.11/build-aux/depcomp
nagios-plugins-2.4.11/build-aux/snippet/
nagios-plugins-2.4.11/build-aux/snippet/_Noreturn.h
nagios-plugins-2.4.11/build-aux/snippet/arg-nonnull.h
nagios-plugins-2.4.11/build-aux/snippet/c++defs.h
nagios-plugins-2.4.11/build-aux/snippet/warn-on-use.h
nagios-plugins-2.4.11/build-aux/test-driver
nagios-plugins-2.4.11/config_test/
nagios-plugins-2.4.11/config_test/Makefile
nagios-plugins-2.4.11/config_test/run_tests
nagios-plugins-2.4.11/config_test/child_test.c
nagios-plugins-2.4.11/gl/
nagios-plugins-2.4.11/gl/m4/
```

```
nagios-plugins-2.4.11/po/fr.gmo
nagios-plugins-2.4.11/po/de.gmo
nagios-plugins-2.4.11/po/nagios-plugins.pot
nagios-plugins-2.4.11/po/stamp-po
nagios-plugins-2.4.11/po/ChangeLog
nagios-plugins-2.4.11/po/LINGUAS
nagios-plugins-2.4.11/release
[ec2-user@ip-172-31-83-157 downloads]$ |
```

Step 23: Again, we need to install the configurations for these files.

cd nagios-plugins-2.4.11 (version may vary)

./configure --with-nagios-user=nagios --with-nagios-group=nagios

```
[ec2-user@ip-172-31-83-157 nagios-plugins-2.4.11]$ ./configure --with-nagios-user=nagios --with-nagios-group=nagios
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for a thread-safe mkdir -p... /usr/bin/mkdir -p
checking for gawk... gawk
checking whether make sets $(MAKE)... yes
checking whether make supports nested variables... yes
checking whether to enable maintainer-specific portions of Makefiles... yes
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
```

```
config.status: creating plugins-scripts/utils.sh
config.status: creating perlmods/Makefile
config.status: creating test.pl
config.status: creating pkg/solaris/pkginfo
config.status: creating po/Makefile.in
config.status: creating config.h
config.status: config.h is unchanged
config.status: executing depfiles commands
config.status: executing libtool commands
config.status: executing po-directories commands
config.status: creating po/POTFILES
config.status: creating po/Makefile
[ec2-user@ip-172-31-83-157 nagios-plugins-2.4.11]$ |
```

Step 24: We need to compile all components of this software according to the instruction in the Makefile. For that, use the commands:

make

sudo make install

```
TI
make[1]: Leaving directory '/home/ec2-user/downloads/nagios-plugins-2.4.11/po'
make[1]: Entering directory '/home/ec2-user/downloads/nagios-plugins-2.4.11'
make[2]: Entering directory '/home/ec2-user/downloads/nagios-plugins-2.4.11'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/home/ec2-user/downloads/nagios-plugins-2.4.11'
make[1]: Leaving directory '/home/ec2-user/downloads/nagios-plugins-2.4.11'
[ec2-user@ip-172-31-83-157 nagios-plugins-2.4.11]$ |
```

Step 25: We need to register the Nagios service with the system, which would make it able to manage the server status. So run the following commands

sudo chkconfig --add nagios

sudo chkconfig nagios on

Step 26: We need to verify the Nagios configuration for any syntax errors or issues before starting or restarting the Nagios service.

```
sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
```

```
[ec2-user@ip-172-31-90-224 nagios-plugins-2.0.3]$ sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg

Nagios Core 4.0.8
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 08-12-2014
License: GPL

Website: http://www.nagios.org
Reading configuration data...
Error in configuration file '/usr/local/nagios/etc/nagios.cfg' - Line 452 (Check result path '/usr/local/nagios/var/spool/checkresults' is not a valid directory)
Error processing main config file!

[ec2-user@ip-172-31-90-224 nagios-plugins-2.0.3]$ sudo mkdir -p /usr/local/nagios/var/spool/checkresults
[ec2-user@ip-172-31-90-224 nagios-plugins-2.0.3]$ sudo chown nagios:nagios /usr/local/nagios/var/spool/checkresults
[ec2-user@ip-172-31-90-224 nagios-plugins-2.0.3]$ sudo chmod 775 /usr/local/nagios/var/spool/checkresults
[ec2-user@ip-172-31-90-224 nagios-plugins-2.0.3]$ sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg

Nagios Core 4.0.8
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 08-12-2014
License: GPL
```

Error: Error in configuration file '/usr/local/nagios/etc/nagios.cfg' - Line 452 (Check result path '/usr/local/nagios/var/spool/checkresults' is not a valid directory)

It is an error in processing main config file!

Solution: Create the missing directory, set the permissions, verify it.

- `sudo mkdir -p /usr/local/nagios/var/spool/checkresults` (Create)
- `sudo chown nagios:nagios /usr/local/nagios/var/spool/checkresults`
- `sudo chmod 775 /usr/local/nagios/var/spool/checkresults` (permissions)

Now rerun the command

```
sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
```

```
Created symlink /etc/systemd/system/multi-user.target.wants/nagios.service → /usr/lib/systemd/system/nagios.service.
[ec2-user@ip-172-31-83-157 nagios-plugins-2.4.11]$ sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg

Nagios Core 4.5.5
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2024-09-17
License: GPL

Website: https://www.nagios.org
Reading configuration data...
  Read main config file okay...
  Read object config files okay...

Running pre-flight check on configuration data...

Checking objects...
  Checked 8 services.
  Checked 1 hosts.
  Checked 1 host groups.
  Checked 0 service groups.
  Checked 1 contacts.
  Checked 1 contact groups.
  Checked 24 commands.
  Checked 5 time periods.
  Checked 0 host escalations.
  Checked 0 service escalations.

Checking for circular paths...
  Checked 1 hosts
  Checked 0 service dependencies
  Checked 0 host dependencies
  Checked 5 timeperiods

Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

Total Warnings: 0
Total Errors: 0

Things look okay - No serious problems were detected during the pre-flight check
[ec2-user@ip-172-31-83-157 nagios-plugins-2.4.11]$
```


Step 27:

sudo service nagios start

```
[ec2-user@ip-172-31-83-157 nagios-plugins-2.4.11]$ sudo service nagios start
Redirecting to /bin/systemctl start nagios.service
[ec2-user@ip-172-31-83-157 nagios-plugins-2.4.11]$ |
```

sudo systemctl status nagios

```
[ec2-user@ip-172-31-90-224 nagios-plugins-2.0.3]$ sudo systemctl restart nagios
Failed to restart nagios.service: Unit nagios.service not found.
[ec2-user@ip-172-31-90-224 nagios-plugins-2.0.3]$ sudo service nagios start
Reloading system: [ OK ]
Starting nagios (via systemctl): [ OK ]
[ec2-user@ip-172-31-90-224 nagios-plugins-2.0.3]$ sudo systemctl status nagios
nagios.service - LSB: Starts and stops the Nagios monitoring server
Loaded: loaded (/etc/rc.d/init.d/nagios; generated)
Active: active (running) since Mon 2024-09-23 09:31:32 UTC; 3min 4s ago
Docs: man:systemd-sysv-generator(8)
Process: 66885 ExecStart=/etc/rc.d/init.d/nagios start (code=exited, status=0/SUCCESS)
Tasks: 6 (limit: 1112)
Memory: 2.4M
CPU: 85ms
CGroup: /system.slice/nagios.service
├─66907 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
├─66909 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
├─66910 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
├─66911 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
├─66912 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
└─66913 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

Sep 23 09:33:21 ip-172-31-90-224.ec2.internal nagios[66907]: Error: Unable to create temp file '/usr/local/nagios/var/nagios.tmpqgZ2SN' for writing status data: Permissi>
Sep 23 09:33:24 ip-172-31-90-224.ec2.internal nagios[66907]: SERVICE ALERT: localhost;HTTP;WARNING;SOFT;1;HTTP WARNING: HTTP/1.1 403 Forbidden - 319 bytes in 0.001 secon>
Sep 23 09:33:31 ip-172-31-90-224.ec2.internal nagios[66907]: Error: Unable to create temp file '/usr/local/nagios/var/nagios.tmpqSMLu4' for writing status data: Permissi>
Sep 23 09:33:51 ip-172-31-90-224.ec2.internal nagios[66907]: Error: Unable to create temp file '/usr/local/nagios/var/nagios.tmpqV8EwB' for writing status data: Permissi>
Sep 23 09:34:01 ip-172-31-90-224.ec2.internal nagios[66907]: Error: Unable to create temp file '/usr/local/nagios/var/nagios.tmpqVotUf' for writing status data: Permissi>
Sep 23 09:34:11 ip-172-31-90-224.ec2.internal nagios[66907]: Error: Unable to create temp file '/usr/local/nagios/var/nagios.tmpqVtUf' for writing status data: Permissi>
Sep 23 09:34:21 ip-172-31-90-224.ec2.internal nagios[66907]: Error: Unable to create temp file '/usr/local/nagios/var/nagios.tmpqHUEUI9' for writing status data: Permissi>
Sep 23 09:34:24 ip-172-31-90-224.ec2.internal nagios[66907]: SERVICE ALERT: localhost;HTTP;WARNING;SOFT;2;HTTP WARNING: HTTP/1.1 403 Forbidden - 319 bytes in 0.001 secon>
Sep 23 09:34:31 ip-172-31-90-224.ec2.internal nagios[66907]: Error: Unable to create temp file '/usr/local/nagios/var/nagios.tmpqpy1CO9' for writing status data: Permissi>
lines 1-26/26 (END)
[ec2-user@ip-172-31-90-224 nagios-plugins-2.0.3]$ ls -ld /usr/local/nagios/var
```

Error:Sep 23 09:34:31 ip-172-31-90-224.ec2.internal nagios[66907]: Error: Unable to create temp file '/usr/local/nagios/var/nagios.tmpqpy1CO9' for writing status data: Permissi>

Solution:

- ls -ld /usr/local/nagios/var
- sudo chown -R nagios:nagios /usr/local/nagios/var
- sudo chmod -R 755 /usr/local/nagios/var
- sudo systemctl restart nagios

Rerun the command

sudo systemctl status nagios

```

[ec2-user@ip-172-31-83-157 nagios-plugins-2.4.11]$ sudo systemctl status nagios
● nagios.service - Nagios Core 4.5.5
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
   Active: active (running) since Sat 2024-09-28 04:18:59 UTC; 12s ago
     Docs: https://www.nagios.org/documentation
   Process: 66052 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg (code=exited, status=0/SUCCESS)
   Process: 66053 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg (code=exited, status=0/SUCCESS)
   Main PID: 66054 (nagios)
    Tasks: 6 (limit: 1112)
   Memory: 5.6M
      CPU: 77ms
   CGroup: /system.slice/nagios.service
           └─66054 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             66055 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             66056 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             66057 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             66058 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             66059 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

Sep 28 04:18:59 ip-172-31-83-157.ec2.internal nagios[66054]: qh: Socket '/usr/local/nagios/var/rw/nagios.qh' successfully initialized
Sep 28 04:18:59 ip-172-31-83-157.ec2.internal nagios[66054]: qh: core query handler registered
Sep 28 04:18:59 ip-172-31-83-157.ec2.internal nagios[66054]: qh: echo service query handler registered
Sep 28 04:18:59 ip-172-31-83-157.ec2.internal nagios[66054]: qh: help for the query handler registered
Sep 28 04:18:59 ip-172-31-83-157.ec2.internal nagios[66054]: wproc: Successfully registered manager as @wproc with query handler
Sep 28 04:18:59 ip-172-31-83-157.ec2.internal nagios[66054]: wproc: Registry request: name=Core Worker 66057;pid=66057
Sep 28 04:18:59 ip-172-31-83-157.ec2.internal nagios[66054]: wproc: Registry request: name=Core Worker 66058;pid=66058
Sep 28 04:18:59 ip-172-31-83-157.ec2.internal nagios[66054]: wproc: Registry request: name=Core Worker 66056;pid=66056
Sep 28 04:18:59 ip-172-31-83-157.ec2.internal nagios[66054]: wproc: Registry request: name=Core Worker 66055;pid=66055
Sep 28 04:18:59 ip-172-31-83-157.ec2.internal nagios[66054]: Successfully launched command file worker with pid 66059
[ec2-user@ip-172-31-83-157 nagios-plugins-2.4.11]$

```

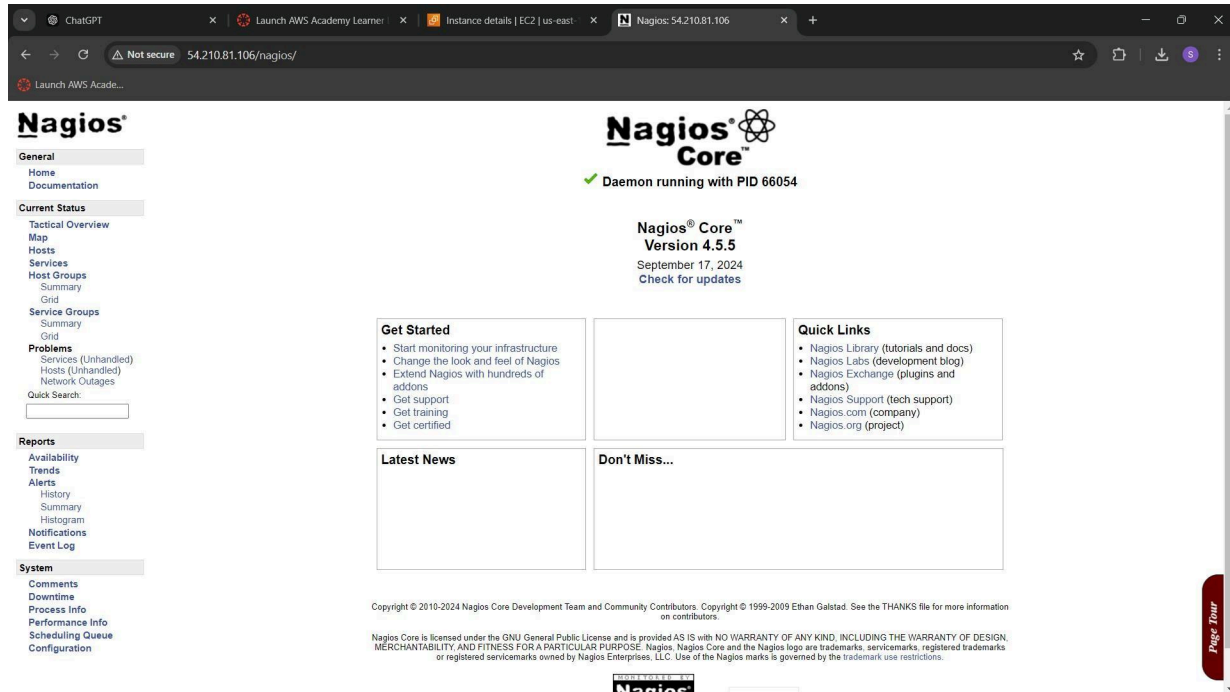
Step 28: Now, go to EC2 instance and click on instance id. Then, click on the copy icon just before the public ip address on public IP.

The screenshot shows the AWS Management Console interface. The left sidebar contains navigation links for EC2 Dashboard, EC2 Global View, Events, Console-to-Code, and various instance management tools. The main content area displays the 'Instance details' for the instance 'i-0e1c706c2051d5121' (nagios-host-27). The instance is in the 'Running' state. Key details include:

- Instance ID:** i-0e1c706c2051d5121 (nagios-host-27)
- Public IPv4 address:** 3.89.111.169 (with a copy icon)
- Private IPv4 address:** 172.31.90.224
- Public IPv4 DNS:** ec2-3-89-111-169.compute-1.amazonaws.com (with a copy icon)
- Instance state:** Running
- Private IP DNS name (IPv4 only):** ip-172-31-90-224.ec2.internal
- Instance type:** t2.micro
- VPC ID:** vpc-05537c9b6c7862468
- Subnet ID:** subnet-0188b4375dc68c2fb
- Instance ARN:** arn:aws:ec2:us-east-1:196908236276:instance/i-0e1c706c2051d5121
- Platform:** Amazon Linux (Inferred)
- Platform details:** Linux/UNIX
- Stop protection:** Disabled
- Launch time:** Mon Sep 23 2024 14:13:55 GMT+0530 (India Standard Time) (about 1 hour)
- AMI ID:** ami-0ebfd941bbafe70c6
- AMI name:** al2023-ami-2023.5.20240916.0-kernel-6.1-x86_64
- Monitoring:** disabled
- Termination protection:** Disabled
- AMI location:** amazon/al2023-ami-2023.5.20240916.0-kernel-6.1-x86_64

The bottom of the console shows the footer with copyright information for Amazon Web Services, Inc. or its affiliates, and links to Privacy, Terms, and Cookie preferences.

Step 29: Open a new tab. In the address bar type `http://<publicipaddress>/nagios`. This would be in the output



Conclusion:

In this experiment, we have learned how to install and configure Nagios Core, Nagios Plugins and NRPE on a Linux machine. We are using an Amazon Linux OS instance configured with the need security rules. We need to make sure that the Nagios-core and Nagios-plugins links that are used are the ones which are up-to date (wget commands). It is needed to extract and configure these files so that no issues are detected while starting the server. Once all the setup is complete, we can start the nagios server. Using the public IP address of the EC2 instance, we can access the nagios dashboard by navigating that IP to nagios.

EXPERIMENT NO. 8

Aim: Create a Jenkins CI/CD Pipeline with SonarQube / GitLab Integration to perform a static analysis of the code to detect bugs, code smells, and security vulnerabilities on a sample Web /Java / Python application.

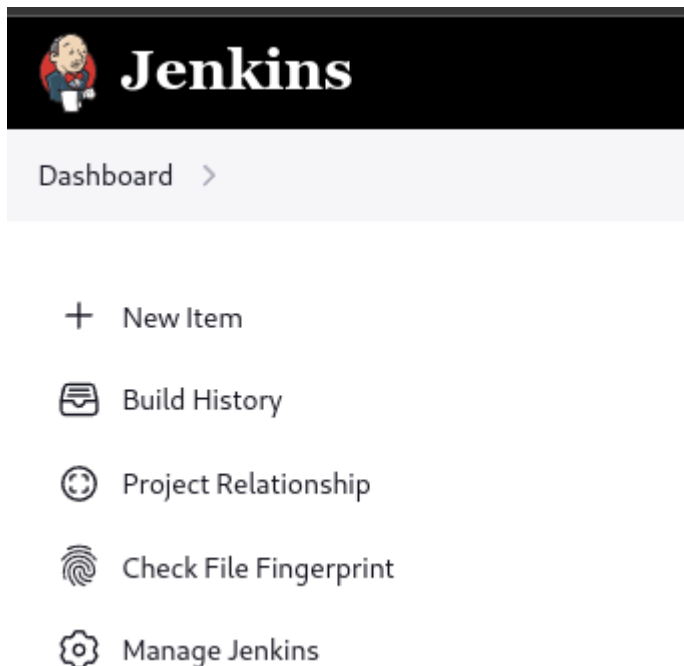
Integrating Jenkins with SonarQube:

Prerequisites:

- Jenkins installed
- Docker Installed (for SonarQube)
- SonarQube Docker Image

Steps to create a Jenkins CI/CD Pipeline and use SonarQube to perform SAST

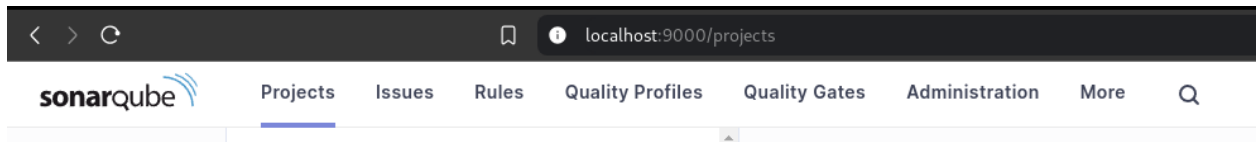
1. Open up Jenkins Dashboard on localhost, port 8080 or whichever port it is at for you.



2. Run SonarQube in a Docker container using this command

```
quantum@machine ~$ sudo docker run -d --name sonarqube -e SONAR_ES_BOOTSTRAP_CHECKS_DISABLE=true -p 9000:9000 sonarqube:latest
```

- Once the container is up and running, you can check the status of SonarQube at localhost port 9000.



- Login to SonarQube using your username and password
- Create a manual project in SonarQube with the name sonarqube-test2

1 of 2

Create a local project

Project display name *

 ✓

Project key *

 ✓

Main branch name *

The name of your project's default branch [Learn More](#)

Cancel

Next

database should be used for evaluation purposes only

ed database will not scale, it will not support upgrading to newer versions of Sonar

Setup the project and come back to Jenkins Dashboard.

- Create a New Item in Jenkins, choose **Pipeline**.

New Item

Enter an item name

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.



Organization Folder

Creates a set of multibranch project subfolders by scanning for repositories.

7. Under Pipeline Script, enter the following -
- ```
node {
 stage('Cloning the GitHub Repo') {
 git 'https://github.com/shazforiot/GOL.git'
 }
 stage('SonarQube analysis') {
 withSonarQubeEnv('sonarqube')
 {
 sh "<PATH_TO_SONARQUBE_FOLDER>/bin//sonar-scanner \
 -D sonar.login=<SonarQube_USERNAME> \
 -D sonar.password=<SonarQube_PASSWORD> \
 -D sonar.projectKey=<Project_KEY> \
 -D sonar.exclusions=vendor/**,resources/**,**/*.java \
 -D sonar.host.url=http://127.0.0.1:9000/"
```

```

}
}
}

```

It is a java sample project which has a lot of repetitions and issues that will be detected by SonarQube.

#### 8. Install sonar-scanner

Now we need to install **sonar-scanner** binary to perform code analysis

To do so, go to

<https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-6.2.0.4584-linux-x64.zip> for linux OS

(<https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-6.2.0.4584-windows-x64.zip> for windows OS)

You will be prompted to download the zip file, download it and extract it and copy the path (absolute path) to <ROOT\_DIRECTORY>/bin/sonar-scanner

For windows path would be

C:\\Users\\<USER\_NAME>\\Downloads\\sonar-scanner-cli-6.2.0.4584-windows-x64\\sonar-scanner-cli-6.2.0.4584-windows-x64\\bin\\sonar-scanner in my case its

"C:\\Users\\ADITYA DUBEY\\Downloads\\sonar-scanner-cli-6.2.0.4584-windows-x64\\sonar-scanner-cli-6.2.0.4584-windows-x64\\bin\\sonar-scanner" // " " for paths with space

Now we need to update the required details in the pipeline script as :

```

pipeline {
 agent any
 stages {
 stage('Cloning the GitHub Repo') {
 steps {
 git 'https://github.com/shazforiot/GOL.git'
 }
 }
 stage('SonarQube analysis') {
 steps {
 withSonarQubeEnv('sonarqube') {configuration
 sh """

```

```

"C:\\Users\\ADITYA DUBEY\\Downloads\\sonar-scanner-cli-6.2.0.4584-windows-x64\\
 sonar-scanner-cli-6.2.0.4584-windows-x64\\bin\\sonar-scanner" ^
 -D sonar.projectKey=sonarqube-test2 ^
 -D sonar.sources=. ^
 -D sonar.exclusions=vendor/**,resources/**,**/*.java ^

```



```
-D sonar.host.url=http://127.0.0.1:9000 ^
-D sonar.login=admin ^
-D sonar.password=my pass
```

```
"""
```

```
}
}
}
}
```

Script ?

```
1 pipeline {
2 agent any
3 stages {
4 stage('Cloning the GitHub Repo') {
5 steps {
6 git 'https://github.com/shazforiot/GOL.git'
7 }
8 }
9 stage('SonarQube Analysis') {
10 steps {
11 withSonarQubeEnv('sonarqube') {
12 bat """
13 "C:/Users/ADITYA DUBEY/Downloads/sonar-scanner-cli-6.2.0.4584-windows-x64/sonar-scanner-6.2.0.4584-windows-x64/bin/sonar"
14 -D sonar.projectKey=SONARQUBE_PROJECT2 ^
15 -D sonar.sources=. ^
16 -D sonar.exclusions=vendor/**,resources/**,**/*.java ^
17 -D sonar.host.url=http://127.0.0.1:9000 ^
18 -D sonar.login=admin ^
19 -D sonar.password=a1b2d4e5i9j10
20 """
21 }
22 }
23 }
24 }
25 }
26
```

## 9. Run the build

Dashboard > SonarQube-exp8 >

 Status

 Changes

 Build Now

 Configure

 Delete Pipeline

 Full Stage View

 Stages

 Rename

 Pipeline Syntax

Check the status of Build



SonarQube analysis for EXP 8

## Stage View



As we can see the SonarQube analysis is completed

10. Check the console output once the build is complete.

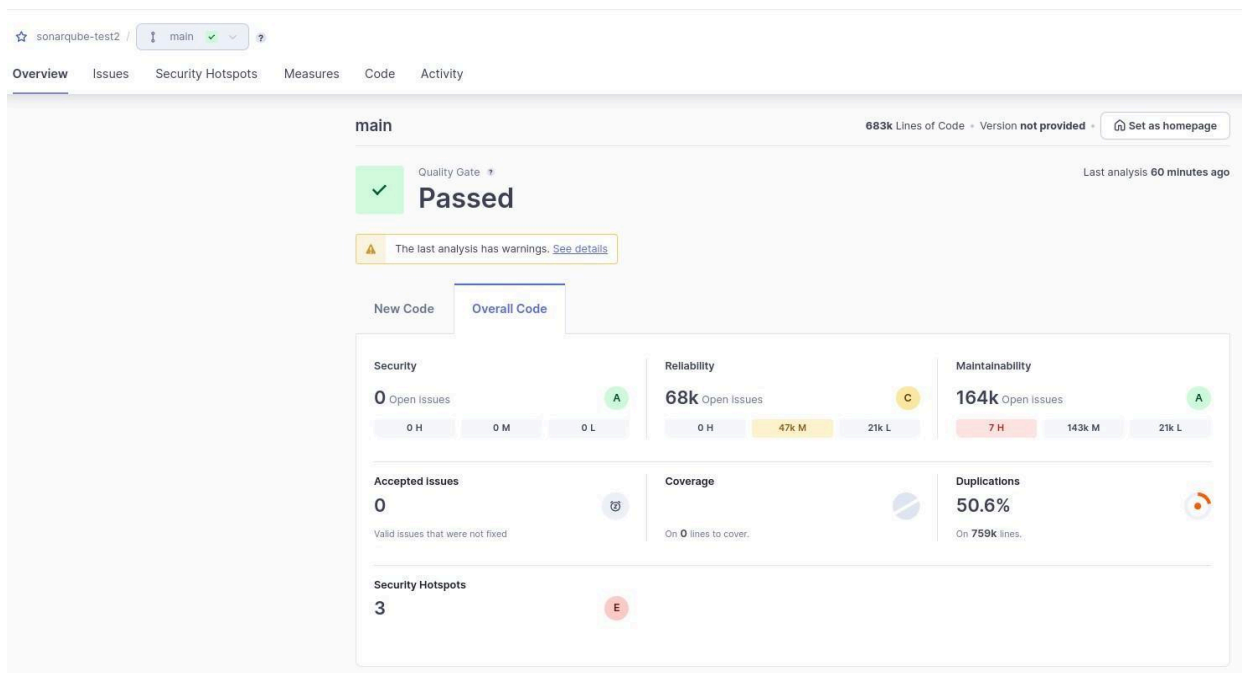
### Console Output

Skipping 4,226 KB.. [Full Log](#)

```
20:11:28.619 WARN Too many duplication references on file gameoflife-web/tools/jmeter/docs/api/org/apache/jmeter/gui/ReportMainFrame.WindowHappenings.html for block at line 296. Keep only the first 100 references.
20:11:28.619 WARN Too many duplication references on file gameoflife-web/tools/jmeter/docs/api/org/apache/jmeter/gui/ReportMainFrame.WindowHappenings.html for block at line 17. Keep only the first 100 references.
20:11:28.619 WARN Too many duplication references on file gameoflife-web/tools/jmeter/docs/api/org/apache/jmeter/gui/ReportMainFrame.WindowHappenings.html for block at line 212. Keep only the first 100 references.
20:11:28.619 WARN Too many duplication references on file gameoflife-web/tools/jmeter/docs/api/org/apache/jmeter/gui/ReportMainFrame.WindowHappenings.html for block at line 215. Keep only the first 100 references.
20:11:28.619 WARN Too many duplication references on file gameoflife-web/tools/jmeter/docs/api/org/apache/jmeter/gui/ReportMainFrame.WindowHappenings.html for block at line 298. Keep only the first 100 references.
20:11:28.619 WARN Too many duplication references on file gameoflife-web/tools/jmeter/docs/api/org/apache/jmeter/gui/ReportMainFrame.WindowHappenings.html for block at line 300. Keep only the first 100 references.
20:11:28.619 WARN Too many duplication references on file gameoflife-web/tools/jmeter/docs/api/org/apache/jmeter/gui/ReportMainFrame.WindowHappenings.html for block at line 215. Keep only the first 100 references.
20:11:28.619 WARN Too many duplication references on file gameoflife-web/tools/jmeter/docs/api/org/apache/jmeter/gui/ReportMainFrame.WindowHappenings.html for block at line 225. Keep only the first 100 references.
```

```
20:11:31.504 INFO CPD Executor CPD calculation finished (done) | time=104775ms
20:11:31.748 INFO SCM revision ID 'ba799ba7e1b576f04a4612322b0412c5e6e1e5e4'
20:11:33.306 INFO Analysis report generated in 1507ms, dir size=127.2 MB
20:11:40.819 INFO Analysis report compressed in 7511ms, zip size=29.6 MB
20:11:42.147 INFO Analysis report uploaded in 1322ms
20:11:42.154 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://127.0.0.1:9000/dashboard?id=sonarqube-test2
20:11:42.154 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
20:11:42.154 INFO More about the report processing at http://127.0.0.1:9000/api/ce/task?id=670aec3c-6c20-460b-ad52-ec6ce041fb1f
20:11:46.802 INFO Analysis total time: 3:42.216 s
20:11:46.835 INFO SonarScanner Engine completed successfully
20:11:48.173 INFO EXECUTION SUCCESS
20:11:48.296 INFO Total time: 3:47.113s
[Pipeline] }
[Pipeline] // withSonarQubeEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline completed.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

11. After that, check the project in SonarQube.



Under different tabs, check all different issues with the code

## 12. Code Problems-

sonarqube-test2 / I main

Overview Issues Security Hotspots Measures Code Activity

My Issues All

Filters

Issues in new code

**Clean Code Attribute**

|                |      |
|----------------|------|
| Consistency    | 197k |
| Intentionality | 14k  |
| Adaptability   | 0    |
| Responsibility | 0    |

Add to selection Ctrl + click

**Software Quality**

|                 |      |
|-----------------|------|
| Security        | 0    |
| Reliability     | 54k  |
| Maintainability | 164k |

**Severity**

|        |      |
|--------|------|
| High   | 0    |
| Medium | 176k |
| Low    | 21k  |

gameoflife-core/build/reports/tests/all-tests.html

190,662 issues 3075d effort

Insert a <DOCTYPE> declaration to before this <html> tag.

Consistency

Reliability

Open Not assigned

L1 - 5min effort - 4 years ago - Bug - Major

Remove this deprecated "width" attribute.

Consistency

Maintainability

Open Not assigned

L8 - 5min effort - 4 years ago - Code Smell - Major

Remove this deprecated "align" attribute.

Consistency

Maintainability

Open Not assigned

L11 - 5min effort - 4 years ago - Code Smell - Major

Remove this deprecated "align" attribute.

Consistency

Maintainability

Open Not assigned

L12 - 5min effort - 4 years ago - Code Smell - Major

Remove this deprecated "size" attribute.

Consistency

## Code Smells

sonarqube-test2 / I main

Overview Issues Security Hotspots Measures Code Activity

Responsibility

|                 |      |
|-----------------|------|
| Security        | 0    |
| Reliability     | 21k  |
| Maintainability | 164k |

**Software Quality**

|                 |      |
|-----------------|------|
| Security        | 0    |
| Reliability     | 21k  |
| Maintainability | 164k |

**Severity**

|        |      |
|--------|------|
| High   | 7    |
| Medium | 143k |
| Low    | 21k  |

**Type**

|               |      |
|---------------|------|
| Bug           | 47k  |
| Vulnerability | 0    |
| Code Smell    | 164k |

Add to selection Ctrl + click

**Scope**

|           |      |
|-----------|------|
| Main code | 164k |
| Test code | 0    |

gameoflife-acceptance-tests/Dockerfile

164,034 issues 1708d effort

Use a specific version tag for the image.

Intentionality

Maintainability

Open Not assigned

L1 - 5min effort - 4 years ago - Code Smell - Major

Surround this variable with double quotes; otherwise, it can lead to unexpected behavior.

Intentionality

Maintainability

Open Not assigned

L12 - 5min effort - 4 years ago - Code Smell - Major

Surround this variable with double quotes; otherwise, it can lead to unexpected behavior.

Intentionality

Maintainability

Open Not assigned

L12 - 5min effort - 4 years ago - Code Smell - Major

Surround this variable with double quotes; otherwise, it can lead to unexpected behavior.

Intentionality

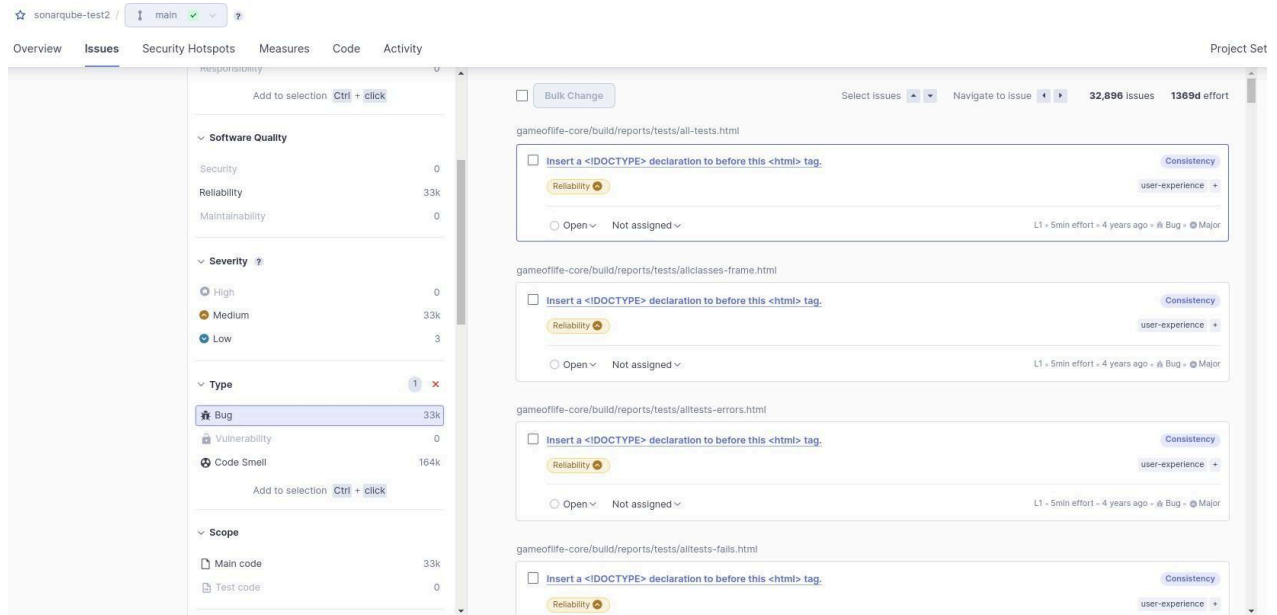
Maintainability

Open Not assigned

L13 - 5min effort - 4 years ago - Code Smell - Major

gameoflife-core/build/reports/tests/all-tests.html

## Bugs



**Conclusion:** We began the experiment with creating a new project in SonarQube and setting up a new Pipeline in Jenkins with proper configuration of pipeline script. Then we installed Sonar Scanner CLI so that Jenkins can do code analysis of Git Repository. We can also configure the pipeline to use the installed Sonar Scanner plugin instead of locally installed Binary of Sonar Scanner. The pipeline ran successfully with all tests passed in SonarQube.

## EXPERIMENT NO. 7

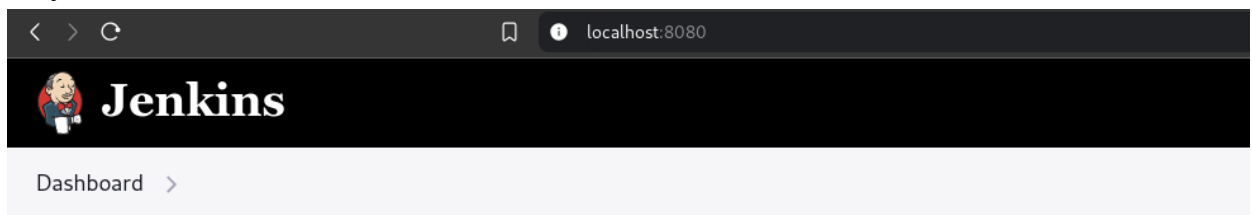
**Aim:** To understand Static Analysis SAST process and learn to integrate Jenkins SAST to SonarQube/GitLab.

Prerequisites:

- Jenkins installed (Java JDK required)
- Docker Installed (for SonarQube)

### Steps to integrate Jenkins with SonarQube

1. Open up Jenkins Dashboard on localhost, port 8080 or whichever port it is at for you.



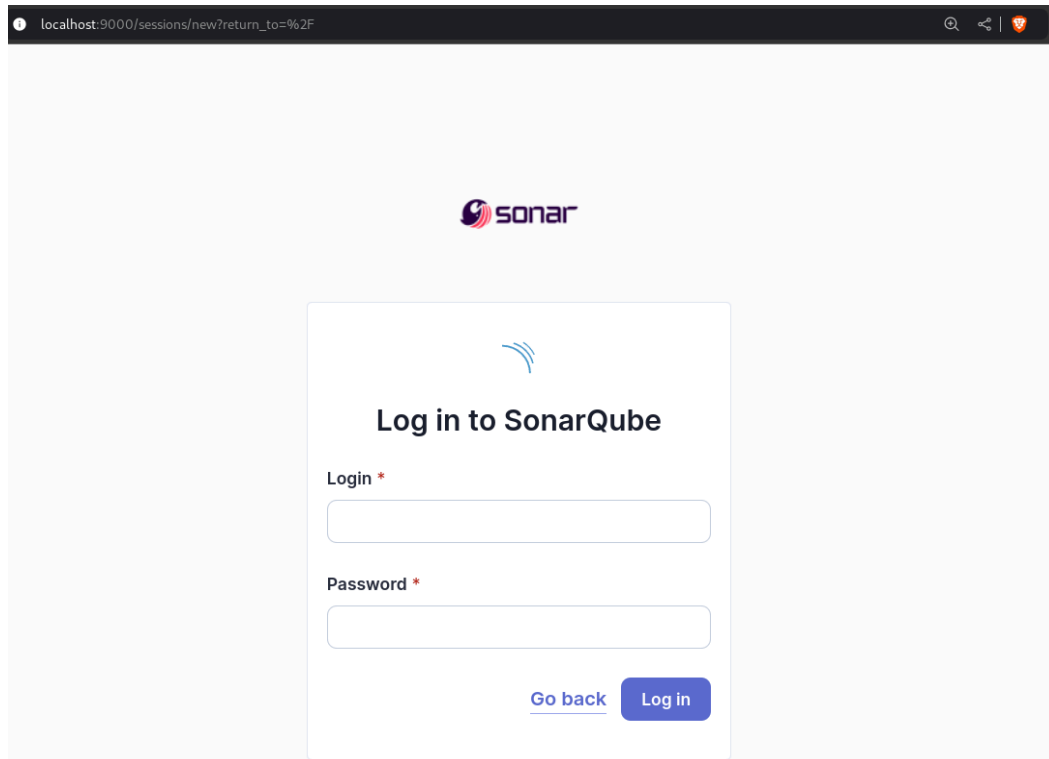
Our jenkins is running on port 8080

2. Run SonarQube in a Docker container using this command  
`sudo docker run -d --name sonarqube -e SONAR_ES_BOOTSTRAP_CHECKS_DISABLE=true -p 9000:9000 sonarqube:latest`

**Warning: run below command only once**

```
quantum@machine ~$ sudo docker run -d --name sonarqube -e SONAR_ES_BOOTSTRAP_CHECKS_DISABLE=true -p 9000:9000 sonarqube:latest
[sudo] password for quantum:
f9c595308e368210e19e099256a47ec1fe44affdc778eb58ccb53174163ce057
```

3. Once the container is up and running, you can check the status of SonarQube at localhost port 9000.



4. Login to SonarQube using username admin and password admin.



localhost:9000/account/reset\_password

## Update your password

 This account should not use the default password.

### Enter a new password

All fields marked with \* are required

**Old Password \***

**New Password \***

**Confirm Password \***

**Update**

After logging, we have to change default password

5. Create a manual project in SonarQube with the name sonarqube Click on **create a local project** on dashboard

First, you need to set up a DevOps platform configuration.

|                                                                                                              |              |                                                                                                                 |              |
|--------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------|--------------|
|  Import from Azure DevOps | <b>Setup</b> |  Import from Bitbucket Cloud | <b>Setup</b> |
|  Import from GitHub       | <b>Setup</b> |  Import from GitLab          | <b>Setup</b> |

Are you just testing or have an advanced use-case? Create a local project.

**Create a local project**

1 of 2

## Create a local project

Project display name \*

SONARQUBE\_PROJECT 

Project key \*

SONARQUBE\_PROJECT 

The project key is a unique identifier for your project. It may contain up to 400 characters. Allowed characters are alphanumeric, '-' (dash), '\_' (underscore), '.' (period) and ':' (colon), with at least one non-digit.

Main branch name \*

main

The name of your project's default branch [Learn More](#) 

Cancel

Next

2 of 2

## Set up project for Clean as You Code

The new code definition sets which part of your code will be considered new code. This helps follow the Clean as You Code methodology. Learn more: [Defining New Code](#) 

### Choose the baseline for new code for this project

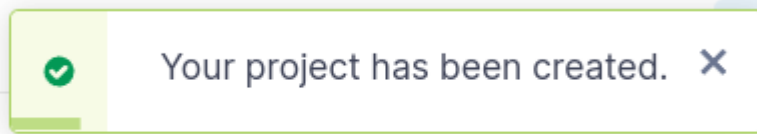
☒ Use the global setting

#### Previous version

Any code that has changed since the previous version is considered new code.

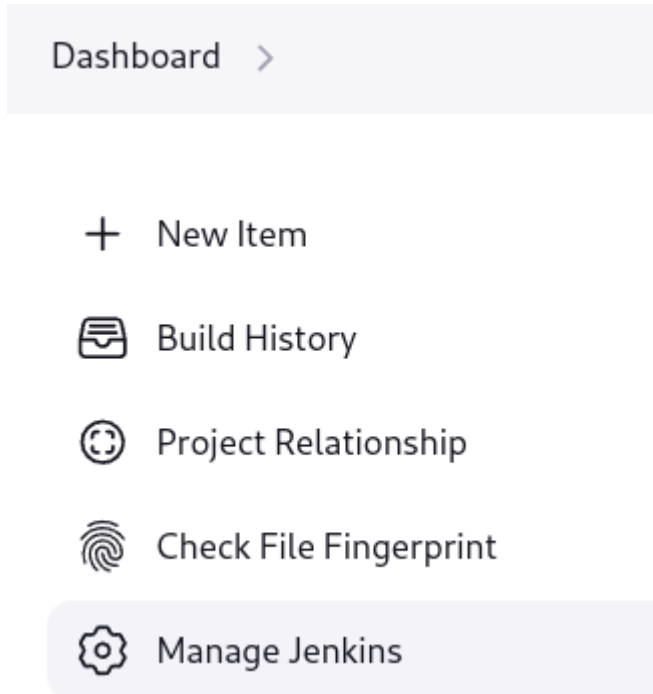
Recommended for projects following regular versions or releases.

We can either use new custom settings for the project or use global settings  
Here I'm using global setting

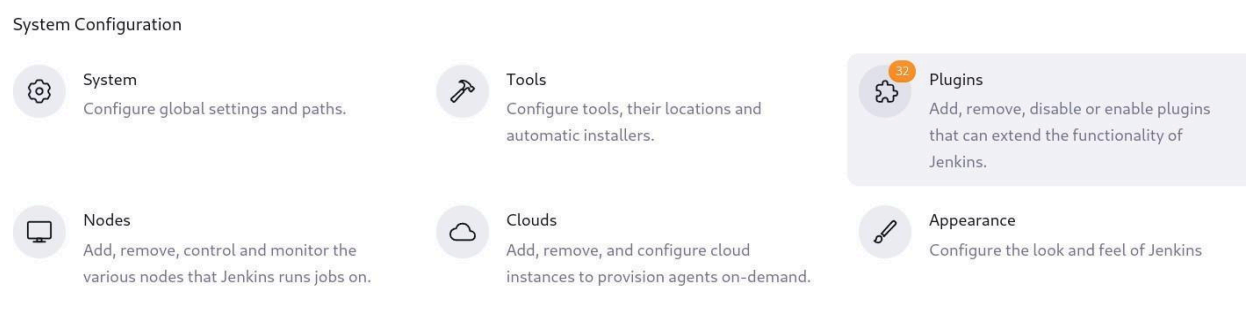


On successful creation of project, we a popup for the same

6. After setting project in sonarqube, go to **Jenkins Dashboard**



Go to Manage Jenkins and search for SonarQube Scanner in Plugins settings and install it.



| Install                             | Name ↓                                                                                                                                                                                                                                    |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | <div><div>SonarQube Scanner 2.17.2</div><div>External Site/Tool Integrations Build Reports</div><div>This plugin allows an easy integration of SonarQube, the open source platform for Continuous Inspection of code quality.</div></div> |
| <input type="checkbox"/>            | <div><div>Sonar Gerrit 388.v9b_f1cb_e42306</div><div>External Site/Tool Integrations</div><div>This plugin allows to submit issues from SonarQube to Gerrit as comments directly.</div></div>                                             |
| <input type="checkbox"/>            | <div><div>SonarQube Generic Coverage 1.0</div><div>TODO</div></div>                                                                                                                                                                       |

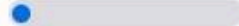
## Download progress

### Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

SonarQube Scanner

Installing



Loading plugin extensions

Pending

→ [Go back to the top page](#)

(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

Our installation is in progress wait for it to download and install packages

## Download progress

### Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

SonarQube Scanner

 Success

Loading plugin extensions

 Success

Plugin installed successfully

7. Under Jenkins dashboard 'Configure System', look for SonarQube Servers and enter the details.

### System Configuration



#### System

Configure global settings and paths.



#### Tools

Configure tools, their locations and automatic installers.



#### Nodes

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.



#### Clouds

Add, remove, and configure cloud instances to provision agents on-demand.

## SonarQube servers

If checked, job administrators will be able to inject a SonarQube server configuration as environment variables in the build.

☒ Environment variables

## SonarQube installations

List of SonarQube installations

Name

sonarqube

Server URL

Default is http://localhost:9000

http://localhost:9000

Server authentication token

SonarQube authentication token. Mandatory when anonymous access is disabled.

- none -

+ Add

8. Search for SonarQube Scanner under Global Tool Configuration. Choose the latest configuration and choose Install automatically.

## System Configuration



**System**  
Configure global settings and paths.



**Tools**  
Configure tools, their locations and automatic installers.



**Nodes**  
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.



**Clouds**  
Add, remove, and configure cloud instances to provision agents on-demand.

## SonarScanner for MSBuild installations

Add SonarScanner for MSBuild

## SonarQube Scanner installations

Add SonarQube Scanner

Click on **Add SonarQube Scanner**

## SonarQube Scanner installations

Add SonarQube Scanner

☰

SonarQube Scanner

✕

Name

sonarqube installer

☒ Install automatically ?

☰

Install from Maven Central

✕

Version

SonarQube Scanner 6.2.0.4584

▼

Add Installer ▼

Select Latest version and save configuration

9. After the configuration, create a New Item in Jenkins, choose a freestyle project.

**New Item**

Enter an item name

SonarQube

Select an item type

**Freestyle project**

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

**Maven project**

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.

**Pipeline**

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

**Multi-configuration project**

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

10. Choose this GitHub repository in Source Code Management.

[https://github.com/shazforiot/MSBuild\\_firstproject.git](https://github.com/shazforiot/MSBuild_firstproject.git) It is a sample hello-world project with no vulnerabilities and issues, just to test the integration

Source Code Management

☐ None

☒ Git ?

Repositories ?

Repository URL ?

Credentials ?

- none -

+ Add

Advanced

Add Repository

Branches to build ?

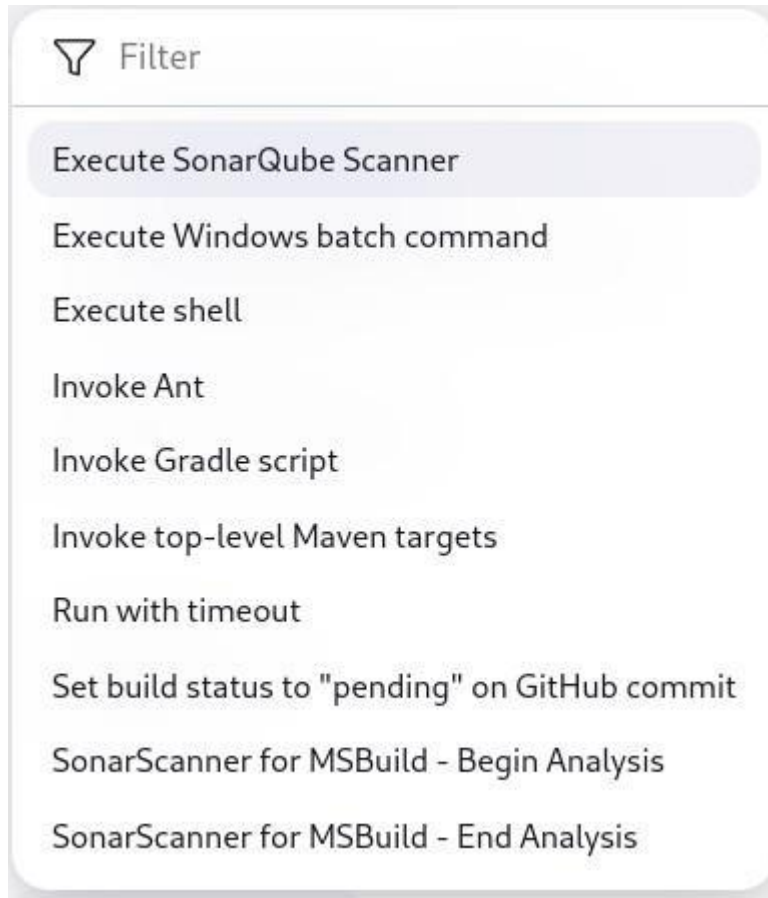
11. Under Build-> Execute SonarQube Scanner

## Build Steps

Add build step

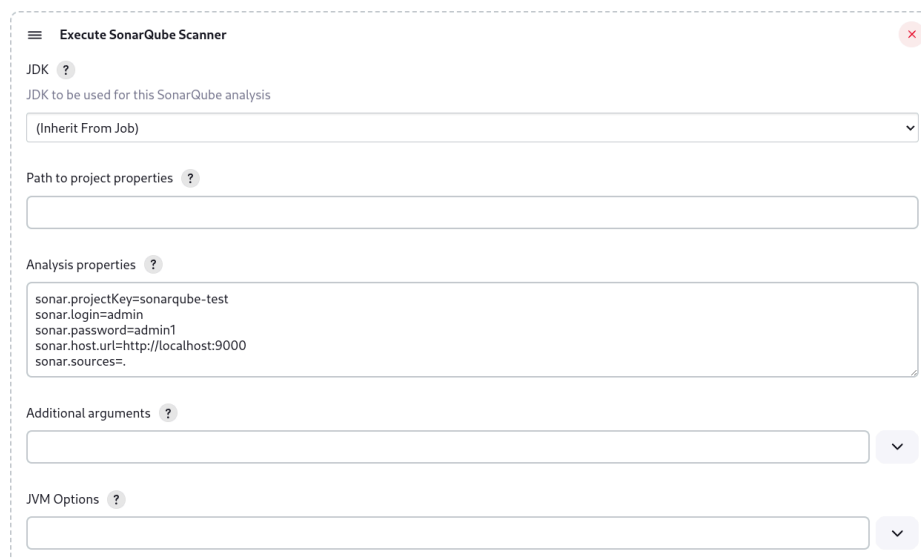
Click on add build steps





Then click on **Execute SonarQube Scanner**

12. Mention the SonarQube Project Key, Login, Password, Source path and Host URL in Analysis properties



13. Go to [http://localhost:9000/project\\_roles?id=<project\\_key>](http://localhost:9000/project_roles?id=<project_key>) and allow Execute Permissions to the Admin user.

| All | Users         | Groups | Search for users or groups... | Administer Issues ?      | Administer Security Hotspots ? | Administer ?             | Execute Analysis ?                  |
|-----|---------------|--------|-------------------------------|--------------------------|--------------------------------|--------------------------|-------------------------------------|
| A   | Administrator | admin  |                               | <input type="checkbox"/> | <input type="checkbox"/>       | <input type="checkbox"/> | <input checked="" type="checkbox"/> |

1 of 1 shown

14. Run The Build.

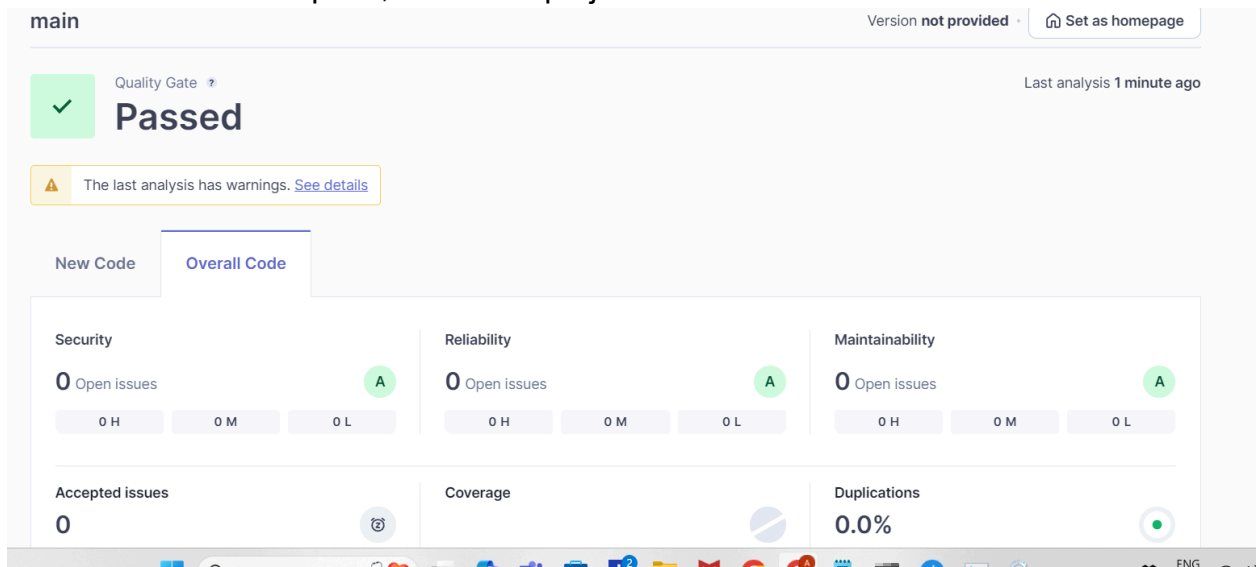
- Status
- Changes
- Workspace
- Build Now
- Configure
- Delete Project
- SonarQube
- Rename

15. Check the console output

```
Running as SYSTEM
Building on the built-in node in workspace C:\Users\ADITYA DUBEY\.jenkins\workspace\Sonarqube
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --resolve-git-dir C:\Users\ADITYA DUBEY\.jenkins\workspace\Sonarqube\.git # timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/shazforiot/MSBuild_firstproject.git # timeout=10
Fetching upstream changes from https://github.com/shazforiot/MSBuild_firstproject.git
> git.exe --version # timeout=10
> git --version # 'git version 2.46.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/shazforiot/MSBuild_firstproject.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision f2bc042c04c6e72427c380bcaee6d6fee7b49adf (refs/remotes/origin/master)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f f2bc042c04c6e72427c380bcaee6d6fee7b49adf # timeout=10
Commit message: "updated"
> git.exe rev-list --no-walk f2bc042c04c6e72427c380bcaee6d6fee7b49adf # timeout=10
```

```
10:30:32.234 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=sonarqube-test
10:30:32.234 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
10:30:32.235 INFO More about the report processing at http://localhost:9000/api/ce/task?id=96795973-9667-4456-a15c-3311dbc9d067
10:30:32.244 INFO Analysis total time: 9.533 s
10:30:32.245 INFO SonarScanner Engine completed successfully
10:30:32.300 INFO EXECUTION SUCCESS
10:30:32.301 INFO Total time: 14.090s
Finished: SUCCESS
```

16. Once the build is complete, check the project in SonarQube.



In this way, we have integrated Jenkins with SonarQube for SAST.

**Conclusion:** We began the experiment with installation of SonarQube Docker Image followed by setting up a new project in SonarQube. Then we installed the SonarQube scanner plugin and then created a new freestyle project in Jenkins with a Git repository for code analysis. Then we configured the Jenkins with appropriate settings to work with SonarQube. Gave permissions to Jenkins to perform code analysis. It is essential to provide correct properties in **Analysis Properties** for Jenkins to run correctly. The Jenkins project ran successfully with all tests passed in SonarQube.

## Creating docker image using terraform

Step 1: Download and install Docker Desktop by visiting <https://www.docker.com>. Run the installer and follow the prompts to complete the installation, then verify by launching Docker Desktop or using the `docker --version` command.

```
C:\Users\Admin>docker
```

```
Usage: docker [OPTIONS] COMMAND
```

```
A self-sufficient runtime for containers
```

### Common Commands:

|         |                                              |
|---------|----------------------------------------------|
| run     | Create and run a new container from an image |
| exec    | Execute a command in a running container     |
| ps      | List containers                              |
| build   | Build an image from a Dockerfile             |
| pull    | Download an image from a registry            |
| push    | Upload an image to a registry                |
| images  | List images                                  |
| login   | Log in to a registry                         |
| logout  | Log out from a registry                      |
| search  | Search Docker Hub for images                 |
| version | Show the Docker version information          |
| info    | Display system-wide information              |

### Management Commands:

|            |                                           |
|------------|-------------------------------------------|
| builder    | Manage builds                             |
| buildx*    | Docker Buildx                             |
| compose*   | Docker Compose                            |
| container  | Manage containers                         |
| context    | Manage contexts                           |
| debug*     | Get a shell into any image or container   |
| desktop*   | Docker Desktop commands (Alpha)           |
| dev*       | Docker Dev Environments                   |
| extension* | Manages Docker extensions                 |
| feedback*  | Provide feedback, right in your terminal! |
| image      | Manage images                             |

```
C:\Users\Admin>docker --version
Docker version 27.1.1, build 6312585
```

Step 2: Now, create a folder named '**Terraform Scripts**' in which we save our different types of scripts which will be further used in this experiment.

Step 3: First, create a new folder named `Docker` inside the `TerraformScripts` folder. Then, open Notepad and create a new file named `docker.tf` within the `Docker` folder. Write the following contents into the `docker.tf` file to create an Ubuntu Linux container. Save the file when done.

Script:

```
terraform {
 required_providers {
 docker = {
 source = "kreuzwerker/docker"
 version = "2.21.0"
 }
 }
}

provider "docker" {
 host = "npipe:////./pipe/docker_engine"
}

Pulls the image
resource "docker_image" "ubuntu" {
 name = "ubuntu:latest"
}

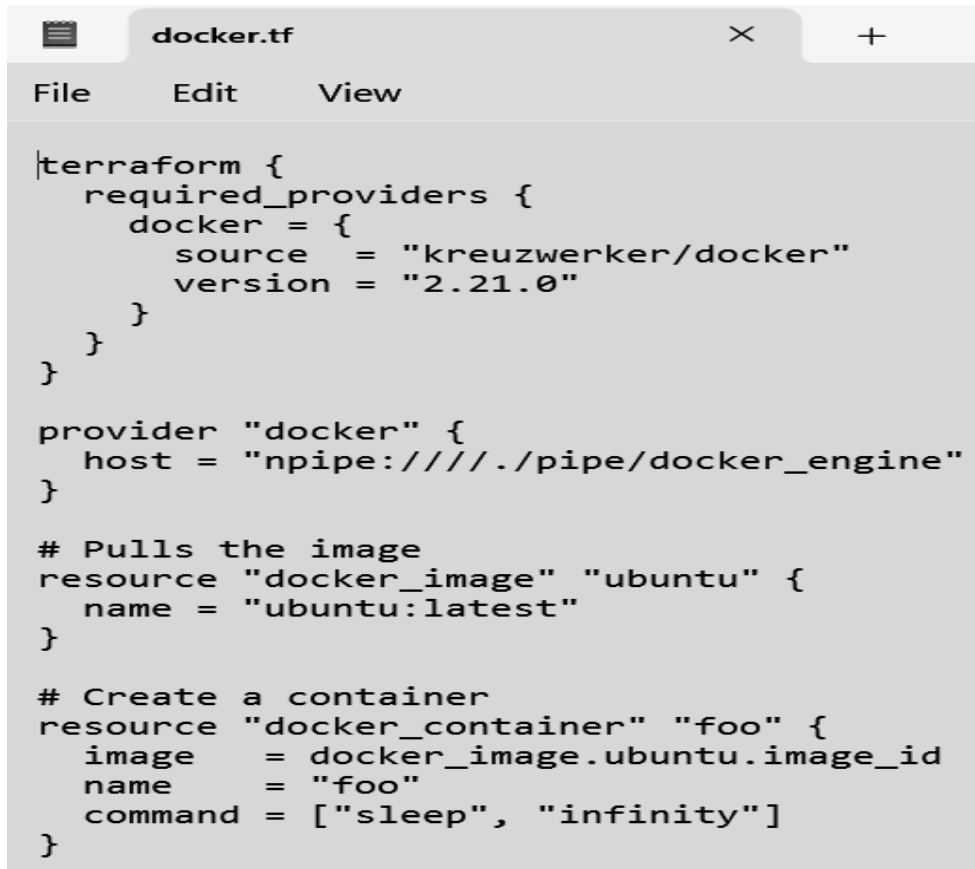
Create a container
resource "docker_container" "foo" {
 image = docker_image.ubuntu.image_id
 name = "foo"
 command = ["sleep", "infinity"]
}
```

This Terraform script configures the Docker provider to communicate with the Docker Engine using a Windows named pipe.

It pulls the latest Ubuntu image from Docker Hub and creates a container named "foo."

The container runs the `sleep infinity` command, which keeps it active indefinitely.

This setup is useful for scenarios where the container needs to remain running continuously.



```
terraform {
 required_providers {
 docker = {
 source = "kreuzwerker/docker"
 version = "2.21.0"
 }
 }
}

provider "docker" {
 host = "npope:///./pipe/docker_engine"
}

Pulls the image
resource "docker_image" "ubuntu" {
 name = "ubuntu:latest"
}

Create a container
resource "docker_container" "foo" {
 image = docker_image.ubuntu.image_id
 name = "foo"
 command = ["sleep", "infinity"]
}
```

Step 4: Execute the `terraform init` command to initialize the working directory, download the necessary provider plugins, and set up the backend for managing Terraform state.

```
PS C:\Users\Admin\Desktop\Terraformscripts\docker> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding kreuzwerker/docker versions matching "2.21.0"...
- Installing kreuzwerker/docker v2.21.0...
- Installed kreuzwerker/docker v2.21.0 (self-signed, key ID BD080C4571C6104C)
Partner and community providers are signed by their developers.
If you'd like to know more about provider signing, you can read about it here:
https://www.terraform.io/docs/cli/plugins/signing.html
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
PS C:\Users\Admin\Desktop\Terraformscripts\docker> |
```

Step 5: Run **terraform plan** to preview the actions Terraform will take to reach the desired state defined in your configuration, including creating, modifying, or deleting resources.

```
PS C:\Users\Admin\Desktop\Terraformscripts\docker> terraform plan
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:  
+ create

Terraform will perform the following actions:

```
docker_container.foo will be created
+ resource "docker_container" "foo" {
 + attach = false
 + bridge = (known after apply)
 + command = [
 + "sleep",
 + "infinity",
]
 + container_logs = (known after apply)
 + entrypoint = (known after apply)
 + env = (known after apply)
 + exit_code = (known after apply)
 + gateway = (known after apply)
 + hostname = (known after apply)
 + id = (known after apply)
 + image = (known after apply)
 + init = (known after apply)
 + ip_address = (known after apply)
 + ip_prefix_length = (known after apply)
 + ipc_mode = (known after apply)
 + log_driver = (known after apply)
 + logs = false
 + must_run = true
 + name = "foo"
 + network_data = (known after apply)
 + read_only = false
```

```
 + read_only = false
 + remove_volumes = true
 + restart = "no"
 + rm = false
 + runtime = (known after apply)
 + security_opts = (known after apply)
 + shm_size = (known after apply)
 + start = true
 + stdin_open = false
 + stop_signal = (known after apply)
 + stop_timeout = (known after apply)
 + tty = false
```

```
 + healthcheck (known after apply)
```

```
 + labels (known after apply)
}
```

```
docker_image.ubuntu will be created
+ resource "docker_image" "ubuntu" {
 + id = (known after apply)
 + image_id = (known after apply)
 + latest = (known after apply)
 + name = "ubuntu:latest"
 + output = (known after apply)
 + repo_digest = (known after apply)
}
```

Plan: 2 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```
PS C:\Users\Admin\Desktop\Terraformscripts\docker> |
```

**Step 6:** Execute “**terraform apply**” to apply the configuration, which will automatically create and run the Ubuntu container based on our configuration.

```
PS C:\Users\Admin\Desktop\Terraformscripts\docker> terraform apply
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:  
+ create

Terraform will perform the following actions:

```
docker_container.foo will be created
+ resource "docker_container" "foo" {
 + attach = false
 + bridge = (known after apply)
 + command = [
 + "sleep",
 + "infinity",
]
 + container_logs = (known after apply)
 + entrypoint = (known after apply)
 + env = (known after apply)
 + exit_code = (known after apply)
 + gateway = (known after apply)
 + hostname = (known after apply)
 + id = (known after apply)
 + image = (known after apply)
 + init = (known after apply)
 + ip_address = (known after apply)
 + ip_prefix_length = (known after apply)
 + ipc_mode = (known after apply)
 + log_driver = (known after apply)
 + logs = false
 + must_run = true
 + name = "foo"
 + network_data = (known after apply)
 + read_only = false
 + remove_volumes = true
```

```
+ read_only = false
+ remove_volumes = true
+ restart = "no"
+ rm = false
+ runtime = (known after apply)
+ security_opts = (known after apply)
+ shm_size = (known after apply)
+ start = true
+ stdin_open = false
+ stop_signal = (known after apply)
+ stop_timeout = (known after apply)
+ tty = false
```

```
+ healthcheck (known after apply)
```

```
+ labels (known after apply)
```

```
}
```

```
docker_image.ubuntu will be created
```

```
+ resource "docker_image" "ubuntu" {
 + id = (known after apply)
 + image_id = (known after apply)
 + latest = (known after apply)
 + name = "ubuntu:latest"
 + output = (known after apply)
 + repo_digest = (known after apply)
}
```

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.  
Only 'yes' will be accepted to approve.

Enter a value: yes

docker\_image.ubuntu: Creating...

docker\_image.ubuntu: Still creating... [10s elapsed]

docker\_image.ubuntu: Still creating... [20s elapsed]

docker\_image.ubuntu: Still creating... [30s elapsed]

docker\_image.ubuntu: Creation complete after 37s [id=sha256:edbfe74c41f8a3501ce542e137cf28ea04dd03e6df8c9d66519b6ad761c2598aubuntu:latest]

docker\_container.foo: Creating...

docker\_container.foo: Creation complete after 2s [id=76c6390ec277dc11f709997c4eb636ee8b45f90723d0f6ec07d85caeda18ead9]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

```
PS C:\Users\Admin\Desktop\Terraformscripts\docker> |
```



Step 7: The command **`docker images`** lists all Docker images stored locally on your system, showing details like repository names, tags, image IDs, and creation dates.

Docker images, Before Executing Apply step:

```
PS C:\Users\Admin\Desktop\Terraformscripts\docker> docker images
```

| REPOSITORY | TAG | IMAGE ID | CREATED | SIZE |
|------------|-----|----------|---------|------|
|------------|-----|----------|---------|------|

Docker images, After Executing Apply step:

```
PS C:\Users\Admin\Desktop\Terraformscripts\docker> docker images
```

| REPOSITORY | TAG    | IMAGE ID    | CREATED     | SIZE   |
|------------|--------|-------------|-------------|--------|
| ubuntu     | latest | edbf74c41f8 | 3 weeks ago | 78.1MB |

Step 8: Execute Terraform destroy to delete the configuration, which will automatically delete the Ubuntu Container.

```
PS C:\Users\Admin\Desktop\Terraformscripts\docker> terraform destroy
docker_image.ubuntu: Refreshing state... [id=sha256:edbf74c41f8a3501ce542e137cf28ea04dd03e6df8c9d66519b6ad761c2598aubuntu:latest]
docker_container.foo: Refreshing state... [id=76c6390ec277dc11f709997c4eb636ee8b45f90723d0f6ec07d85caeda18ead9]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

docker_container.foo will be destroyed
- resource "docker_container" "foo" {
 - attach = false -> null
 - command = [
 - "sleep",
 - "infinity",
] -> null
 - cpu_shares = 0 -> null
 - dns = [] -> null
 - dns_opts = [] -> null
 - dns_search = [] -> null
 - entrypoint = [] -> null
 - env = [] -> null
 - gateway = "172.17.0.1" -> null
 - group_add = [] -> null
 - hostname = "76c6390ec277" -> null
 - id = "76c6390ec277dc11f709997c4eb636ee8b45f90723d0f6ec07d85caeda18ead9" -> null
 - image = "sha256:edbf74c41f8a3501ce542e137cf28ea04dd03e6df8c9d66519b6ad761c2598a" -> null
 - init = false -> null
 - ip_address = "172.17.0.2" -> null
 - ip_prefix_length = 16 -> null
 - ipc_mode = "private" -> null
 - links = [] -> null
 - log_driver = "json-file" -> null
 - log_opts = {} -> null
 - logs = false -> null
 - max_retry_count = 0 -> null
 - memory = 0 -> null
 - memory_swap = 0 -> null
 - must_run = true -> null
 - name = "foo" -> null
 - network_data = [
 - {
 - gateway = "172.17.0.1"
 - global_ipv6_prefix_length = 0
 - ip_address = "172.17.0.2"
 - ip_prefix_length = 16
 - network_name = "bridge"
 }
]
 # (2 unchanged attributes hidden)
```

```

- network_name = "bridge"
 # (2 unchanged attributes hidden)
},
] -> null
- network_mode = "bridge" -> null
- privileged = false -> null
- publish_all_ports = false -> null
- read_only = false -> null
- remove_volumes = true -> null
- restart = "no" -> null
- rm = false -> null
- runtime = "runc" -> null
- security_opts = [] -> null
- shm_size = 64 -> null
- start = true -> null
- stdin_open = false -> null
- stop_timeout = 0 -> null
- storage_opts = {} -> null
- sysctls = {} -> null
- tmpfs = {} -> null
- tty = false -> null
 # (8 unchanged attributes hidden)
}

docker_image.ubuntu will be destroyed
- resource "docker_image" "ubuntu" {
 - id = "sha256:edbfe74c41f8a3501ce542e137cf28ea04dd03e6df8c9d66519b6ad761c2598aubuntu:latest" -> null
 - image_id = "sha256:edbfe74c41f8a3501ce542e137cf28ea04dd03e6df8c9d66519b6ad761c2598a" -> null
 - latest = "sha256:edbfe74c41f8a3501ce542e137cf28ea04dd03e6df8c9d66519b6ad761c2598a" -> null
 - name = "ubuntu:latest" -> null
 - repo_digest = "ubuntu@sha256:8a37d68f4f73ebf3d4efafbcf66379bf3728902a8038616808f04e34a9ab63ee" -> null
}

```

Plan: 0 to add, 0 to change, 2 to destroy.

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.  
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

docker\_container.foo: Destroying... [id=76c6390ec277dc11f709997c4eb636ee8b45f90723d0f6ec07d85caeda18ead9]

docker\_container.foo: Destruction complete after 1s

docker\_image.ubuntu: Destroying... [id=sha256:edbfe74c41f8a3501ce542e137cf28ea04dd03e6df8c9d66519b6ad761c2598aubuntu:latest]

docker\_image.ubuntu: Destruction complete after 1s

Destroy complete! Resources: 2 destroyed.

PS C:\Users\Admin\Desktop\Terraformscripts\docker> |

### Step 9: Docker images After Executing Destroy step

```

PS C:\Users\Admin\Desktop\Terraformscripts\docker> docker images
REPOSITORY TAG IMAGE ID CREATED SIZE

```

**Aim:** To understand terraform lifecycle, and core concepts/terminologies, and install it on a Linux Machine or Windows.

**Terraform:** Terraform is an open-source Infrastructure as Code (IaC) tool developed by HashiCorp. It allows you to define, provision, and manage infrastructure across various cloud providers using a simple declarative configuration language. Terraform's key features include automation of infrastructure deployments, version control for infrastructure changes, and the ability to manage both cloud and on-premises resources in a consistent manner.

**Step 1:**Visit the website:

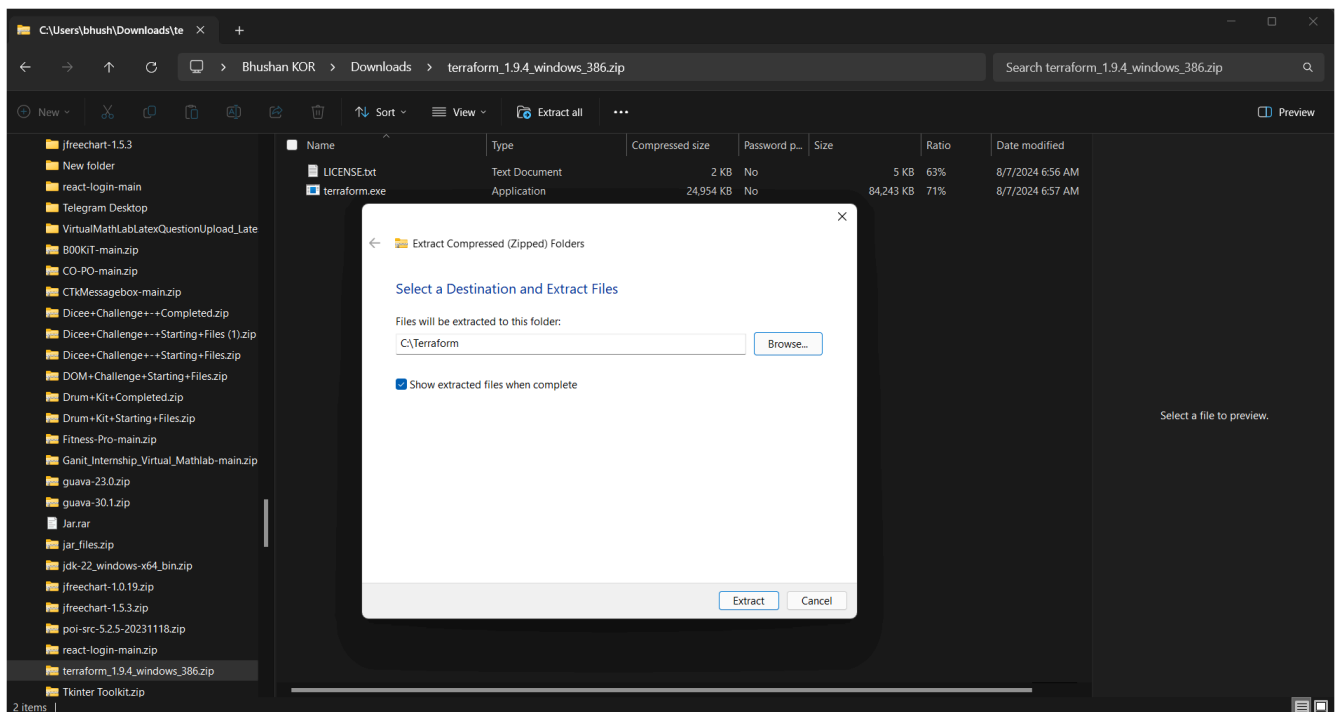
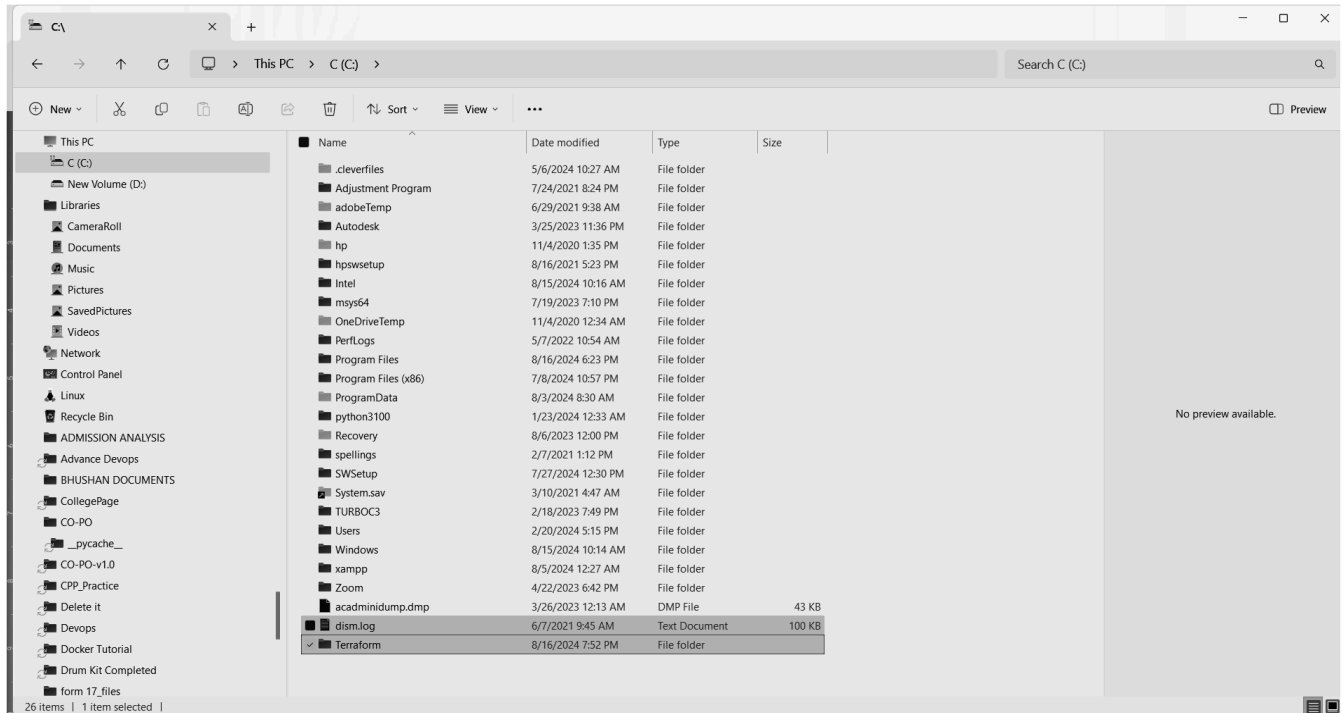
[https://developer.hashicorp.com/terraform/install?product\\_intent=terraform](https://developer.hashicorp.com/terraform/install?product_intent=terraform) and install the terraform according to your system. Here we are downloading it for Windows.

The screenshot shows the Terraform installation page on the HashiCorp developer website. The page is titled "Terraform" and has a navigation bar with links to "Install", "Tutorials", "Documentation", "Registry", and "Try Cloud". The main content area is divided into sections for "Windows" and "Linux". Under "Windows", there is a "Binary download" section with a "Download" button. Under "Linux", there is a "Package manager" section with a "Download" button. The page also includes a sidebar with links to "Terraform Home", "Install Terraform", "Operating Systems", "macOS", "Windows", "Linux", "FreeBSD", "OpenBSD", "Solaris", "Release information", "Next steps", and "Resources". A footer at the bottom states: "We use cookies & other similar technology to collect data to improve your experience on our site, as described in our [Privacy Policy](#) and [Cookie Policy](#). Manage Preferences ACCEPT".

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Step 2:Done Downloaded Successfully Now create the folder named Terraform in C: directory and extract the file in C:\Terraform.



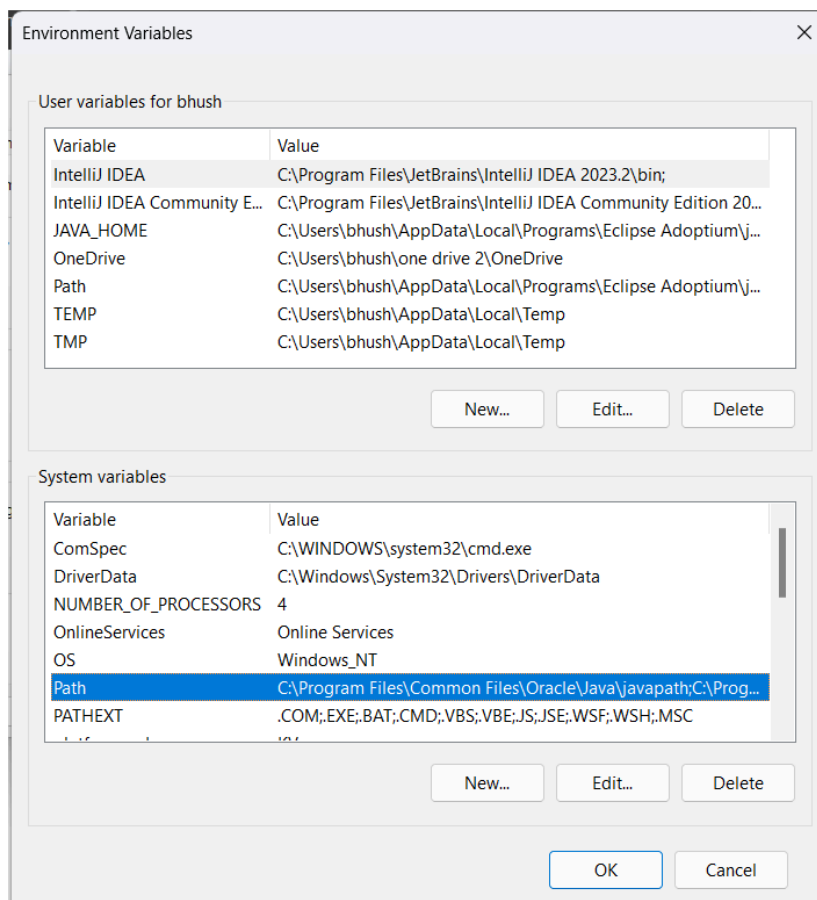
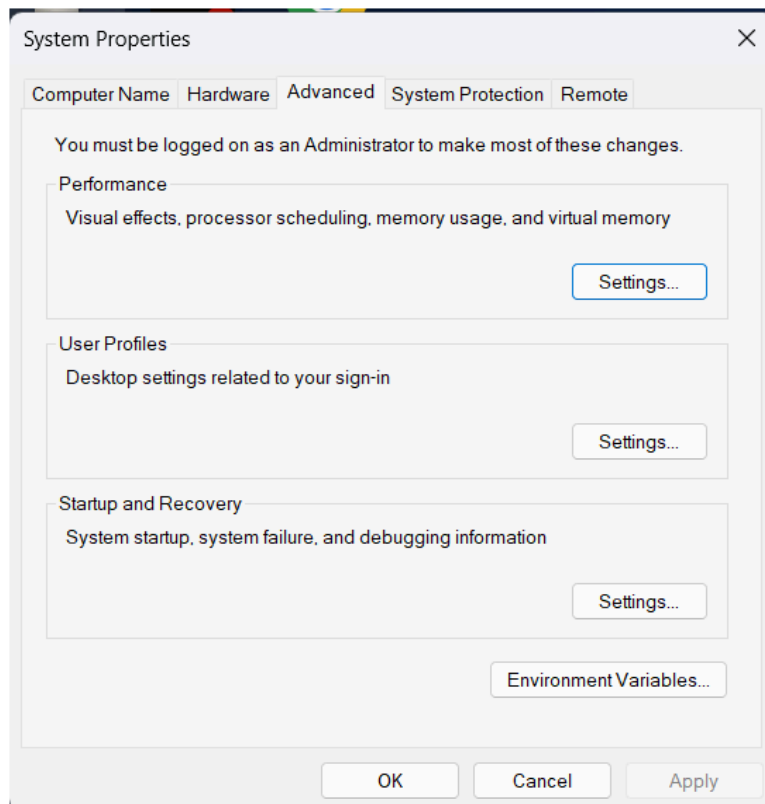
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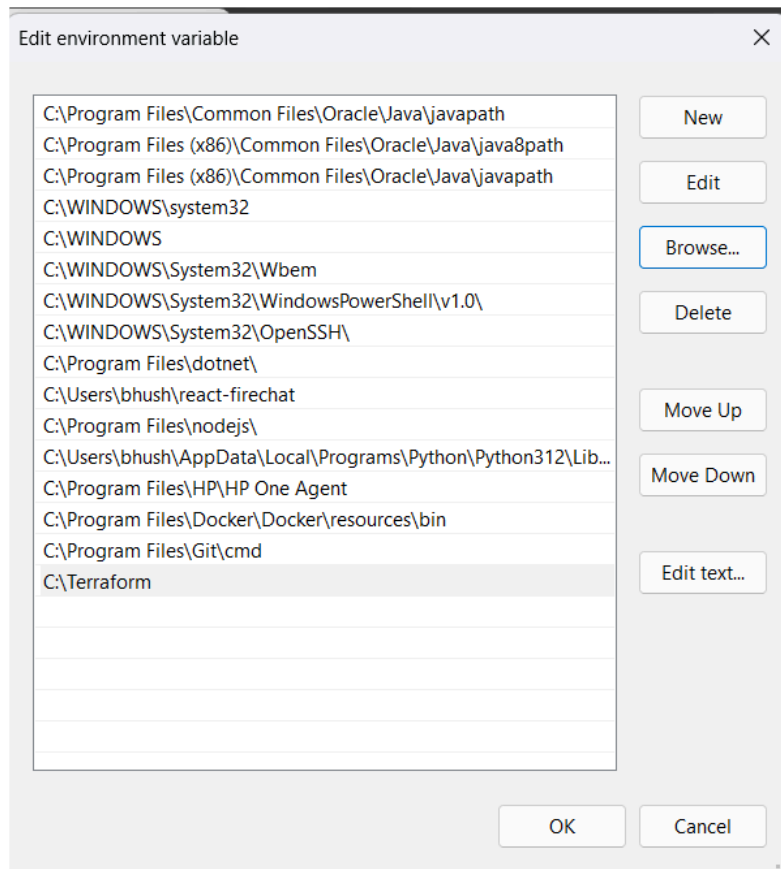
Year:2024-2025

Step 3. Edit the environment variable. In the path of system variables add the path of terraform by clicking edit. After that save and close it.

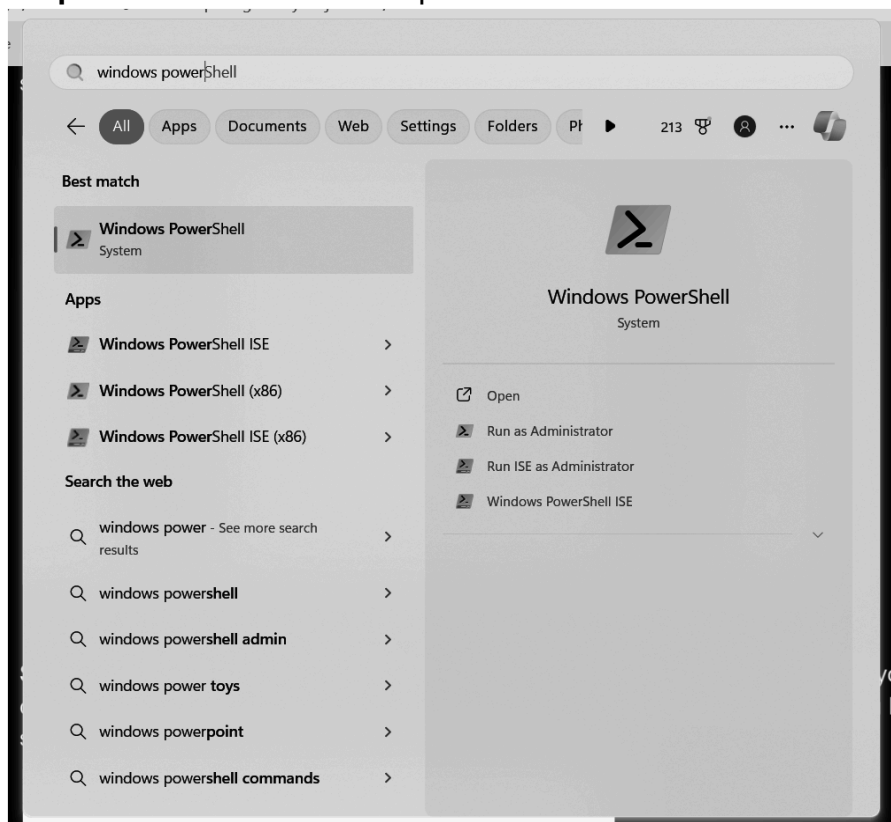


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**Step 4:** Search for Windows power shell and Run it as administrator.



**Step 5:** Now run the command terraform in Windoes Powershell.

```

Administrator: Windows PowerShell (x86)
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> terraform
Usage: terraform [global options] <subcommand> [args]

The available commands for execution are listed below.
The primary workflow commands are given first, followed by
less common or more advanced commands.

Main commands:
 init Prepare your working directory for other commands
 validate Check whether the configuration is valid
 plan Show changes required by the current configuration
 apply Create or update infrastructure
 destroy Destroy previously-created infrastructure

All other commands:
 console Try Terraform expressions at an interactive command prompt
 fmt Reformat your configuration in the standard style
 force-unlock Release a stuck lock on the current workspace
 get Install or upgrade remote Terraform modules
 graph Generate a Graphviz graph of the steps in an operation
 import Associate existing infrastructure with a Terraform resource
 login Obtain and save credentials for a remote host
 logout Remove locally-stored credentials for a remote host
 metadata Metadata related commands
 output Show output values from your root module
 providers Show the providers required for this configuration
 refresh Update the state to match remote systems
 show Show the current state or a saved plan
 state Advanced state management
 taint Mark a resource instance as not fully functional
 test Execute integration tests for Terraform modules
 untaint Remove the 'tainted' state from a resource instance
 version Show the current Terraform version
 workspace Workspace management

Global options (use these before the subcommand, if any):
 -chdir=DIR Switch to a different working directory before executing the
 given subcommand.
 -help Show this help output, or the help for a specified subcommand.
 -version An alias for the "version" subcommand.
PS C:\WINDOWS\system32>

```

**Step 6:** Congratulations You have successfully installed Terraform in your system.

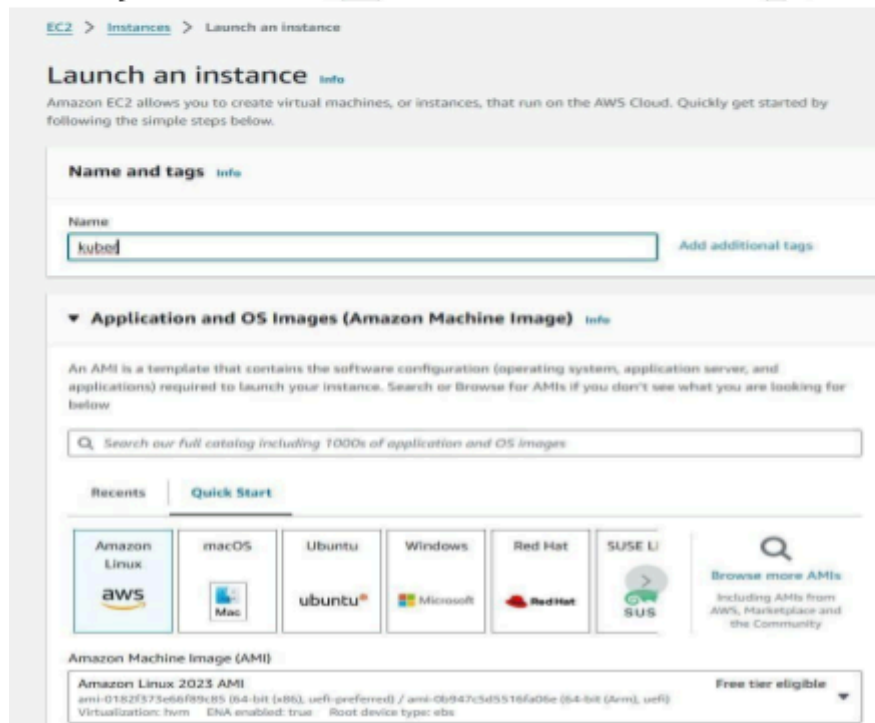
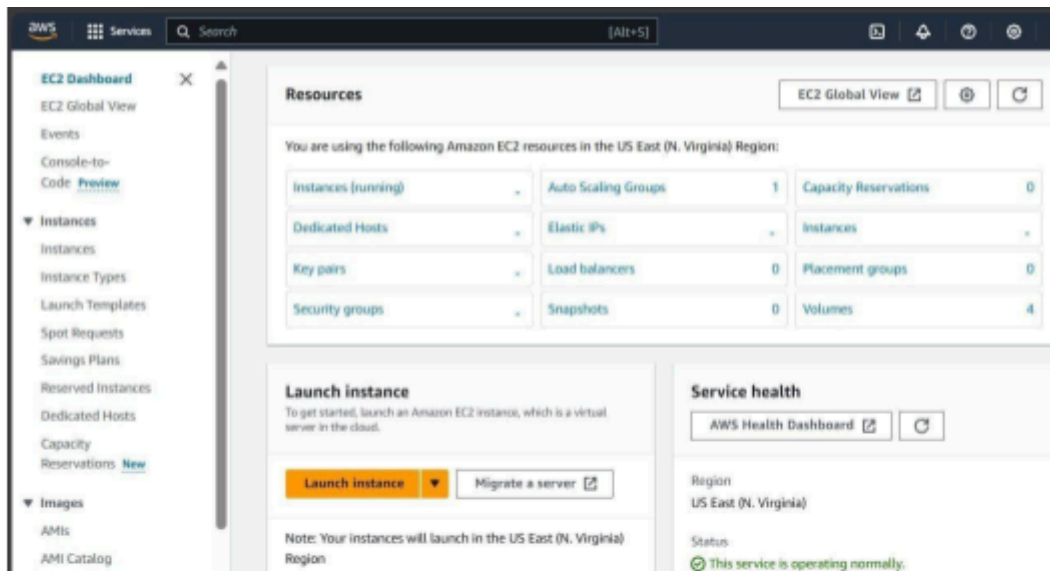
## Experiment 4

**Aim:** To install Kubectl and execute Kubectl commands to manage the Kubernetes cluster and deploy Your First Kubernetes Application.

### Procedure:

#### 1. Creation Of EC-2 instance

- Create an EC2 AWS Linux instance on AWS .also edit the Security Group Inbound Rules to allow SSH. then select the t2.micro instance type





**Network settings** [Info](#) [Edit](#)

Network [Info](#)  
vpc-0381e49e607677b63

Subnet [Info](#)  
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)  
Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)  
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-8' with the following rules:

- ☒ Allow SSH traffic from  [Helps you connect to your instance](#)
- ☒ Allow HTTPS traffic from the internet [To set up an endpoint, for example when creating a web server](#)
- ☒ Allow HTTP traffic from the internet [To set up an endpoint, for example when creating a web server](#)

**Summary**

Number of instances [Info](#)

Software Image (AMI)  
Amazon Linux 2023 AMI 2023.5.2...[read more](#)  
ami-0182f373e66f89c85

Virtual server type (instance type)  
t2.micro

Firewall (security group)  
New security group

Storage (volumes)  
1 volume(s) - 8 GiB

**Free tier:** In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which

[Cancel](#) [Launch instance](#) [Review commands](#)

**Instances (6)** [Info](#) Last updated less than a minute ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#) [1](#) [Settings](#)

| <input type="checkbox"/> | Name <a href="#">↗</a> | Instance ID         | Instance state <a href="#">↗</a>            | Instance type <a href="#">↗</a> | Status check      | Alarm status                                  | Availability Zone <a href="#">↗</a> | Public IPv4  |
|--------------------------|------------------------|---------------------|---------------------------------------------|---------------------------------|-------------------|-----------------------------------------------|-------------------------------------|--------------|
| <input type="checkbox"/> | worker-1_sb            | i-078ed31a706f6f589 | Running <a href="#">🔍</a> <a href="#">🔍</a> | t2.micro                        | 2/2 checks passed | <a href="#">View alarms</a> <a href="#">+</a> | us-east-1a                          | ec2-54-89... |
| <input type="checkbox"/> | worker-2_sb            | i-05da8301287468d59 | Running <a href="#">🔍</a> <a href="#">🔍</a> | t2.micro                        | 2/2 checks passed | <a href="#">View alarms</a> <a href="#">+</a> | us-east-1f                          | ec2-3-237... |
| <input type="checkbox"/> | kuber                  | i-097d2af538d0ca45a | Pending <a href="#">🔍</a> <a href="#">🔍</a> | t2.micro                        | -                 | <a href="#">View alarms</a> <a href="#">+</a> | us-east-1f                          | ec2-3-237... |

- Thus Kuber named -instance gets created. Then click on Id of that instance then click on connect button you will see this:

[EC2](#) > [Instances](#) > [i-09dbca91fa3edcaea](#) > [Connect to instance](#)

**Connect to instance** [Info](#)  
Connect to your instance i-09dbca91fa3edcaea (kuber) using any of these options

[EC2 Instance Connect](#) [Session Manager](#) [SSH client](#) [EC2 serial console](#)

**Port 22 (SSH) is open to all IPv4 addresses**  
Port 22 (SSH) is currently open to all IPv4 addresses, indicated by 0.0.0.0/0 in the inbound rule in [your security group](#). For increased security, consider restricting access to only the EC2 Instance Connect service IP addresses for your Region: 18.206.107.24/29. [Learn more](#).

Instance ID  
[🔍](#) i-09dbca91fa3edcaea (kuber)

Connection Type

☒ Connect using EC2 Instance Connect  
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 address.

☐ Connect using EC2 Instance Connect Endpoint  
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IPv4 address  
[🔍](#) 54.211.131.109

Username  
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.  
 [✕](#)

**Note:** In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

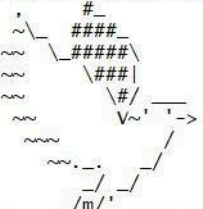
[Cancel](#) [Connect](#)

- Then go into SSH client where you will get this command  
Chmod 400 “keyname.pem”  
ssh -i <keyname>.pem ubuntu@<public\_ip\_address> copy it and then connect it and run the following command for establishing connection.(I have entered this command on git bash where i entered in downloads where server.pem is stored then as the key is not accessible hence we need to change its mode using chmod 400 “key name.pem”. Then use the given command for making connections).

```
ADMIN@DESKTOP-LPV2RP5 MINGW64 ~
$ cd Downloads/

ADMIN@DESKTOP-LPV2RP5 MINGW64 ~/Downloads
$ chmod 400 "server.pem"

ADMIN@DESKTOP-LPV2RP5 MINGW64 ~/Downloads
$ ssh -i "server.pem" ec2-user@ec2-3-237-37-35.compute-1.amazonaws.com
The authenticity of host 'ec2-3-237-37-35.compute-1.amazonaws.com (3.237.37.35)'
can't be established.
ED25519 key fingerprint is SHA256:uhsLQC4WN+6i3np6iMFwW+vGF8aGkTrm89ryXQ4wTlg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-237-37-35.compute-1.amazonaws.com' (ED25519) t
o the list of known hosts.
```



```
#_
####_
#####\
###|
#/
V~ -> https://aws.amazon.com/linux/amazon-linux-2023
m/
```

```
Last login: Sat Sep 14 11:47:26 2024 from 18.206.107.28
[ec2-user@ip-172-31-65-181 ~]$
```

## 2. Installation of Docker

1. For installation of Docker into the machines run the following command: `sudo yum install docker -y`

```
[ec2-user@ip-172-31-65-181 ~]$ sudo yum install docker -y
Last metadata expiration check: 0:08:49 ago on Sat Sep 14 11:42:25 2024.
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
docker x86_64 25.0.6-1.amzn2023.0.2 amazonlinux 44 M
Installing dependencies:
containerd x86_64 1.7.20-1.amzn2023.0.1 amazonlinux 35 M
iptables-libs x86_64 1.8.8-3.amzn2023.0.2 amazonlinux 401 k
iptables-nft x86_64 1.8.8-3.amzn2023.0.2 amazonlinux 183 k
libcgroup x86_64 3.0-1.amzn2023.0.1 amazonlinux 75 k
libnetfilter_conntrack x86_64 1.0.8-2.amzn2023.0.2 amazonlinux 58 k
libnftnl x86_64 1.0.1-19.amzn2023.0.2 amazonlinux 30 k
libnftnl x86_64 1.2.2-2.amzn2023.0.2 amazonlinux 84 k
pigz x86_64 2.5-1.amzn2023.0.3 amazonlinux 83 k
runc x86_64 1.1.13-1.amzn2023.0.1 amazonlinux 3.2 M
=====

Transaction Summary
=====
Install 10 Packages

Total download size: 84 M
Installed size: 317 M
Downloading Packages:
(1/10): iptables-libs-1.8.8-3.amzn2023.0.2.x86_ 4.5 MB/s | 401 kB 00:00
(2/10): iptables-nft-1.8.8-3.amzn2023.0.2.x86_6 4.5 MB/s | 183 kB 00:00
(3/10): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm 3.9 MB/s | 75 kB 00:00
(4/10): libnetfilter_conntrack-1.0.8-2.amzn2023 2.6 MB/s | 58 kB 00:00
(5/10): libnftnl-1.0.1-19.amzn2023.0.2.x86_ 1.2 MB/s | 30 kB 00:00
(6/10): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rp 2.2 MB/s | 84 kB 00:00
(7/10): pigz-2.5-1.amzn2023.0.3.x86_64.rpm 2.8 MB/s | 83 kB 00:00
(8/10): runc-1.1.13-1.amzn2023.0.1.x86_64.rpm 30 MB/s | 3.2 MB 00:00
(9/10): containerd-1.7.20-1.amzn2023.0.1.x86_64 38 MB/s | 35 MB 00:00
(10/10): docker-25.0.6-1.amzn2023.0.2.x86_64.rp 34 MB/s | 44 MB 00:01

Total 62 MB/s | 84 MB 00:01

created symlink /etc/systemd/system/sockets.target.wants/docker.socket -> /usr/lib/systemd/system/docker.socket.

Verifying : containerd-1.7.20-1.amzn2023.0.1.x86_64
Verifying : docker-25.0.6-1.amzn2023.0.2.x86_64
Verifying : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
Verifying : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
Verifying : libcgroup-3.0-1.amzn2023.0.1.x86_64
Verifying : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
Verifying : libnftnl-1.0.1-19.amzn2023.0.2.x86_64
Verifying : libnftnl-1.2.2-2.amzn2023.0.2.x86_64
Verifying : pigz-2.5-1.amzn2023.0.3.x86_64
Verifying : runc-1.1.13-1.amzn2023.0.1.x86_64

Installed:
containerd-1.7.20-1.amzn2023.0.1.x86_64 docker-25.0.6-1.amzn2023.0.2.x86_64 iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
libcgroup-3.0-1.amzn2023.0.1.x86_64 libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
pigz-2.5-1.amzn2023.0.3.x86_64 runc-1.1.13-1.amzn2023.0.1.x86_64 libnftnl-1.0.1-19.amzn2023.0.2.x86_64

Complete!
```

- Then, configure cgroup in a daemon.json file by using following commands  
`cd /etc/docker`  
`cat <<EOF | sudo tee /etc/docker/daemon.json`

```
{
"exec-opts":
["native.cgroupdriver=systemd"],
"log-driver": "json-file",
"log-opts": {
"max-size": "100m"
},
"storage-driver": "overlay2"
```

```
}
EOF
```

```
[ec2-user@ip-172-31-65-181 ~]$ cd /etc/docker
```

```
[ec2-user@ip-172-31-65-181 docker]$ cat <<EOF | sudo tee /etc/docker/daemon.json
{
 "exec-opts":
 ["native.cgroupdriver=systemd"],
 "log-driver": "json-file",
 "log-opts": {
 "max-size": "100m"
 },
 "storage-driver": "overlay2"
}
EOF
{
 "exec-opts":
 ["native.cgroupdriver=systemd"],
 "log-driver": "json-file",
 "log-opts": {
 "max-size": "100m"
 },
 "storage-driver": "overlay2"
}
```

- Then after this run the following command to enable and start docker and also to load the daemon.json file.

```
sudo systemctl enable docker
sudo systemctl daemon-reload
sudo systemctl restart docker
```

```
[ec2-user@ip-172-31-81-216 docker]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-81-216 docker]$ sudo systemctl daemon-reload
[ec2-user@ip-172-31-81-216 docker]$ sudo systemctl restart docker
```

- docker -v

```
[ec2-user@ip-172-31-65-181 docker]$ docker -v
Docker version 25.0.5, build 5dc9bcc
```

### 3. Then Install Kubernetes with the following command.

- SELinux needs to be disabled before configuring kubelet thus run the following command  
sudo setenforce 0

```
sudo sed -i 's/^SELINUX=enforcing$/SELINUX=permissive/' /etc/selinux/config
```

```
[ec2-user@ip-172-31-65-181 docker]$ sudo setenforce 0
[ec2-user@ip-172-31-65-181 docker]$ sudo sed -i 's/^SELINUX=enforcing$/SELINUX=permissive/' /etc/selinux/config
```

- Here We are adding kubernetes using the repository whose command is given below. cat <<EOF | sudo tee /etc/yum.repos.d/kubernetes.repo  
[kubernetes]

```
name=Kubernetes
baseurl=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/
enabled=1
gpgcheck=1
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key
exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni
EOF
```

```
[ec2-user@ip-172-31-65-181 docker]$ sudo setenforce 0
[ec2-user@ip-172-31-65-181 docker]$ sudo sed -i 's/^SELINUX=enforcing$/SELINUX=permissive/' /etc/selinux/config
[ec2-user@ip-172-31-65-181 docker]$ cat <<EOF | sudo tee /etc/yum.repos.d/kubernetes.repo
[kubernetes]
name=Kubernetes baseurl=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/ enabled=1
gpgcheck=1 gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni
EOF
[kubernetes]
name=Kubernetes baseurl=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/ enabled=1
gpgcheck=1 gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni
[ec2-user@ip-172-31-65-181 docker]$
```

- After that Run following command to make the updation and also to install kubelet ,kubeadm, kubectl: `sudo yum update`

```
[ec2-user@ip-172-31-65-181 docker]$ sudo yum update
Invalid configuration value: gpgcheck=1 gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni in /etc/yum.repos.d/kubernetes.repo; invalid boolean value '1 gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni'
Error: Cannot find a valid baseurl for repo: kubernetes
Ignoring repositories: kubernetes
Last metadata expiration check: 0:19:14 ago on Sat Sep 14 11:42:25 2024.
Dependencies resolved.
Nothing to do.
Complete!
```

```
sudo yum install -y kubelet kubeadm kubectl --disableexcludes=kubernetes
```

```
Complete!
[ec2-user@ip-172-31-81-216 ~]$ sudo yum install -y kubelet kubeadm kubectl --dis
ableexcludes=kubernetes
Last metadata expiration check: 0:00:35 ago on Sat Sep 14 07:38:02 2024.
Dependencies resolved.
```

| Package                  | Arch   | Version               | Repository  | Size  |
|--------------------------|--------|-----------------------|-------------|-------|
| Installing:              |        |                       |             |       |
| kubeadm                  | x86_64 | 1.30.5-150500.1.1     | kubernetes  | 10 M  |
| kubectl                  | x86_64 | 1.30.5-150500.1.1     | kubernetes  | 10 M  |
| kubelet                  | x86_64 | 1.30.5-150500.1.1     | kubernetes  | 17 M  |
| Installing dependencies: |        |                       |             |       |
| conntrack-tools          | x86_64 | 1.4.6-2.amzn2023.0.2  | amazonlinux | 208 k |
| cri-tools                | x86_64 | 1.30.1-150500.1.1     | kubernetes  | 8.6 M |
| kubernetes-cni           | x86_64 | 1.4.0-150500.1.1      | kubernetes  | 6.7 M |
| libnetfilter_cthelper    | x86_64 | 1.0.0-21.amzn2023.0.2 | amazonlinux | 24 k  |
| libnetfilter_cttimeout   | x86_64 | 1.0.0-19.amzn2023.0.2 | amazonlinux | 24 k  |
| libnetfilter_queue       | x86_64 | 1.0.5-2.amzn2023.0.2  | amazonlinux | 30 k  |

#### Transaction Summary

Install 9 Packages

Total download size: 53 M

Installed size: 292 M

Downloading Packages:

```
(1/9): libnetfilter_cthelper-1.0.0-21.amzn2023. 436 kB/s | 24 kB 00:00
(2/9): libnetfilter_queue-1.0.5-2.amzn2023.0.2. 1.5 MB/s | 30 kB 00:00
(3/9): libnetfilter_cttimeout-1.0.0-19.amzn2023 309 kB/s | 24 kB 00:00
(4/9): conntrack-tools-1.4.6-2.amzn2023.0.2.x86 2.3 MB/s | 208 kB 00:00
(5/9): cri-tools-1.30.1-150500.1.1.x86_64.rpm 34 MB/s | 8.6 MB 00:00
(6/9): kubectl-1.30.5-150500.1.1.x86_64.rpm 21 MB/s | 10 MB 00:00
(7/9): kubeadm-1.30.5-150500.1.1.x86_64.rpm 17 MB/s | 10 MB 00:00
(8/9): kubelet-1.30.5-150500.1.1.x86_64.rpm 32 MB/s | 17 MB 00:00
(9/9): kubernetes-cni-1.4.0-150500.1.1.x86_64.r 21 MB/s | 6.7 MB 00:00
```

```
Verifying : kubeadm-1.30.5-150500.1.1.x86_64 7/9
Verifying : kubectl-1.30.5-150500.1.1.x86_64 7/9
Verifying : kubelet-1.30.5-150500.1.1.x86_64 8/9
Verifying : kubernetes-cni-1.4.0-150500.1.1.x86_64 9/9
```

Installed:

```
conntrack-tools-1.4.6-2.amzn2023.0.2.x86_64
cri-tools-1.30.1-150500.1.1.x86_64
kubeadm-1.30.5-150500.1.1.x86_64
kubectl-1.30.5-150500.1.1.x86_64
kubelet-1.30.5-150500.1.1.x86_64
kubernetes-cni-1.4.0-150500.1.1.x86_64
libnetfilter_cthelper-1.0.0-21.amzn2023.0.2.x86_64
libnetfilter_cttimeout-1.0.0-19.amzn2023.0.2.x86_64
libnetfilter_queue-1.0.5-2.amzn2023.0.2.x86_64
```

Complete!

```
[ec2-user@ip-172-31-81-216 ~]$ |
```

- After installing Kubernetes, we need to configure internet options to allow bridging.
  - sudo swapoff -a
  - echo "net.bridge.bridge-nf-call-iptables=1" | sudo tee -a /etc/sysctl.conf
  - sudo sysctl -p

```
[ec2-user@ip-172-31-81-216 ~]$ sudo swapoff -a
[ec2-user@ip-172-31-81-216 ~]$ echo "net.bridge.bridge-nf-call-iptables=1" | sud
o tee -a /etc/sysctl.conf
net.bridge.bridge-nf-call-iptables=1
[ec2-user@ip-172-31-81-216 ~]$ sudo sysctl -p
net.bridge.bridge-nf-call-iptables = 1
```

## 4. Initialize the Kubecuster

```
sudo kubeadm init --pod-network-cidr=10.244.0.0/16
```

```
[ec2-user@ip-172-31-80-126 docker]$ sudo kubeadm init --pod-network-cidr=10.244.0.0/16
I0913 10:32:44.629146 26680 version.go:256] remote version is much newer: v1.31.0; falling back to: stable-1.30
[init] Using Kubernetes version: v1.30.4
[preflight] Running pre-flight checks
```



Your Kubernetes control-plane has initialized successfully!

To start using your cluster, you need to run the following as a regular user:

```
mkdir -p $HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

Alternatively, if you are the root user, you can run:

```
export KUBECONFIG=/etc/kubernetes/admin.conf
```

You should now deploy a pod network to the cluster.

Run "kubectl apply -f [podnetwork].yaml" with one of the options listed at:  
<https://kubernetes.io/docs/concepts/cluster-administration/addons/>

Then you can join any number of worker nodes by running the following on each as root:

```
kubeadm join 172.31.26.174:6443 --token pv0yyi.xh1lqhclfjr50pt8 \
--discovery-token-ca-cert-hash sha256:8293b2f6d29de466bd859007f5adbcdb3a
ecb0c446ba09033d32a5846b3d434f
```

- copy the token and save for future use .  
kubeadm join 172.31.26.174:6443 --token pv0yyi.xh1lqhclfjr50pt8  
\--discovery-token-ca-cert-hash  
sha256:8293b2f6d29de466bd859007f5adbcdb3aecb0c446ba09033d32a5846b  
3d434f
- Copy the mkdir and chown commands from the top and execute  
them mkdir -p \$HOME/.kube  
sudo cp -i /etc/kubernetes/admin.conf \$HOME/.kube/config  
sudo chown \$(id -u):\$(id -g) \$HOME/.kube/config

```
[ec2-user@ip-172-31-80-126 docker]$ nc
[ec2-user@ip-172-31-80-126 docker]$ mkdir -p $HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

- Then, add a common networking plugin called flannel as mentioned in the code.

```
kubectl apply -f
```

```
https://raw.githubusercontent.com/coreos/flannel/master/Documentation/kube-flannel.yml
```

```
[ec2-user@ip-172-31-26-174 docker]$ kubectl apply -f https://raw.githubusercontent.com/coreos/flannel/master/Documentation/kube-flannel.yml
namespace/kube-flannel created
clusterrole.rbac.authorization.k8s.io/flannel created
clusterrolebinding.rbac.authorization.k8s.io/flannel created
serviceaccount/flannel created
configmap/kube-flannel-cfg created
daemonset.apps/kube-flannel-ds created
```

5. Now that the cluster is up and running, we can deploy our nginx

server on this cluster. Apply deployment using this following command:

```
kubectl apply -f
```

```
https://k8s.io/examples/pods/simple-pod.yaml
```

```
[ec2-user@ip-172-31-26-174 docker]$ kubectl apply -f https://k8s.io/examples/pods/simple-pod.yaml
pod/nginx created
```

Then use kubectl get nodes to check whether the pod gets created or not.

```
[ec2-user@ip-172-31-26-174 docker]$ kubectl get pods
NAME READY STATUS RESTARTS AGE
nginx 0/1 Pending 0 12s
```

To convert state from pending to running use following command:

kubectl describe pod nginx This command will help to describe the pods

it gives reason for failure as it shows the untolerated taints which need to be untainted.



```
[ec2-user@ip-172-31-26-174 ~]$ kubectl describe pod nginx
Name: nginx
Namespace: default
Priority: 0
Service Account: default
Node: <none>
Labels: <none>
Annotations: <none>
Status: Pending
IP: <none>
IPs: <none>
Containers:
 nginx:
 Image: nginx:1.14.2
 Port: 80/TCP
 Host Port: 0/TCP
 Environment: <none>
 Mounts:
 /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-k4lj6 (ro)
```

```
Conditions:
 Type Status
 PodScheduled False
Volumes:
 kube-api-access-k4lj6:
 Type: Projected (a volume that contains injected data from multiple sources)
 TokenExpirationSeconds: 3607
 ConfigMapName: kube-root-ca.crt
 ConfigMapOptional: <nil>
 DownwardAPI: true
QoS Class: BestEffort
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
 node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
 Type Reason Age From Message
 ---- -
 Warning FailedScheduling 7s default-scheduler 0/1 nodes are available: 1 node(s) had untolerated taint {node-role.kubernetes.io/control-plane: }. Preemption: 0/1 nodes are available: 1 Preemption is not helpful for scheduling.
[ec2-user@ip-172-31-26-174 ~]$ kubectl taint nodes --all node-role.kubernetes.io/control-plane-
node/ip-172-31-26-174.ec2.internal untainted
```

6. Now check pod status is is running

```
[ec2-user@ip-172-31-26-174 ~]$ kubectl get pods
NAME READY STATUS RESTARTS AGE
nginx 1/1 Running 1 (6s ago) 90s
```

7. Lastly, mention the port you want to host. Here i have used localhost 8081 then check it.

```
kubectl port-forward nginx 8081:80
```

```
[ec2-user@ip-172-31-26-174 ~]$ kubectl port-forward nginx 8081:80
Forwarding from 127.0.0.1:8081 -> 80
Forwarding from [::1]:8081 -> 80
```

### **8. Verify your deployment**

Open up a new terminal and ssh to your EC2 instance.

Then, use this curl command to check if the Nginx server is running.

```
curl --head http://127.0.0.1:8080
```

If the response is 200 OK and you can see the Nginx server name, your deployment was successful. We have successfully deployed our Nginx server on our EC2 instance.

### **Conclusion:**

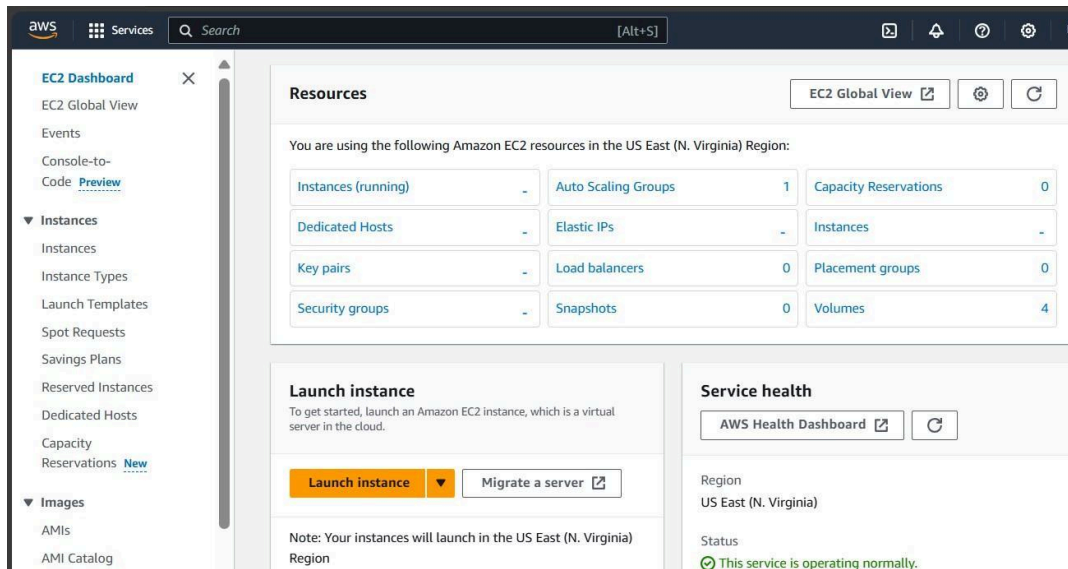
First, I successfully launched an AWS EC2 instance running Amazon Linux. After that, I installed Docker and Kubernetes on the instance. Following the installation, I initialized the Kubernetes cluster, which provided me with a token, along with chown and mkdir commands. I then executed both the mkdir and chown commands successfully. Next, I installed the Flannel networking plugin without any issues. Initially, there was an error while deploying Nginx, but after correcting it, I successfully deployed Nginx using a simple-pod.yml file. I confirmed its deployment with the get pods command and hosted it locally on <http://localhost:8081>, which worked as expected.

## EXPERIMENT NO. 3

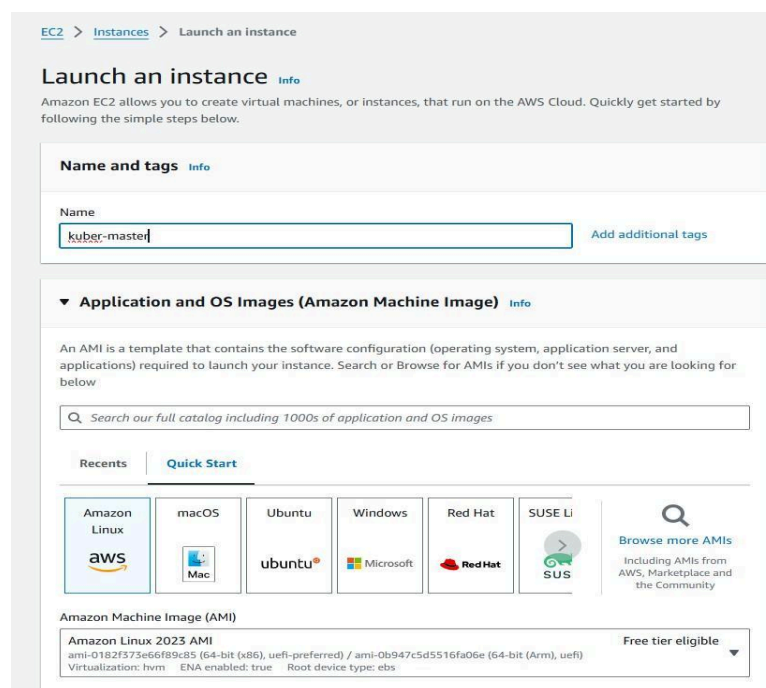
**Aim:** To understand the Kubernetes Cluster Architecture, install and Spin Up a Kubernetes Cluster on Linux Machines/Cloud.

### Procedure:

#### 1. Creation Of Instance



Search EC-2 instance. Then create three EC-2 instances and choose Amazon Linux as OS and also allow ssh traffic from anywhere.



Description

Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-0182f373e66f89c85

Verified provider

▼ Instance type

Info | Get advice

Instance type

t3.medium  
Family: t3 2 vCPU 4 GiB Memory Current generation: true  
On-Demand SUSE base pricing: 0.0979 USD per Hour  
On-Demand Windows base pricing: 0.06 USD per Hour  
On-Demand Linux base pricing: 0.0416 USD per Hour  
On-Demand RHEL base pricing: 0.0704 USD per Hour

☐ All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

server

Create new key pair

▼ Network settings

Info

Edit

Network

Info

vpc-051bba342b5626898

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called **launch-wizard-31** with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance.

Anywhere  
0.0.0.0/0

☒ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

▼ Configure storage

Info

Advanced

1x

8

GiB

gp3

Root volume (Not encrypted)

Free tier eligible customers can get up to 30 GiB of EBS General Purpose (SSD) or Magnetic storage

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.5.2...read more  
ami-0182f373e66f89c85

Virtual server type (instance type)

t3.medium

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Launch instance

Review commands

| <input type="checkbox"/> | Name  | Instance ID         | Instance state                                                                            |                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | kube-master                                                                            | i-00aa79ac09d7462c0 |  Running |   |
| <input type="checkbox"/> | kube-worker1                                                                           | i-0bab86cd3fbfcb40a |  Running |   |
| <input type="checkbox"/> | kube-worker2                                                                           | i-00dcfd302ffd80dda |  Running |   |

- ```
Sadneya@DESKTOP-IEPNL3D MINGW64 ~ (master)
$ cd downloads

Sadneya@DESKTOP-IEPNL3D MINGW64 ~/downloads (master)
$ chmod 400 "server.pem"

Sadneya@DESKTOP-IEPNL3D MINGW64 ~/downloads (master)
$ ssh -i "server.pem" ec2-user@ec2-54-174-206-93.compute-1.amazonaws.com
The authenticity of host 'ec2-54-174-206-93.compute-1.amazonaws.com (54.174.206.93)' can't be established.
ED25519 key fingerprint is SHA256:T+tsGyI15gAvUvjeAZ7GjDIWXHOaI4EPF5g5oICrkoQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-174-206-93.compute-1.amazonaws.com' (ED25519)
to the list of known hosts.

#_
~\####_ Amazon Linux 2023
~~\#####\
~~\#####\
~~\###|
~~\#/\_--> https://aws.amazon.com/linux/amazon-linux-2023
~~V~'~>
~~~
~~~.~.
~~\_/
~~/_/m/'_>
```

2. Installation Of Docker on three machines

- For installation of Docker into all three machines run the following command:
`sudo yum install docker -y`

```
[ec2-user@ip-172-31-87-114 ~]$ sudo yum install docker -y
Last metadata expiration check: 0:06:20 ago on Fri Sep 13 03:20:22 2024.
Dependencies resolved.
=====
Package                        Arch      Version                               Repository    Size
=====
Installing:
docker                        x86_64    25.0.6-1.amzn2023.0.2               amazonlinux   44 M
Installing dependencies:
containerd                   x86_64    1.7.20-1.amzn2023.0.1               amazonlinux   35 M
iptables-libs                x86_64    1.8.8-3.amzn2023.0.2               amazonlinux   401 k
iptables-nft                 x86_64    1.8.8-3.amzn2023.0.2               amazonlinux   183 k
libcgroup                   x86_64    3.0-1.amzn2023.0.1                 amazonlinux    75 k
libnetfilter_conntrack       x86_64    1.0.8-2.amzn2023.0.2               amazonlinux    58 k
libnfnetlink                 x86_64    1.0.1-19.amzn2023.0.2              amazonlinux    30 k
libnftnl                     x86_64    1.2.2-2.amzn2023.0.2               amazonlinux    84 k
pigz                         x86_64    2.5-1.amzn2023.0.3                 amazonlinux    83 k
runc                         x86_64    1.1.13-1.amzn2023.0.1              amazonlinux   3.2 M

Transaction Summary
=====
Install 10 Packages

Total download size: 84 M
Installed size: 317 M
Downloading Packages:
(1/10): iptables-libs-1.8.8-3.amzn2023.0.2.x86_ 3.6 MB/s | 401 kB    00:00
(2/10): iptables-nft-1.8.8-3.amzn2023.0.2.x86_6 4.6 MB/s | 183 kB    00:00

Installed:
containerd-1.7.20-1.amzn2023.0.1.x86_64
docker-25.0.6-1.amzn2023.0.2.x86_64
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
libcgroup-3.0-1.amzn2023.0.1.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64
libnftnl-1.2.2-2.amzn2023.0.2.x86_64
pigz-2.5-1.amzn2023.0.3.x86_64
runc-1.1.13-1.amzn2023.0.1.x86_64

Complete!
```

- Then, configure cgroup in a daemon.json file by using following commands
`cd /etc/docker`
`cat <<EOF | sudo tee /etc/docker/daemon.json`

```
{
"exec-opts":
["native.cgroupdriver=systemd"],
"log-driver": "json-file",
"log-opts": {
"max-size": "100m"
},
}
```

```
"storage-driver": "overlay2"
}
EOF
```

```
[ec2-user@ip-172-31-87-114 ~]$ cd /etc/docker
[ec2-user@ip-172-31-87-114 docker]$ cat <<EOF | sudo tee /etc/docker/daemon.json
{
"exec-opts": ["native.cgroupdriver=systemd"],
"log-driver": "json-file",
"log-opts": {
"max-size": "100m"
},
"storage-driver": "overlay2"
}
EOF
{
"exec-opts": ["native.cgroupdriver=systemd"],
"log-driver": "json-file",
"log-opts": {
"max-size": "100m"
},
"storage-driver": "overlay2"
}
```

- Then after this run the following command to enable and start docker and also to load the daemon.json file.

```
sudo systemctl enable docker
sudo systemctl daemon-reload
sudo systemctl restart docker
```

```
[ec2-user@ip-172-31-80-126 docker]$ sudo systemctl enable docker
sudo systemctl daemon-reload
sudo systemctl restart docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
```

- Then check the version of docker installed. docker -v

```
[ec2-user@ip-172-31-80-126 docker]$ docker -v
Docker version 25.0.5, build 5dc9bcc
```

3. Installation Of Kubernetes on three machines

- SELinux needs to be disabled before configuring kubelet thus run the following command
`sudo setenforce 0`

```
sudo sed -i 's/^SELINUX=enforcing$/SELINUX=permissive/' /etc/selinux/config
```

```
[ec2-user@ip-172-31-80-126 docker]$ sudo setenforce 0  
sudo sed -i 's/^SELINUX=enforcing$/SELINUX=permissive/' /etc/selinux/config
```

- Here We are adding kubernetes using the repository whose command is given below.
`cat <<EOF | sudo tee /etc/yum.repos.d/kubernetes.repo`

```
[kubernetes]
```

```
name=Kubernetes
```

```
baseurl=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/
```

```
enabled=1
```

```
gpgcheck=1
```

```
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key
```

```
exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni
```

```
EOF
```

```
[ec2-user@ip-172-31-80-126 docker]$ cat <<EOF | sudo tee /etc/yum.repos.d/kubernetes.repo  
[kubernetes]  
name=Kubernetes  
baseurl=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/  
enabled=1  
gpgcheck=1  
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key  
exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni  
EOF  
[kubernetes]  
name=Kubernetes  
baseurl=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/  
enabled=1  
gpgcheck=1  
gpgkey=https://pkgs.k8s.io/core:/stable:/v1.30/rpm/repodata/repomd.xml.key  
exclude=kubelet kubeadm kubectl cri-tools kubernetes-cni
```

- After that Run following command to make the update and also to install kubelet, kubeadm, kubectl: `sudo yum update`

```
[ec2-user@ip-172-31-80-126 docker]$ sudo yum update  
Kubernetes  
Dependencies resolved.  
Nothing to do.  
Complete!
```


sudo yum install -y kubelet kubeadm kubectl --disableexcludes=kubernetes

```
[ec2-user@ip-172-31-80-126 docker]$ sudo yum install -y kubelet kubeadm kubectl --disableexcludes=kubernetes
Last metadata expiration check: 0:00:10 ago on Fri Sep 13 10:31:17 2024.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
kubeadm	x86_64	1.30.5-150500.1.1	kubernetes	10
kubectl	x86_64	1.30.5-150500.1.1	kubernetes	10
kubelet	x86_64	1.30.5-150500.1.1	kubernetes	17 M
Installing dependencies:				
conntrack-tools	x86_64	1.4.6-2.amzn2023.0.2	amazonlinux	208 k
cri-tools	x86_64	1.30.1-150500.1.1	kubernetes	8.6 M
kubernetes-cni	x86_64	1.4.0-150500.1.1	kubernetes	6.7 M
libnetfilter_cthelper	x86_64	1.0.0-21.amzn2023.0.2	amazonlinux	24 k
libnetfilter_cttimeout	x86_64	1.0.0-19.amzn2023.0.2	amazonlinux	24 k
libnetfilter_queue	x86_64	1.0.5-2.amzn2023.0.2	amazonlinux	30 k
Transaction Summary				
Install 9 Packages				

```
Total
Kubernetes
Importing GPG key 0x9A296436:
  Userid : "Isv:kubernetes OBS Project <isv:kubernetes@build.opensuse.org>"
  Fingerprint: DE15 B144 86CD 3778 9E87 6E1A 2346 54DA 9A29 6436
  From : https://pkgs.k8s.io/core/stable/v1.30/rpm/repodata/repomd.xml.key
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing :
  Installing : kubernetes-cni-1.4.0-150500.1.1.x86_64
  Installing : cri-tools-1.30.1-150500.1.1.x86_64
  Installing : libnetfilter_queue-1.0.5-2.amzn2023.0.2.x86_64
  Installing : libnetfilter_cttimeout-1.0.0-19.amzn2023.0.2.x86_64
  Installing : libnetfilter_cthelper-1.0.0-21.amzn2023.0.2.x86_64
  Installing : conntrack-tools-1.4.6-2.amzn2023.0.2.x86_64
  Running scriptlet: conntrack-tools-1.4.6-2.amzn2023.0.2.x86_64
  Installing : kubelet-1.30.5-150500.1.1.x86_64
  Running scriptlet: kubelet-1.30.5-150500.1.1.x86_64
  Installing : kubeadm-1.30.5-150500.1.1.x86_64
  Installing : kubectl-1.30.5-150500.1.1.x86_64
  Running scriptlet: kubectl-1.30.5-150500.1.1.x86_64
  Verifying : conntrack-tools-1.4.6-2.amzn2023.0.2.x86_64
  Verifying : libnetfilter_cthelper-1.0.0-21.amzn2023.0.2.x86_64
  Verifying : libnetfilter_cttimeout-1.0.0-19.amzn2023.0.2.x86_64
  Verifying : libnetfilter_queue-1.0.5-2.amzn2023.0.2.x86_64
  Verifying : cri-tools-1.30.1-150500.1.1.x86_64
  Verifying : kubeadm-1.30.5-150500.1.1.x86_64
  Verifying : kubectl-1.30.5-150500.1.1.x86_64
  Verifying : kubelet-1.30.5-150500.1.1.x86_64
  Verifying : kubernetes-cni-1.4.0-150500.1.1.x86_64
Installed:
  conntrack-tools-1.4.6-2.amzn2023.0.2.x86_64      cri-tools-1.30.1-150500.1.1.x86_64      kubeadm-1.30.5-150500.1.1.x86_64
  kubectl-1.30.5-150500.1.1.x86_64                kubelet-1.30.5-150500.1.1.x86_64      kubernetes-cni-1.4.0-150500.1.1.x86_64
  libnetfilter_cthelper-1.0.0-21.amzn2023.0.2.x86_64  libnetfilter_cttimeout-1.0.0-19.amzn2023.0.2.x86_64  libnetfilter_queue-1.0.5-2.amzn2023.0.2.x86_64
Complete!
```

- After installing Kubernetes, we need to configure internet options to allow bridging.
 - sudo swapoff -a
 - echo "net.bridge.bridge-nf-call-iptables=1" | sudo tee -a /etc/sysctl.conf
 - sudo sysctl -p

```
[ec2-user@ip-172-31-80-126 docker]$ sudo swapoff -a
echo "net.bridge.bridge-nf-call-iptables=1" | sudo tee -a /etc/sysctl.conf
sudo sysctl -p
net.bridge.bridge-nf-call-iptables=1
net.bridge.bridge-nf-call-iptables = 1
```

4. Perform this ONLY on the Master machine

- Initialize kubernetes by typing below command

```
sudo kubeadm init --pod-network-cidr=10.244.0.0/16 --ignore-preflight-errors=all
```

```
[ec2-user@ip-172-31-80-126 docker]$ sudo kubeadm init --pod-network-cidr=10.244.0.0/16
I0913 10:32:44.629146 26680 version.go:256] remote version is much newer: v1.31.0; falling back to: stable-1.30
[init] Using Kubernetes version: v1.30.4
[preflight] Running pre-flight checks

Your Kubernetes control-plane has initialized successfully!

To start using your cluster, you need to run the following as a regular user:

  mkdir -p $HOME/.kube
  sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
  sudo chown $(id -u):$(id -g) $HOME/.kube/config

Alternatively, if you are the root user, you can run:

  export KUBECONFIG=/etc/kubernetes/admin.conf

You should now deploy a pod network to the cluster.
Run "kubectl apply -f [podnetwork].yaml" with one of the options listed at:
  https://kubernetes.io/docs/concepts/cluster-administration/addons/

Then you can join any number of worker nodes by running the following on each as root:

kubeadm join 172.31.80.126:6443 --token jhtgwo.4qv2vtxrcf6nvgpk \
  --discovery-token-ca-cert-hash sha256:766e48546942419274bcd18c370d2492f6e49dac9f98890804362194690f0f4a
```

- So after initialization you will get token at the end for joining master and worker. Like here I got this :(save this token as it is required later.Then you can join any number of worker nodes by running the following on each as root.)

```
kubeadm join 172.31.80.126:6443 --token jhtgwo.4qv2vtxrcf6nvgpk\
--discovery-token-ca-cert-hash
sha256:766e48546942419274bcd18c370d2492f6e49dac9f98890804362194690f0f4a
```

- Also,Copy the mkdir and chown commands from the top and execute them mkdir -p \$HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf \$HOME/.kube/config
sudo chown \$(id -u):\$(id -g) \$HOME/.kube/config

```
[ec2-user@ip-172-31-80-126 docker]$ mkdir -p $HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

- Then, add a common networking plugin called flannel file as mentioned in the code.
kubectl apply -f
<https://raw.githubusercontent.com/coreos/flannel/master/Documentation/kube-flannel.yml>

```
[ec2-user@ip-172-31-80-126 docker]$ kubectl apply -f https://raw.githubusercontent.com/coreos/flannel/master/Documentation/kube-flannel.yml
namespace/kube-flannel created
clusterrole.rbac.authorization.k8s.io/flannel created
clusterrolebinding.rbac.authorization.k8s.io/flannel created
serviceaccount/flannel created
configmap/kube-flannel-cfg created
daemonset.apps/kube-flannel-ds created
```

- Now to Check the created pod use this command `kubectl get pods`

5. Perform this ONLY on the worker machines

Paste the below command on all 2 worker machines

- `sudo yum install iproute-tc -y`
- `sudo systemctl enable kubelet`
- `sudo systemctl restart kubelet`

Now use this

```
kubeadm join 172.31.80.126:6443 --token jhtgwo.4qv2vtxrcf6nvgpk\
--discovery-token-ca-cert-hash
sha256:766e48546942419274bcd18c370d2492f6e49dac9f98890804362194690f0f4a
```

(Optional To check the status of pods executed these commands:

`Kubectl get pods -n kube-system` :gives status of all pods

`Kubectl get daemonstat -n kube-system`: gives status of pod named daemonstat

```
[ec2-user@ip-172-31-87-114 docker]$ kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
coredns-55cb58b774-fx12f           1/1     Running   0           100s
coredns-55cb58b774-xn14v           1/1     Running   0           100s
etcd-ip-172-31-87-114.ec2.internal 1/1     Running   1 (2m45s ago) 75s
kube-apiserver-ip-172-31-87-114.ec2.internal 1/1     Running   1 (2m15s ago) 2m11s
kube-controller-manager-ip-172-31-87-114.ec2.internal 0/1     CrashLoopBackOff 1 (8s ago) 70s
kube-proxy-4dv8m                   1/1     Running   2 (26s ago) 100s
kube-scheduler-ip-172-31-87-114.ec2.internal 1/1     Running   1 (2m45s ago) 76s
[ec2-user@ip-172-31-87-114 docker]$ kubectl get daemonset -n kube-system
NAME    DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELECTOR   AGE
kube-proxy 1          1         1       1             1           kubernetes.io/os=linux 3m
```

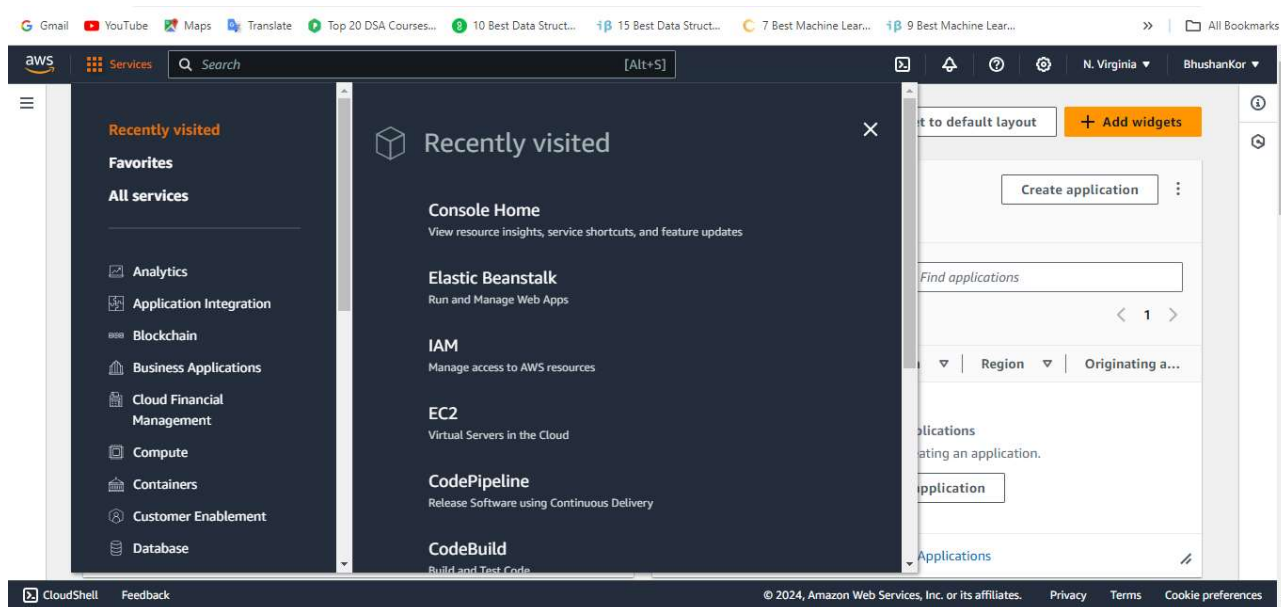
)

Now to see whether master and workers get connected successfully or not run **kubectl get nodes** command on master machine

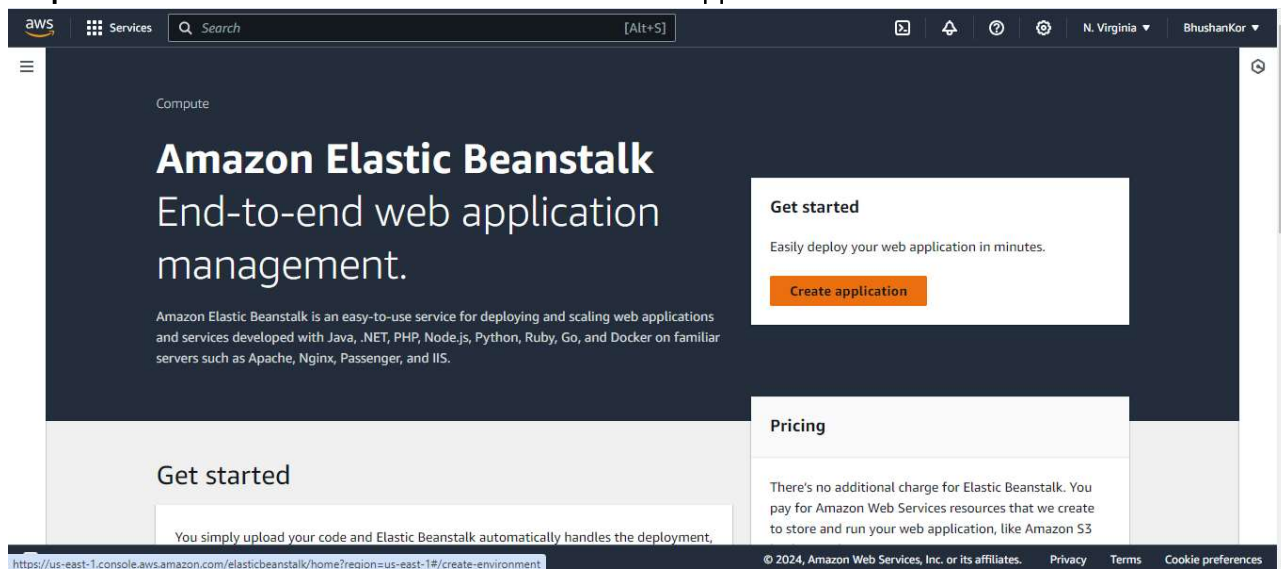
```
[ec2-user@ip-172-31-87-114 docker]$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-172-31-87-114.ec2.internal      Ready     control-plane 3m21s v1.30.5
```

Conclusion: In these EC-2 instance created successfully on AWS Linux. Then I installed docker ,kuberneted and then kubelet ,kubeadm, kubectl.Then on Master machine ,I initailized the kubernetes which given me the token which will be used for connection of master and workers.then on slave I installed iproute and enabled and restarted kubelet then i enter the token which i got from master but there was an issue in joint.that is why on output i just got of only one pc mater on performing command `kubectl get nodes`.

Step 1: Login to your AWS console. Search for Elastic Beanstalk in the searchbar near services.



Step 2: Go to Elastic Beanstalk and click on Create Application



Step 3: Enter the name of your application. Scroll down and in the platform, select platform as PHP. Keep the application code as Sample Application. Set the instance to single instance. Click on NEXT

Step 1
Configure environment

Step 2
Configure service access

Step 3 - optional
Set up networking, database, and tags

Step 4 - optional
Configure instance traffic and scaling

Step 5 - optional
Configure updates, monitoring, and logging

Step 6
Review

Configure environment [Info](#)

Environment tier [Info](#)

Amazon Elastic Beanstalk has two types of environment tiers to support different types of web applications.

☒ **Web server environment**
Run a website, web application, or web API that serves HTTP requests. [Learn more](#)

☐ **Worker environment**
Run a worker application that processes long-running workloads on demand or performs tasks on a schedule. [Learn more](#)

Application information [Info](#)

Application name

Maximum length of 100 characters.

▶ Application tags (optional)

Environment information [Info](#)

Choose the name, subdomain and description for your environment. These cannot be changed later.

Environment name

Must be from 4 to 40 characters in length. The name can contain only letters, numbers, and hyphens. It can't start or end with a hyphen. This name must be unique within a region in your account.

Domain

 .us-east-1.elasticbeanstalk.com

Environment description

Platform [Info](#)

Platform type

☒ **Managed platform**
Platforms published and maintained by Amazon Elastic Beanstalk. [Learn more](#)

☐ **Custom platform**
Platforms created and owned by you. This option is unavailable if you have no platforms.

Platform

Platform branch

Platform branch

PHP 8.3 running on 64bit Amazon Linux 2023

Platform version

4.3.2 (Recommended)

Application code

info

☒ Sample application

☐ Existing version

Application versions that you have uploaded.

☐ Upload your code

Upload a source bundle from your computer or copy one from Amazon S3.

Presets

info

Start from a preset that matches your use case or choose custom configuration to unset recommended values and use the service's default values.

Configuration presets

☒ Single instance (free tier eligible)

☐ Single instance (using spot instance)

☐ High availability

☐ High availability (using spot and on-demand instances)

☐ Custom configuration

Cancel

Next

Step 4 : Use an existing service role and choose whatever service role is available on your account.

Step 1
[Configure environment](#)

Step 2
Configure service access

Step 3 - optional
[Set up networking, database, and tags](#)

Step 4 - optional
[Configure instance traffic and scaling](#)

Step 5 - optional
[Configure updates, monitoring, and logging](#)

Step 6
[Review](#)

Configure service access Info

Service access
IAM roles, assumed by Elastic Beanstalk as a service role, and EC2 instance profiles allow Elastic Beanstalk to create and manage your environment. Both the IAM role and instance profile must be attached to IAM managed policies that contain the required permissions. [Learn more](#)

Service role
☐ Create and use new service role
☒ Use an existing service role
Existing service roles
 Choose an existing IAM role for Elastic Beanstalk to assume as a service role. The existing IAM role must have the required IAM managed policies.

EC2 key pair
 Select an EC2 key pair to securely log in to your EC2 instances. [Learn more](#)

EC2 instance profile
 Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.

Cancel Skip to review Previous **Next**

Step 5 : Review the settings that you have set up for your application and submit your application.

Step 1
[Configure environment](#)

Step 2
[Configure service access](#)

Step 3 - optional
[Set up networking, database, and tags](#)

Step 4 - optional
[Configure instance traffic and scaling](#)

Step 5 - optional
[Configure updates, monitoring, and logging](#)

Step 6
Review

Review Info

Step 1: Configure environment

Environment information

Environment tier	Application name
Web server environment	kaushalapp
Environment name	Application code
Kaushalapp-env	Sample application
Platform	
amazon-elasticbeanstalk-ec2-1:platform/PHP 8.3 running on 64bit Amazon Linux 2023/4.3.2	

Step 2: Configure service access

Service access Info
 Configure the service role and EC2 instance profile that Elastic Beanstalk uses to manage your environment. Choose an EC2 key pair to securely log in to your EC2 instances.

Service role	EC2 key pair	EC2 instance profile
arn:aws:iam::010928205712:role/kaushalrole1	k11	kaushalrole1

Step 3: Set up networking, database, and tags

Networking, database, and tags Info

Deactivated	7	false
Updates		
Managed updates	Deployment batch size	Deployment batch size type
Activated	100	Percentage
Command timeout	Deployment policy	Health threshold
600	AllAtOnce	Ok
Ignore health check	Instance replacement	
false	false	
Platform software		
Lifecycle	Log streaming	Allow URL fopen
false	Deactivated	On
Display errors	Document root	Max execution time
Off	-	60
Memory limit	Zlib output compression	Proxy server
256M	Off	nginx
Logs retention	Rotate logs	Update level
7	Deactivated	minor
X-Ray enabled		
Deactivated		
Environment properties		
Key Value No environment properties There are no environment properties defined		
Cancel Previous Submit		

Environment successfully launched.

Elastic Beanstalk > Environments > Kaushalsapp-env

Kaushalsapp-env

Info

Actions

Upload and deploy

Environment overview

Health

Warning

Domain

Kaushalsapp-env.eba-2ex22i22.us-east-1.elasticbeanstalk.com

Environment ID

e-5rjwzivmxy

Application name

kaushalsapp

Platform

Change version

Platform

PHP 8.3 running on 64bit Amazon Linux 2023/4.3.2

Running version

-

Platform state

Supported

Events

Health

Logs

Monitoring

Alarms

Managed updates

Tags

Events (10)

Info

Filter events by text, property or value

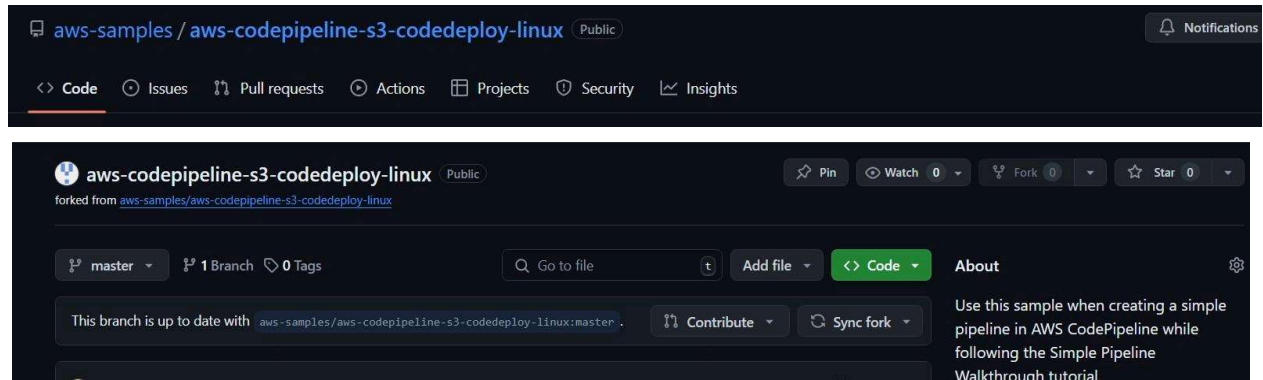
< 1 >

Time	Type	Details
August 16, 2024 21:46:10 (UTC+5:30)	INFO	Successfully launched environment: Kaushalsapp-env
August 16, 2024 21:46:08 (UTC+5:30)	INFO	Application available at Kaushalsapp-env.eba-2ex22i22.us-east-1.elasticbeanstalk.com
August 16, 2024 21:45:52 (UTC+5:30)	INFO	Instance deployment completed successfully.
August 16, 2024 21:45:47 (UTC+5:30)	INFO	Instance deployment: You didn't include a 'composer.json' file in your source bundle. The deployment didn't install Composer dependencies.
August 16, 2024 21:45:33 (UTC+5:30)	INFO	Environment health has transitioned to Pending. Initialization in progress (running for 12 seconds). There are no instances.

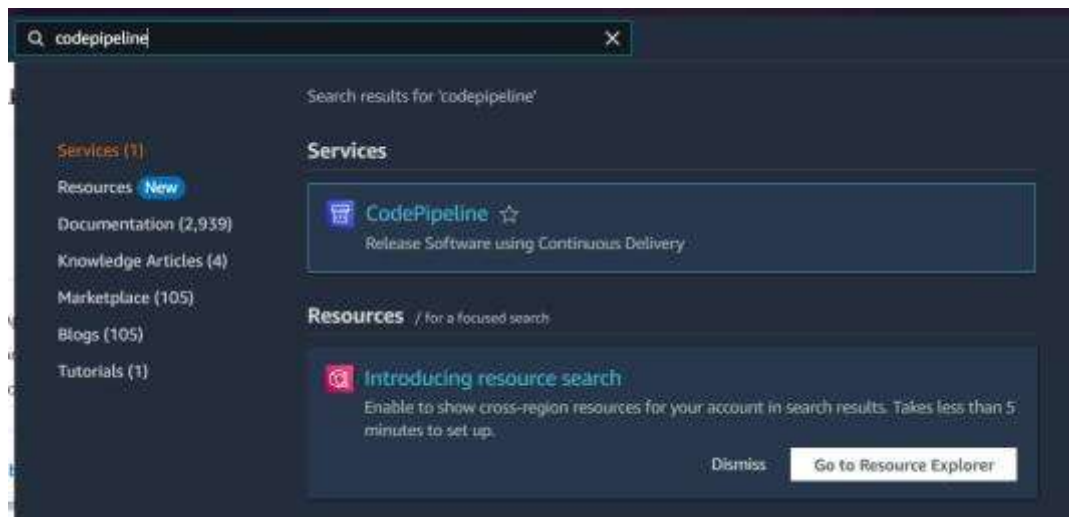
© 2024 Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookies

Step 6: Go to the github link below. This is a github with a sample code for deploying a file on AWS CodePipeline. Fork this repository into your personal github.

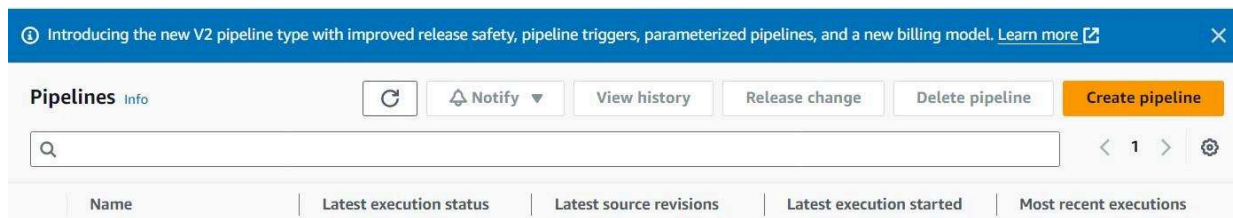
<https://github.com/aws-samples/aws-codepipeline-s3-codedeploy-linux>



Step 7: Search CodePipeline in the services tab and click on it.



Step 8: Click on Create Pipeline.



Step 9: Give a name to your Pipeline. A new service role would be created with the name of the pipeline

Step 1 of 5

Pipeline settings

Pipeline name
Enter the pipeline name. You cannot edit the pipeline name after it is created.

kaushalpipeline

No more than 100 characters.

Pipeline type

i You can no longer create V1 pipelines through the console. We recommend you use the V2 pipeline type with improved release safety, pipeline triggers, parameterized pipelines, and a new billing model.

Execution mode
Choose the execution mode for your pipeline. This determines how the pipeline is run.

☐ Superseded
A more recent execution can overtake an older one. This is the default.

☒ Queued (Pipeline type V2 required)
Executions are processed one by one in the order that they are queued.

☐ Parallel (Pipeline type V2 required)
Executions don't wait for other runs to complete before starting or finishing.

Service role

☒ New service role
Create a service role in your account

☐ Existing service role
Choose an existing service role from your account

Role name

AWSCodePipelineServiceRole-us-east-1-kaushalpipeline

Type your service role name

☒ Allow AWS CodePipeline to create a service role so it can be used with this new pipeline.

Step 10 : Select a source provider (as Github (Version 2)). Click on Connect to Github to connect your github.

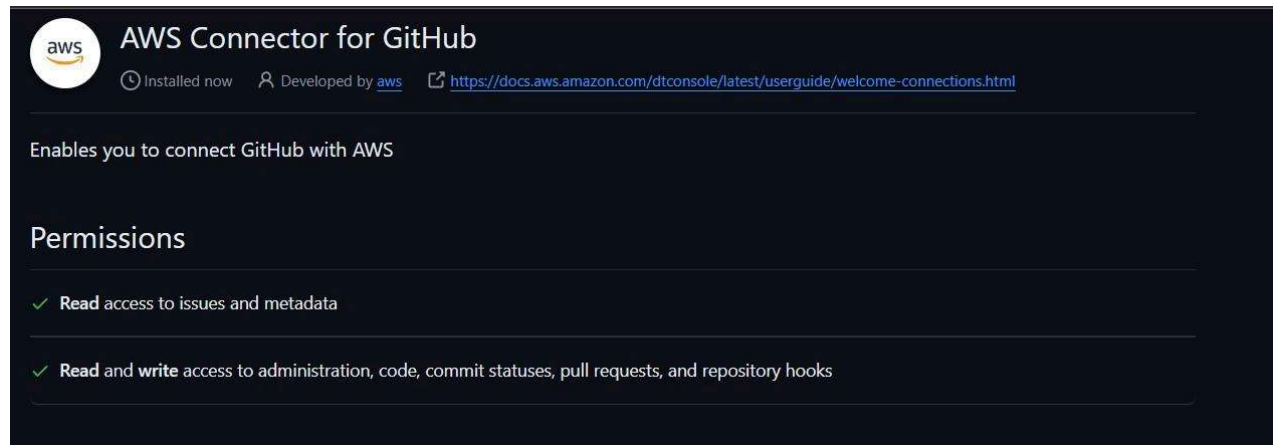
The screenshot shows the 'Add source stage' screen in the AWS CodePipeline console. The breadcrumb trail is 'Developer Tools > CodePipeline > Pipelines > Create new pipeline'. The left sidebar shows a sequence of steps: Step 1 (Choose pipeline settings), Step 2 (Add source stage), Step 3 (Add build stage), Step 4 (Add deploy stage), Step 5, and Review. The main content area is titled 'Add source stage' with an 'Info' icon and 'Step 2 of 5'. Under the 'Source' section, the 'Source provider' is set to 'GitHub (Version 2)'. A blue information box states: 'New GitHub version 2 (app-based) action: To add a GitHub version 2 action in CodePipeline, you create a connection, which uses GitHub Apps to access your repository. Use the options below to choose an existing connection or create a new one. Learn more'. The 'Connection' section prompts to 'Choose an existing connection that you have already configured, or create a new one and then return to this task.' It features a search input field, a 'Connecting' button, and a 'Repository name' search input field.

Step 11: Give a name to your GitHub app Connection and click on Connect. This will give you a prompt to either to select a GitHub app or to install a new app. If it is your first time, click on Install a new app

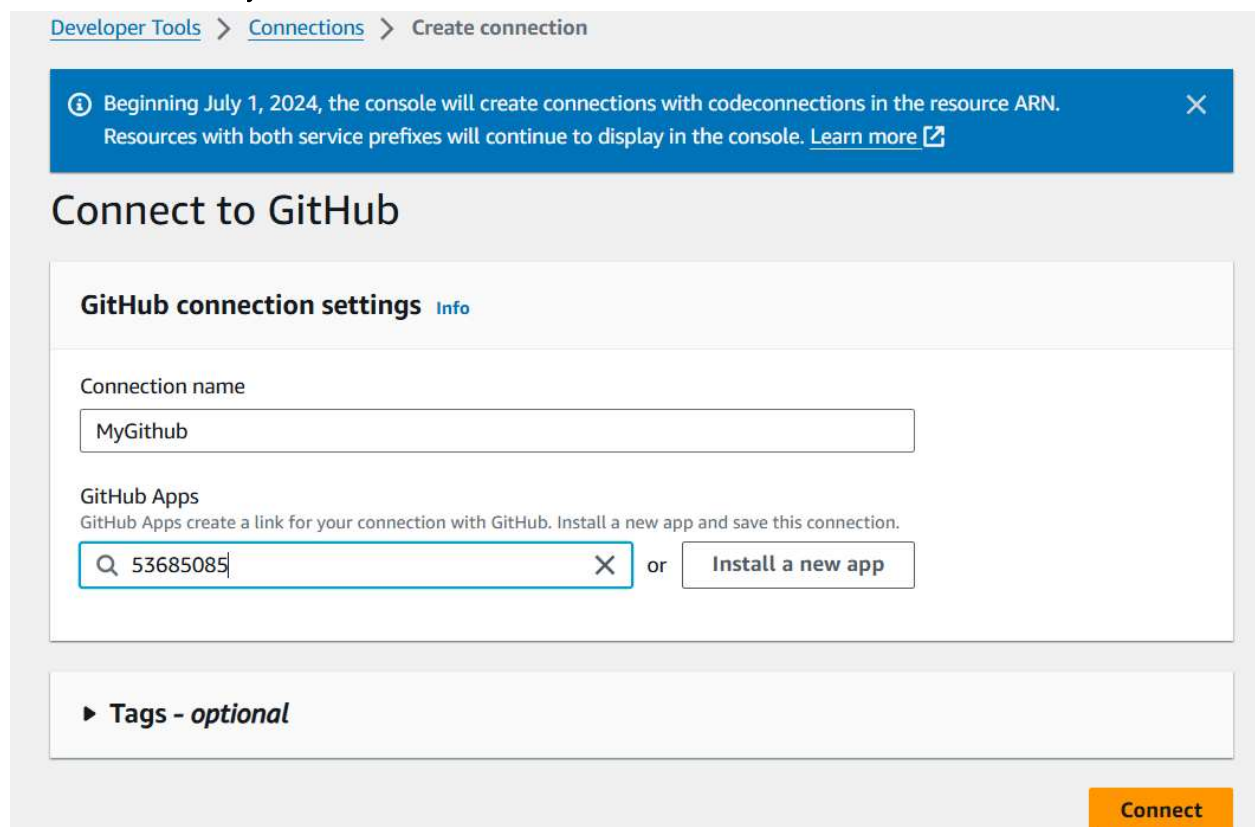
The screenshot shows the 'Create connection' screen in the AWS CodePipeline console. The breadcrumb trail is 'Developer Tools > Connections > Create connection'. The main heading is 'Create a connection' with an 'Info' icon. Below it, the section 'Create GitHub App connection' has an 'Info' icon. A 'Connection name' input field contains the text 'MyGithub'. A section titled 'Tags - optional' is expanded. At the bottom right, there is a yellow 'Connect to GitHub' button.

The screenshot shows a 'Connect to GitHub' dialog box overlaid on the 'Create connection' screen. At the top, a blue banner states: 'Beginning July 1, 2024, the console will create connections with codecommit@aws.com in the resource ARN. Resources with both service profiles will continue to display in the console. Learn more >>'. The dialog has a title 'Connect to GitHub' and a subtitle 'GitHub connection settings'. It contains a 'Connection name' input field with 'MyGithub'. Below it, the 'GitHub App' section shows a search input and an 'Install a new app' button. At the bottom right of the dialog is a yellow 'Connect' button.

Step 12 : This will direct you to install AWS Connector On Your GitHub. Install it to your account and give it its permissions.



Step 13: After the app is set up, it gives the number in the text field. Click on Connect. After clicking on connect, the link is shown in the connection field and AWS shows that GitHub connection is ready to use



Step 1: Choose pipeline settings

Step 2: **Add source stage**

Step 3: Add build stage

Step 4: Add deploy stage

Step 5: Review

Add Source Stage Info

Step 2 of 5

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (Version 2) ▼

 **New GitHub version 2 (app-based) action**
To add a GitHub version 2 action in CodePipeline, you create a connection, which uses GitHub Apps to access your repository. Use the options below to choose an existing connection or create a new one. [Learn more](#)

Connection
Choose an existing connection that you have already configured, or create a new one and then return to this task.

🔍 ✕ or [Connect to GitHub](#)

 **Ready to connect**
Your GitHub connection is ready for use.

Repository name

Step 14: Select the repository that you had forked to your GitHub. After that select the branch on which the files are present (default is Master).

Step 1: Choose pipeline settings

Step 2: **Add source stage**

Step 3: Add build stage

Step 4: Add deploy stage

Step 5: Review

Add Source Stage Info

Step 2 of 5

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (Version 2) ▼

 **New GitHub version 2 (app-based) action**
To add a GitHub version 2 action in CodePipeline, you create a connection, which uses GitHub Apps to access your repository. Use the options below to choose an existing connection or create a new one. [Learn more](#)

Connection
Choose an existing connection that you have already configured, or create a new one and then return to this task.

🔍 ✕ or [Connect to GitHub](#)

 **Ready to connect**
Your GitHub connection is ready for use.

Repository name

Step 15: Set the Trigger type as no filter. This would allow it to the website to update as soon as some change is made in the github.

☒ **CodePipeline default**
AWS CodePipeline uses the default zip format for artifacts in the pipeline. Does not include Git metadata about the repository.

☐ **Full clone**
AWS CodePipeline passes metadata about the repository that allows subsequent actions to do a full Git clone. Only supported for AWS CodeBuild actions.

Trigger

Trigger type
Choose the trigger type that starts your pipeline.

☒ **No filter**
Starts your pipeline on any push and clones the HEAD.

☐ **Specify filter**
Starts your pipeline on a specific filter and clones the exact commit. Pipeline type V2 is required.

☐ **Do not detect changes**
Don't automatically trigger the pipeline.

i You can add additional sources and triggers by editing the pipeline after it is created.

Cancel Previous Next

Step 16: Skip the build stage and go to Deploy. Select the deploy provider as AWS Elastic Beanstalk and Input Artifact as SourceArtifact. The application name would be the name of your Elastic Beanstalk. Then click on next.

Step 1
Choose pipeline settings

Step 2
Add source stage

Step 3
Add build stage

Step 4
Add deploy stage

Step 5
Review

Add build stage Info

Step 3 of 5

Build - optional

Build provider
This is the tool of your build project. Provide build artifact details like operating system, build spec file, and output file names.

Cancel Previous Skip build stage Next

Choose pipeline settings Step 4 of 5

Step 2
Add source stage

Step 3
Add build stage

Step 4
Add deploy stage

Step 5
Review

You cannot skip this stage
Pipelines must have at least two stages. Your second stage must be either a build or deployment stage. Choose a provider for either the build stage or deployment stage.

Deploy

Deploy provider
Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

AWS Elastic Beanstalk

Region
US East (N. Virginia)

Input artifacts
Choose an input artifact for this action. [Learn more](#)

SourceArtifact

No more than 100 characters

Application name
Choose an application that you have already created in the AWS Elastic Beanstalk console. Or create an application in the AWS Elastic Beanstalk console and then return to this task.

Q kaushalsapp X

Environment name
Choose an environment that you have already created in the AWS Elastic Beanstalk console. Or create an environment in the AWS Elastic Beanstalk console and then return to this task.

Q Kaushalsapp-env X

☐ Configure automatic rollback on stage failure

Cancel Previous Next

Step 17: Check all the information and click on create Pipeline.

Trigger type
No filter

Step 3: Add build stage

Build action provider

Build stage
No build

Step 4: Add deploy stage

Deploy action provider

Deploy action provider
AWS Elastic Beanstalk

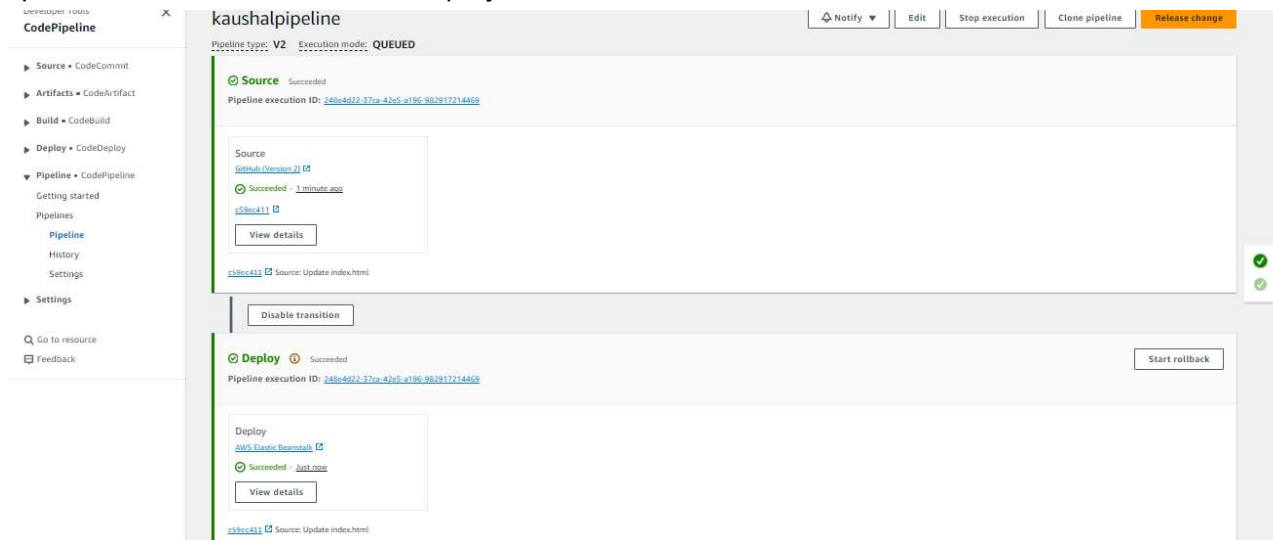
ApplicationName
kaushalsapp

EnvironmentName
Kaushalsapp-env

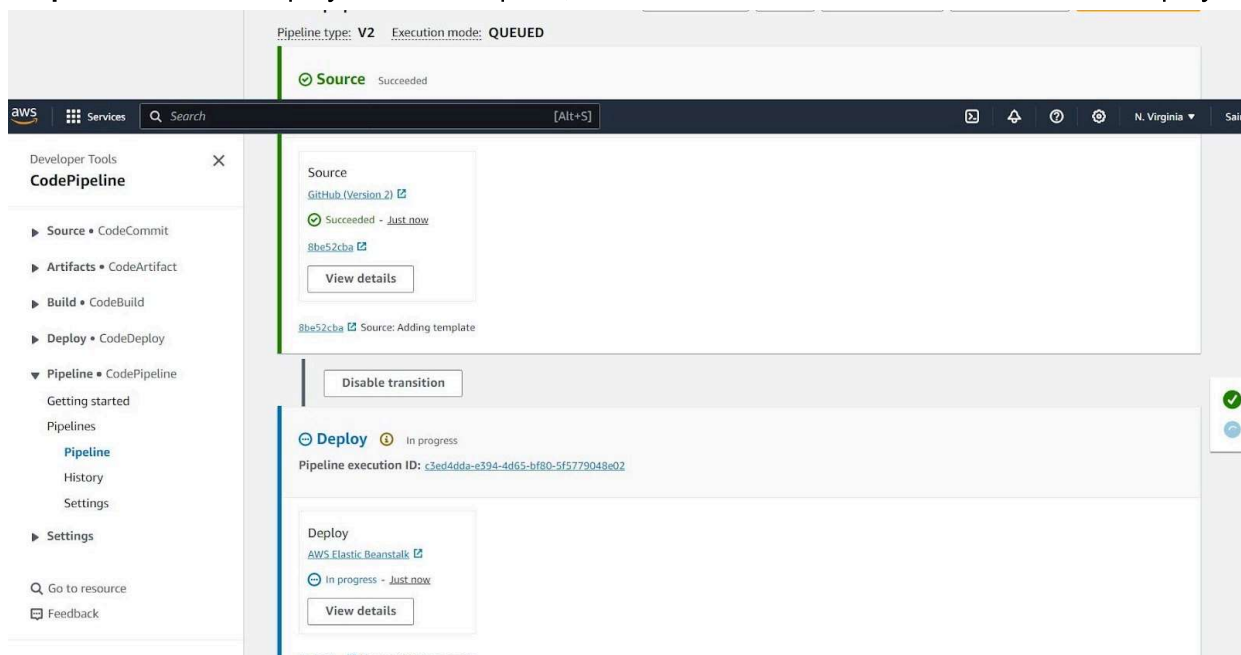
Configure automatic rollback on stage failure
Disabled

Cancel Previous Create pipeline

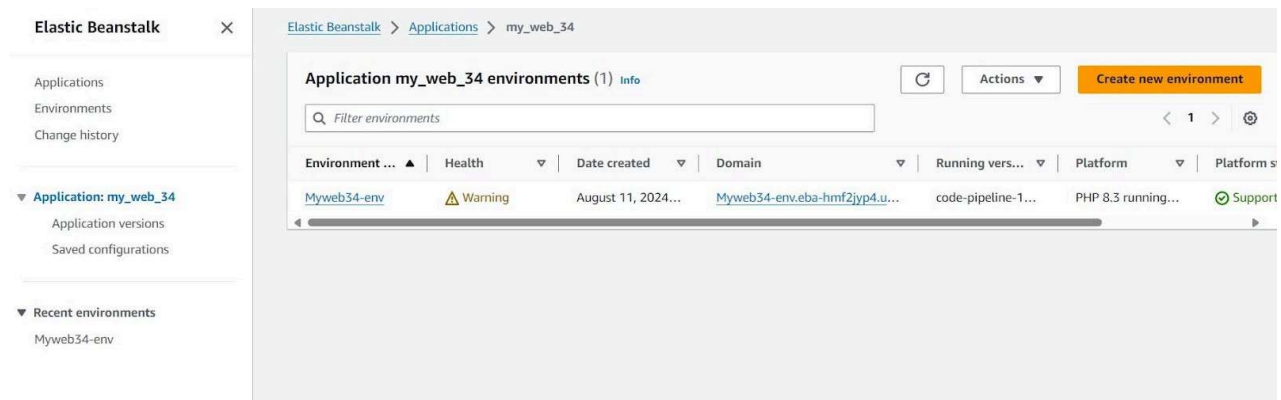
Step 18: If the pipeline is successfully deployed, this screen comes up where the source is set up and then it is transitioned to deploy.



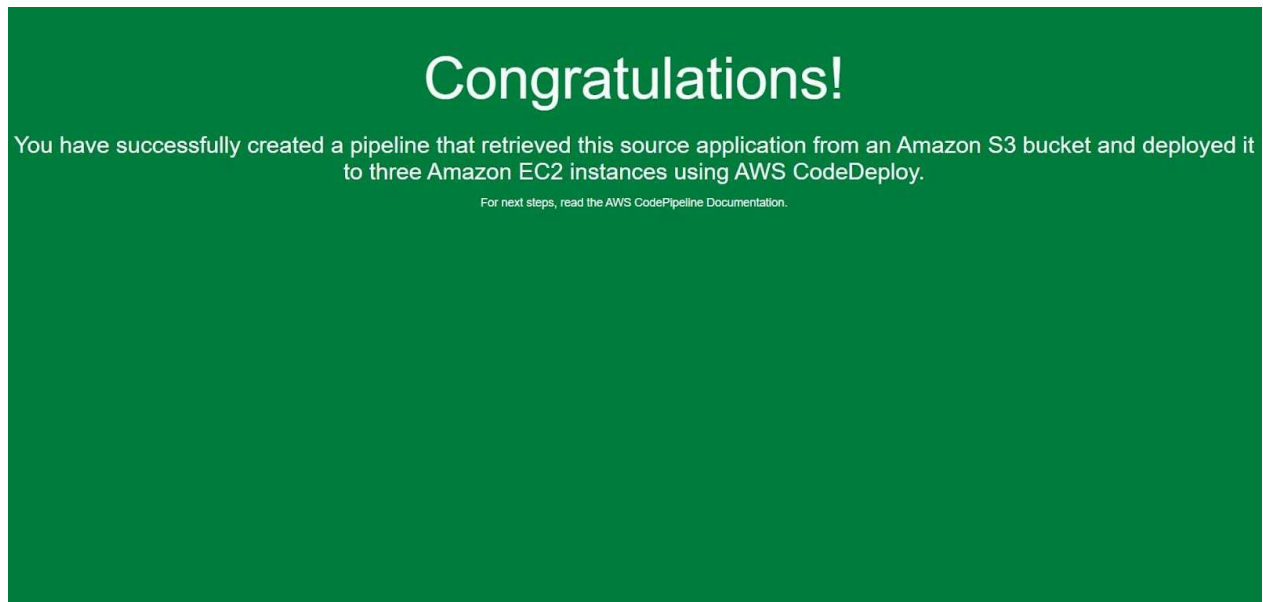
Step 19: Once the deployment is complete, click on the AWS Elastic Beanstalk under Deploy.



Step 20: This will redirect you to the application screen of Elastic Beanstalk. Click on the link shown under Domain.



Step 21: This will successfully show the sample website hosted.



Step 22: Now, we make some changes to the index.html file in the github.

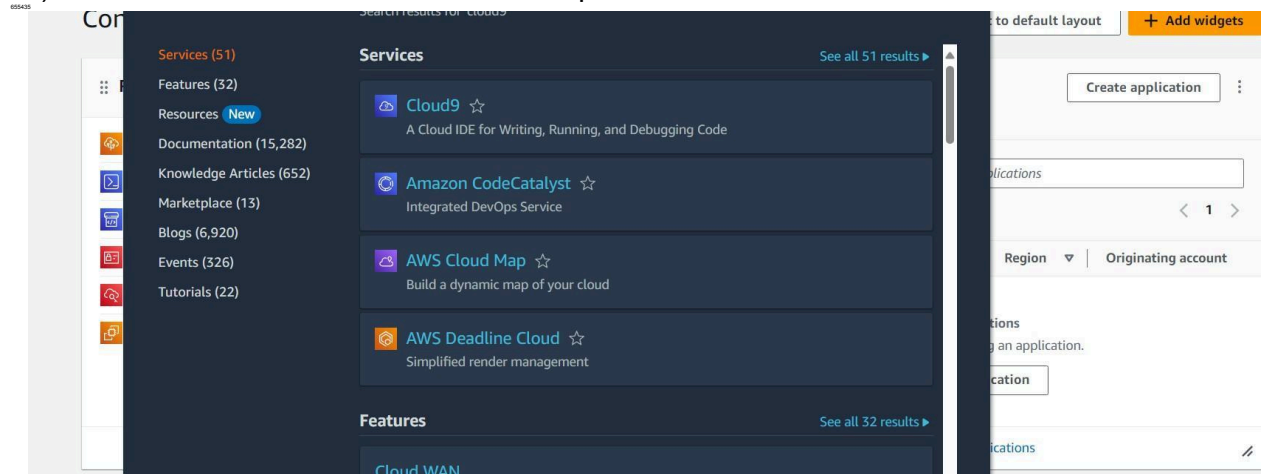
For eg: If you make some changes to the <h2>tag.

Once the changes are committed, when the website is refreshed, the changes can be seen.



Step 1: Set up Cloud9 environment.

1) Search Cloud9 in the services tab and open it

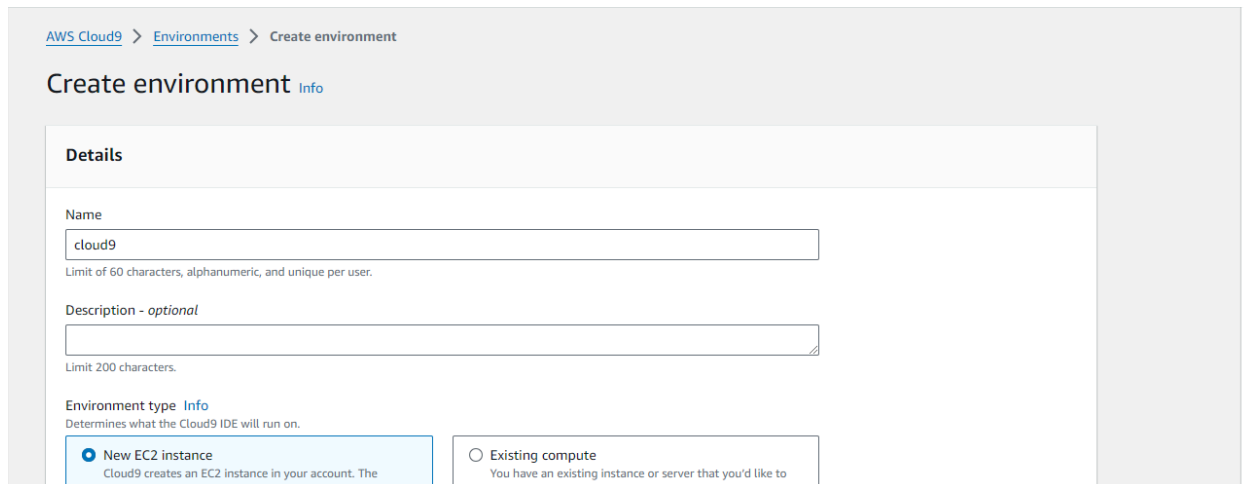


2) Click on Create Environment.

→ Give a name to your Cloud9 Environment. You can add a description if needed.

→ Select the option new EC2 instance if you do not have one ready for the environment. Give the specifications of that EC2 instance ahead

→ On the AWS Academy account, if we select AWS System Manager(SSM) in Network settings, it gives an error as the account does not have permissions to use the setting. So we select Secure Shell (SSH). After that click on Create



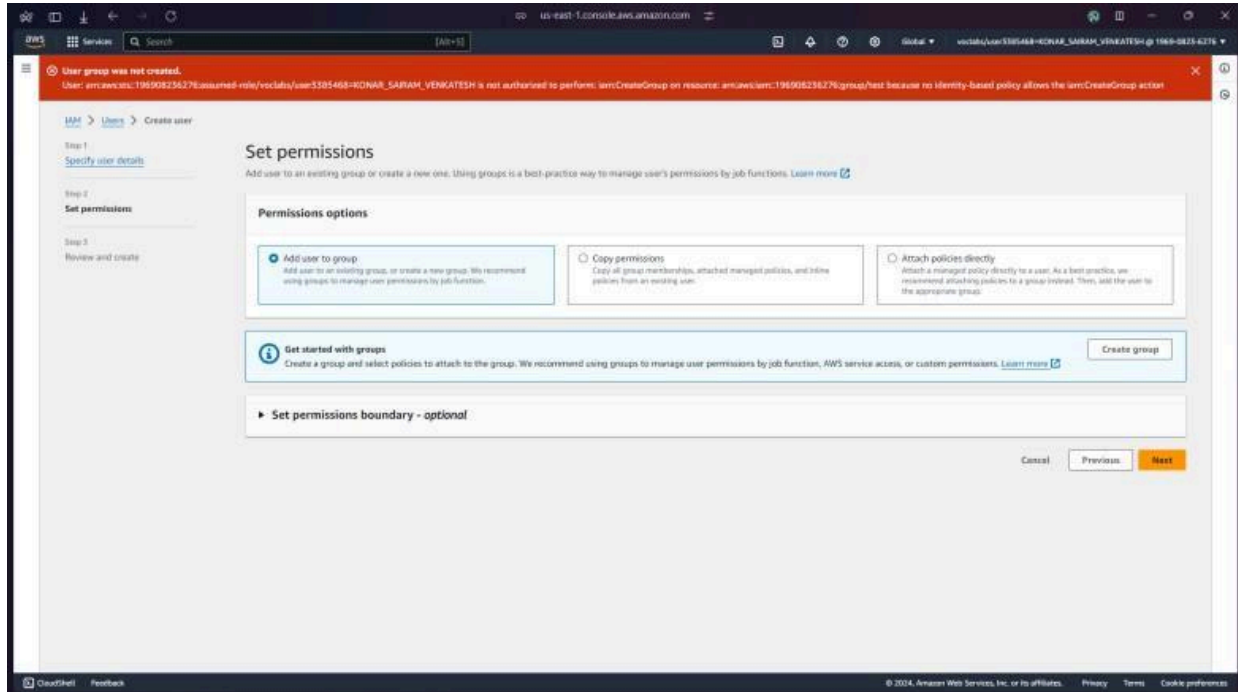
The screenshot shows the AWS Cloud9 console interface. On the left, there is a sidebar with 'AWS Cloud9' at the top, followed by 'My environments', 'Shared with me', 'All account environments', and 'Documentation'. The main content area has a breadcrumb trail 'AWS Cloud9 > Environments > cloud9'. At the top right of the main area are buttons for 'Delete' and 'Open in Cloud9'. Below this is a 'Details' section with an 'Edit' button. The details are organized into three columns: Name (cloud9), Description (-), and Environment type (EC2 instance); Owner ARN (arn:aws:sts::945236649319:assumed-role/voclabs/user5400206-GALAV_KAUSHAL_SAHEBRAO); Status (Ready); and Lifecycle status (Created). Below the details are tabs for 'EC2 instance', 'Network settings', and 'Tags'. The 'EC2 instance' tab is active, showing an 'EC2 instance' section with a 'Manage EC2 instance' button. The instance details include ARN (arn:aws:cloud9:us-east-1:945236649319:environment:1834653dbc8c4ff8a0528c1c62c19835), Instance type (t2.micro (1 GiB RAM + 1 vCPU)), Platform (Amazon Linux 2023), and Storage (EBS only).

6) The environment is being created.

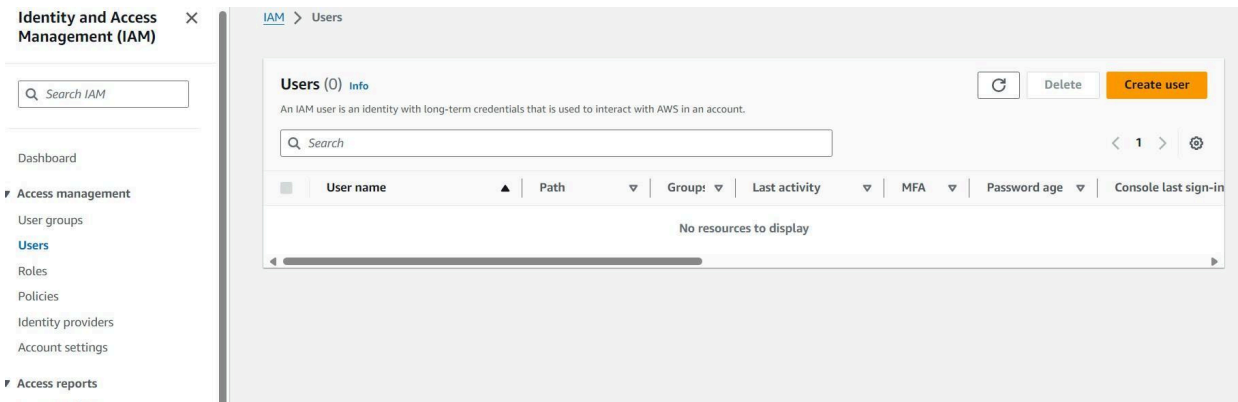
The screenshot shows the AWS Cloud9 console interface. On the left, there is a sidebar with 'AWS Cloud9' at the top, followed by 'My environments', 'Shared with me', 'All account environments', and 'Documentation'. The main content area has a breadcrumb trail 'AWS Cloud9 > Environments'. At the top right of the main area are buttons for 'Delete', 'View details', 'Open in Cloud9', and 'Create environment'. Below this is a section titled 'Environments (1)' with a search bar labeled 'My environments'. Below the search bar is a table with columns: Name, Cloud9 IDE, Environment type, Connection, Permission, and Owner ARN. The table contains one entry: 'cloud9' with a status of 'Open', Environment type of 'EC2 instance', Connection of 'Secure Shell (SSH)', Permission of 'Owner', and Owner ARN of 'arn:aws:sts::945236649319:assumed-role/voclabs/user5400206-GALAV_KAUSHAL_SAHEBRAO'.

Step 2: Creating IAM user.

When we go to add user to a group, the AWS Academy account throws an error as we do not have the permissions to create a group. So we have to use our personal AWS account for this part. ‘



1) Search IAM on the services search bar and open it. Click on Create User.



2) Give a username to your user and click Next.

The screenshot shows the 'Specify user details' step in the AWS IAM console. The breadcrumb trail is 'IAM > Users > Create user'. The left sidebar shows 'Step 1 Specify user details' as the active step, with 'Step 2 Set permissions' and 'Step 3 Review and create' below it. The main heading is 'Specify user details'. Under the 'User details' section, the 'User name' field contains 'kaushal'. Below the field, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and =, +, @, _ - (hyphen)'. There is an unchecked checkbox for 'Provide user access to the AWS Management Console - optional' with a note: 'If you're providing console access to a person, it's a best practice to manage their access in IAM Identity Center.' A blue information box contains a note: 'If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. Learn more'. At the bottom right are 'Cancel' and 'Next' buttons.

3) Select add User to Group. If there are no user groups on your accounts, you will have to create one. Click on Create Group.

The screenshot shows the 'Set permissions' step in the AWS IAM console. The breadcrumb trail is 'IAM > Users > Create user'. The left sidebar shows 'Step 1 Specify user details' and 'Step 2 Set permissions' as active steps, with 'Step 3 Review and create' below it. The main heading is 'Set permissions'. Below the heading, a note states: 'Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. Learn more'. Under the 'Permissions options' section, there are three radio button options: 'Add user to group' (selected), 'Copy permissions', and 'Attach policies directly'. Each option has a brief description. Below these options is a blue information box with the heading 'Get started with groups', a description: 'Create a group and select policies to attach to the group. We recommend using groups to manage user permissions by job function, AWS service access, or custom permissions. Learn more', and a 'Create group' button. At the bottom, there is a section for 'Set permissions boundary - optional'. At the bottom right are 'Cancel', 'Previous', and 'Next' buttons.

4) Give a name to your user group. Then click on Create User Group.

User group name

Enter a meaningful name to identify this group.

kaushalgroup

Maximum 128 characters. Use alphanumeric and '+','=','@','_.' characters.

Permissions policies (955) [Refresh](#) [Create policy](#)

Filter by Type

< 1 2 3 4 5 6 7 ... 48 > [Filter icon](#)

☐	Policy name ↗	Type	Use...	Description
☐	AdministratorAccess	AWS managed ...	None	Provides full access to AWS service:
☐	AdministratorAcce...	AWS managed	None	Grants account administrative perm
☐	AdministratorAcce...	AWS managed	None	Grants account administrative perm
☐	AlexaForBusinessD...	AWS managed	None	Provide device setup access to Alex
☐	AlexaForBusinessF...	AWS managed	None	Grants full access to AlexaForBusin
☐	AlexaForBusinessG...	AWS managed	None	Provide gateway execution access t

[Cancel](#) [Create user group](#)

5) The group is created and shown under the groups area, select the group by clicking on the checkbox. Then click Next.

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Step 2
Set permissions

Permissions options

☒ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1) [Refresh](#) [Create group](#)

< 1 > [Filter icon](#)

☐	Group name ↗	Users	Attached policies ↗	Created
☐	kaushalgroup	0	-	2024-08-11 ()

► **Set permissions boundary - optional**

[Cancel](#) [Previous](#) [Next](#)

6) Review all the Information, then click on Create user.

The screenshot shows the 'Review and create' step in the AWS IAM console. The left sidebar indicates 'Step 3: Review and create'. The main content area is divided into three sections:

- User details:** A table with three columns: 'User name' (value: kaushal), 'Console password type' (value: None), and 'Require password reset' (value: No).
- Permissions summary:** A section with a table header: 'Name', 'Type', and 'Used as'. The table body is empty, with a message 'No resources'.
- Tags - optional:** A section explaining that tags are key-value pairs for AWS resources. It states 'No tags associated with the resource.' and includes an 'Add new tag' button.

7) Open User Groups tab from the left side option. Click on the name of your group.

The screenshot shows the 'User groups' page in the AWS IAM console. The left sidebar shows 'Identity and Access Management (IAM)' with a search bar and a list of options: Dashboard, Access management (expanded), User groups (selected), Users, Roles, Policies, and Identity providers. The main content area shows 'User groups (1)' with a search bar and a table:

☐	Group name	Users	Permissions	Creation time
☐	kaushalgroup	0	Not defined	2 minutes ago

8) Go to permissions and click on Add permissions. Click on Attach Policies.

The screenshot shows the 'Permissions' page for the 'kaushalgroup' in the AWS IAM console. The left sidebar is the same as in the previous screenshot. The main content area shows the 'Summary' section with details: 'User group name: kaushalgroup', 'Creation time: August 11, 2024, 18:14 (UTC+05:30)', and 'ARN: arn:aws:iam::011528263357:group'. Below this is the 'Permissions policies (0)' section with a search bar, a 'Filter by Type' dropdown (set to 'All types'), and a table header: 'Policy name', 'Type', and 'Attached entities'. The table body is empty, with a message 'No resources to display'.

9) Search for AWSCloud9EnvironmentMember, select it and click on Attach policies

► Current permissions policies (0)

Other permission policies (1/953)

You can attach up to 10 managed policies to this user group. All of the users in this group inherit the attached permissions.

Filter by Type

Q AWSCloud9 All types 4 matches

	Policy name	Type	Used as	Description
☐	AWSCloud9Administrator	AWS managed	None	Provides administrator access to AWS Clo...
☒	AWSCloud9EnvironmentMember	AWS managed	None	Provides the ability to be invited into AW...
☐	AWSCloud9SSMInstanceProfile	AWS managed	Permissions policy (1)	This policy will be used to attach a role o...
☐	AWSCloud9User	AWS managed	None	Provides permission to create AWS Cloud...

Cancel Attach policies

10) The policies have been attached

Dashboard

▼ Access management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings

▼ Access reports

- Access Analyzer
- External access
- Unused access
- Analyzer settings
- Credential report
- Organization activity

Summary Edit

User group name: kaushalgroup

Creation time: August 11, 2024, 18:14 (UTC+05:30)

ARN: arn:aws:iam::011528263337:group

Users Permissions Access Advisor

Permissions policies (1) info

You can attach up to 10 managed policies.

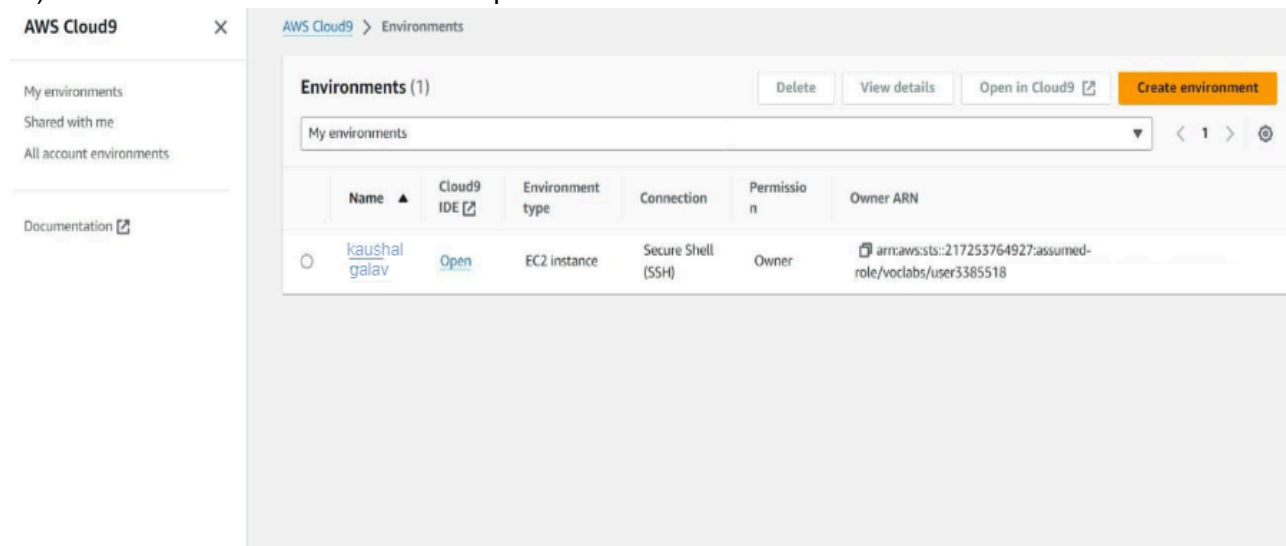
Filter by Type

Q Search All types 1

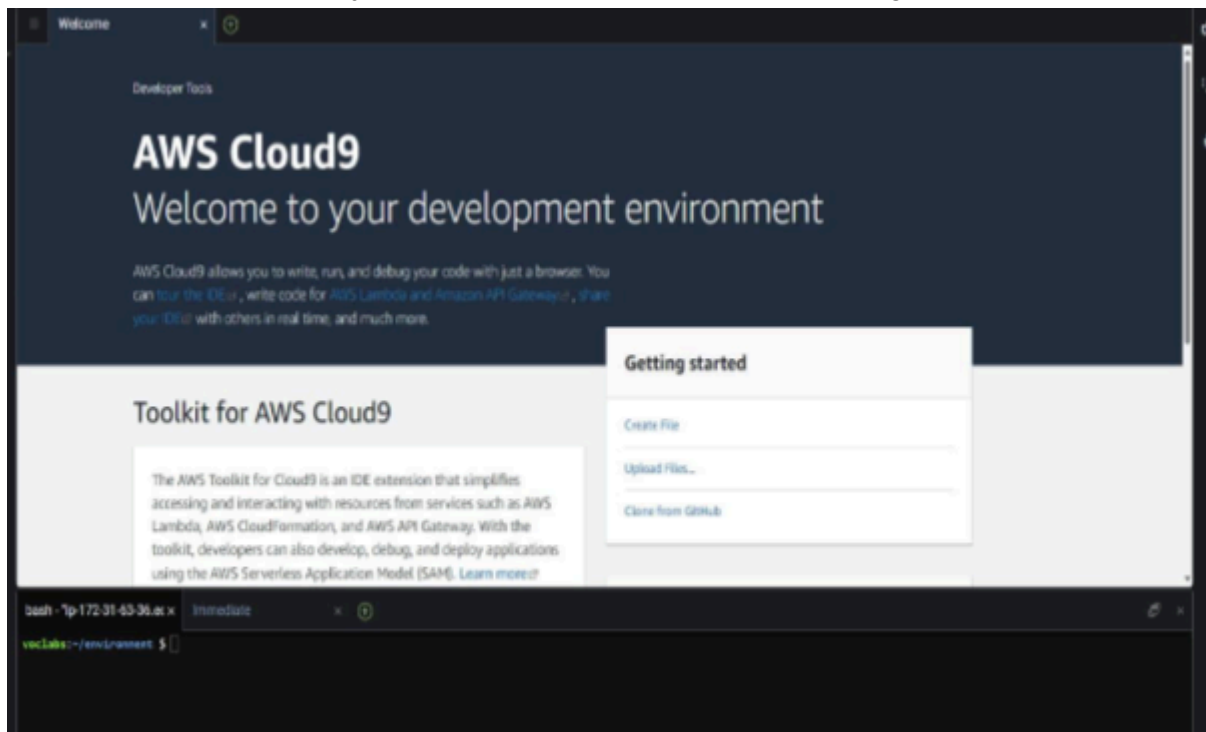
	Policy name	Type	Attached entities
☐	AWSCloud9EnvironmentMember	AWS managed	1

Step 3: Working on Cloud9 IDE

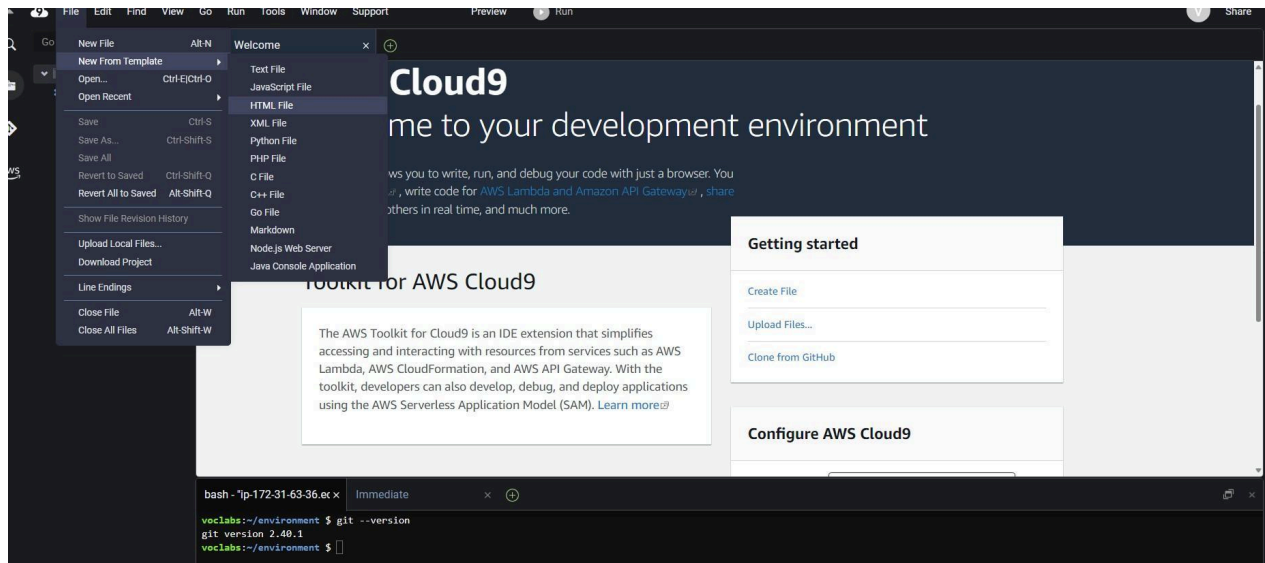
1) Go to Cloud9 services. Click on Open under Cloud9 IDE.



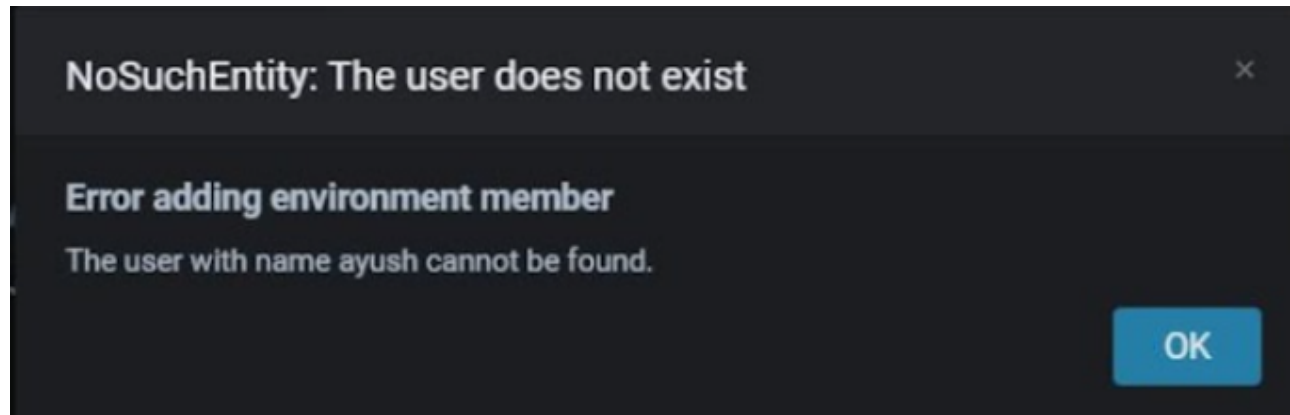
2) This is the Cloud9 IDE interface. The major part of the screen is the coding IDE. There is a command console just below it. For example, the command `git --version` is run



3) To add a file, click on file. For this experiment, we are to add an HTML file. So go to File → New From Template → HTML file. This gives a basic HTML template on the coding



Make a basic website on the HTML template and save it. After saving, on the toolbar towards the far right, click on Share. Then put the username that you had put during creating IAM user. Here, it gives an error as Cloud9 was created on the academy account where creating an IAM group is not available, meanwhile on the personal account, the services of Cloud9 have been deprecated. So currently, it is not possible to integrate the cloud9 and IAM parts of the experiment.



Static Hosting:

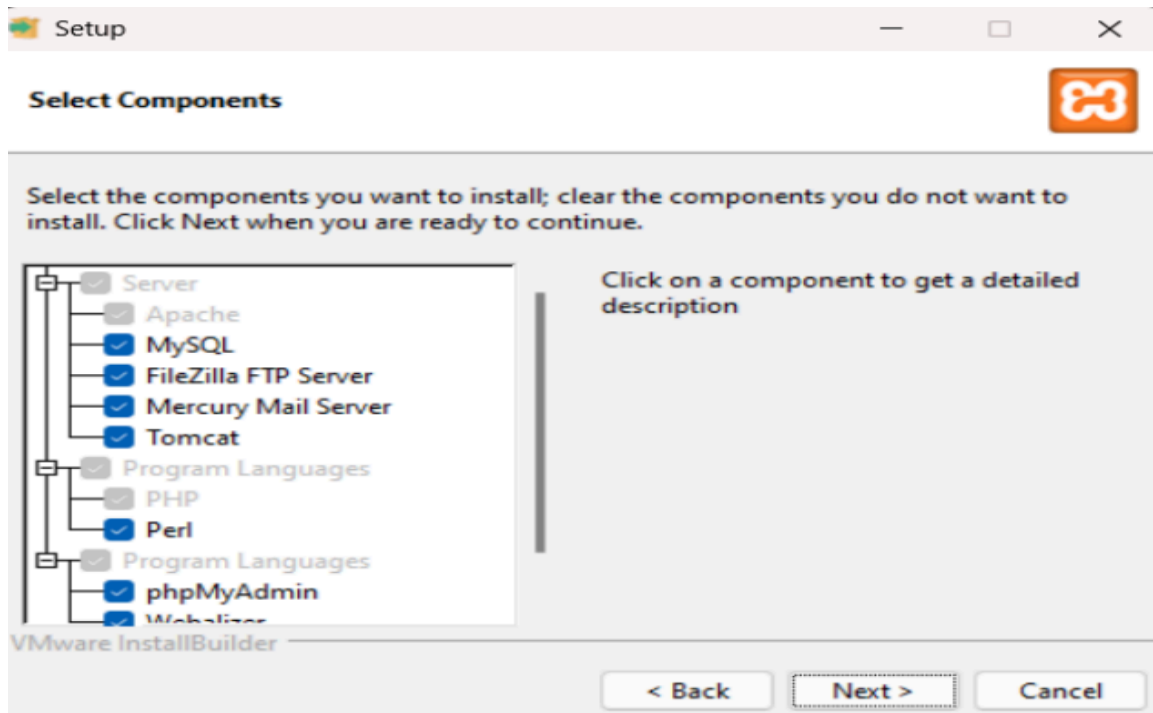
1) On local server (XAMPP)

Step 1: Install XAMPP from <https://www.apachefriends.org/>

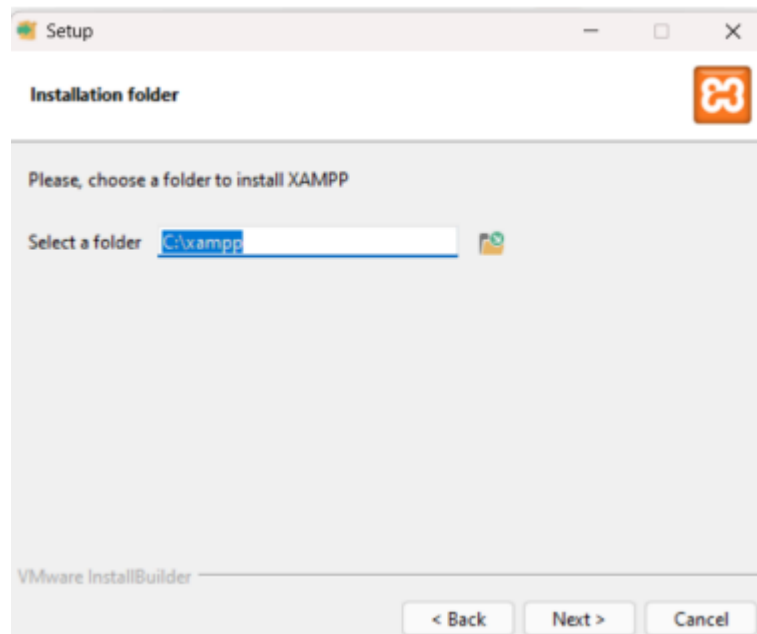
1) Select your OS. It will automatically start downloading.



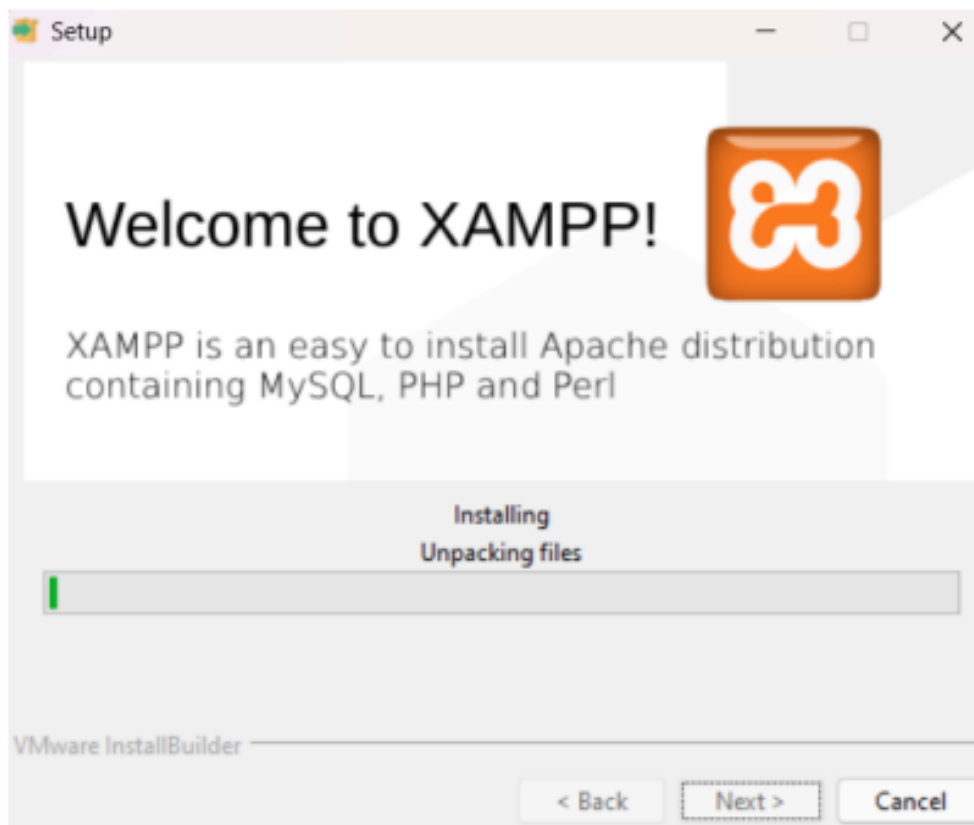
2) Open the setup file. Select all the required components and click next



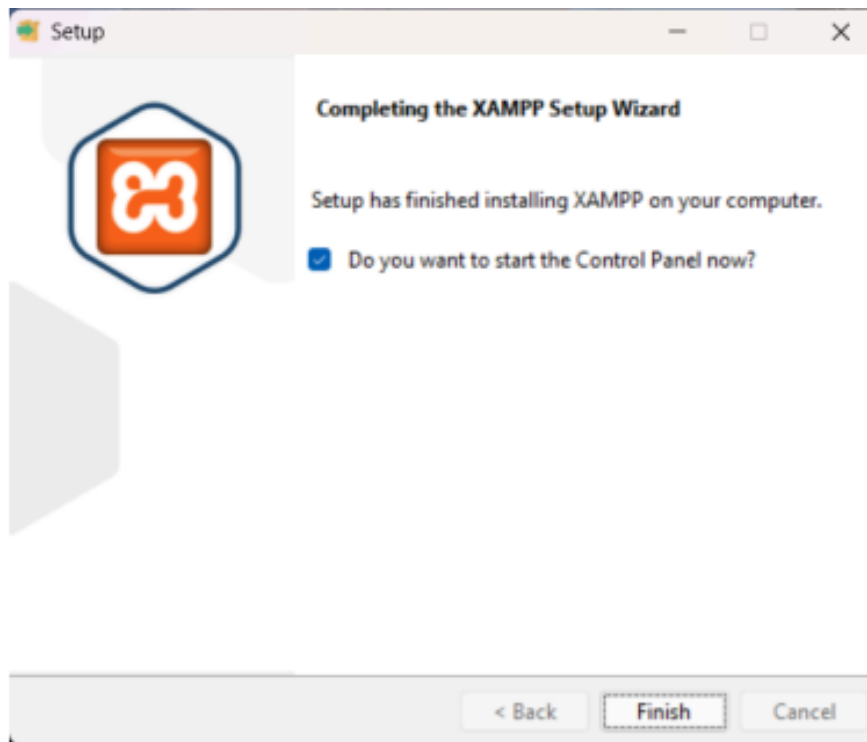
3) Choose the folder to install XAMPP in. Make sure the folder is empty. Click next



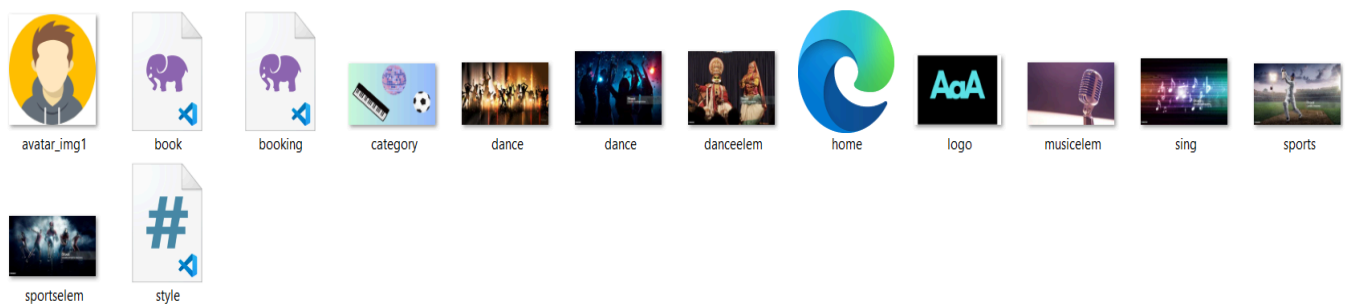
4) Select the language, click next. XAMPP starts to install



5) The installation is complete. Click Finish



Step 2: Setup a file that is to be hosted on the server. Make sure the file has extension .php



Step 3: Go to the directory where XAMPP was installed. Go to htdocs folder. Place your folder in this directory.

dashboard	05-08-2024 00:16	File folder	
file	07-01-2024 23:08	File folder	
img	05-08-2024 00:16	File folder	
webalizer	05-08-2024 00:16	File folder	
xampp	05-08-2024 00:16	File folder	
applications	15-06-2022 21:37	Microsoft Edge HT...	4 KB
bitnami	15-06-2022 21:37	CSS Source File	1 KB
favicon	16-07-2015 21:02	ICO File	31 KB
index	16-07-2015 21:02	PHP Source File	1 KB

Step 4: Open XAMPP Control Panel, start the Apache service (Required) and mySQL service (if needed)

XAMPP Control Panel v3.3.0 [Compiled: Apr 6th 2021]

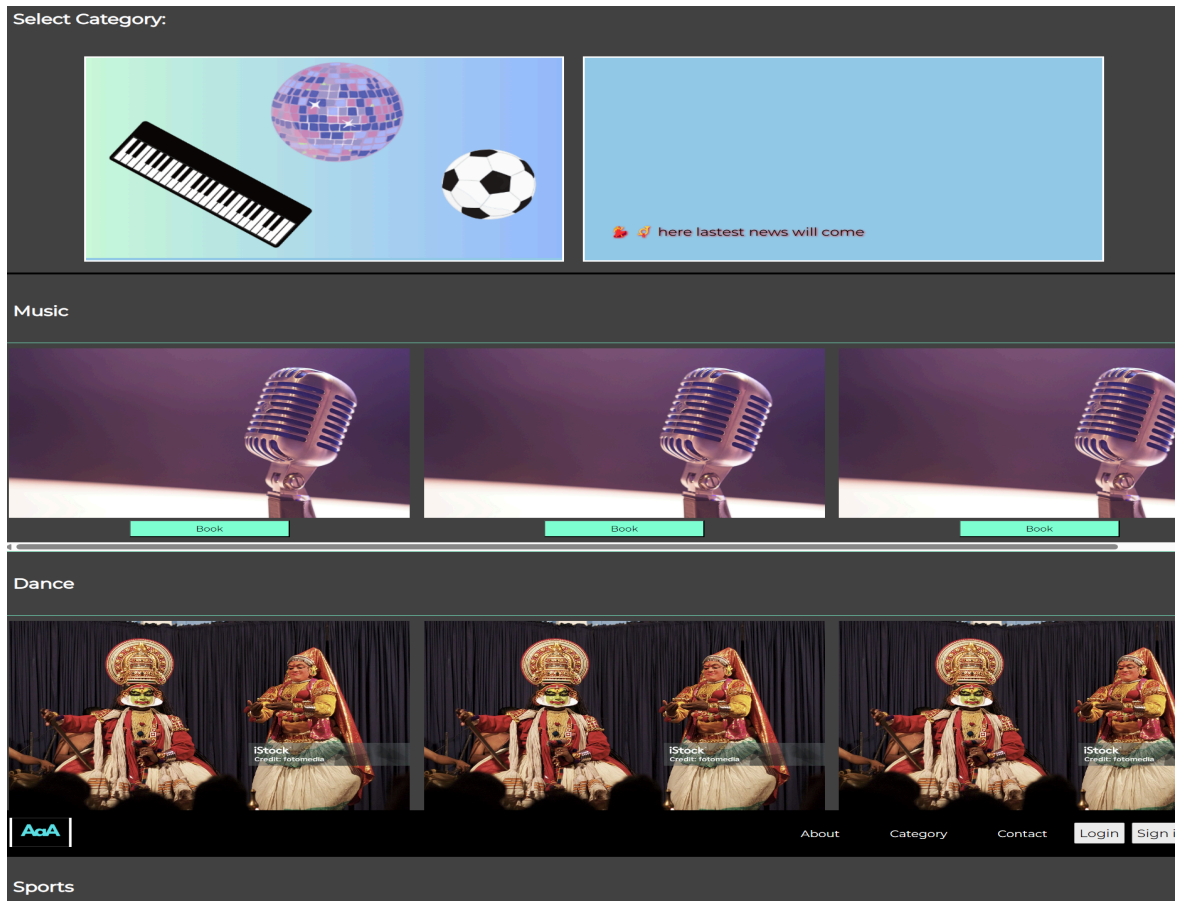
XAMPP Control Panel v3.3.0

[Config](#)
[Netstat](#)
[Shell](#)
[Explorer](#)
[Services](#)
[Help](#)
[Quit](#)

Service	Module	PID(s)	Port(s)	Actions
	Apache	19328 16548	80, 443	Stop Admin Config Logs
	MySQL			Start Admin Config Logs
	FileZilla			Start Admin Config Logs
	Mercury			Start Admin Config Logs
	Tomcat	8196	8080	Stop Admin Config Logs

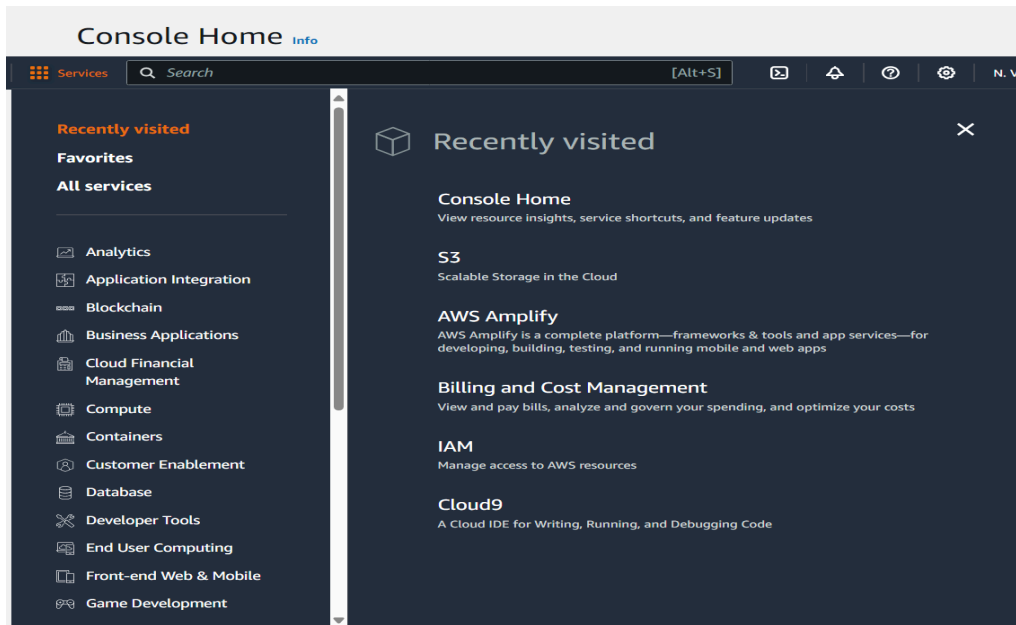
00:32:18 [mysql] Press the Logs button to view error logs and check
00:32:18 [mysql] the Windows Event Viewer for more clues
00:32:18 [mysql] If you need more help, copy and post this
00:32:18 [mysql] entire log window on the forums
00:32:40 [Apache] Attempting to stop Apache (PID: 2636)
00:32:40 [Apache] Attempting to stop Apache (PID: 13212)
00:32:41 [Apache] Status change detected: stopped
00:32:42 [Apache] Attempting to start Apache app...
00:32:42 [Apache] Status change detected: running

Step 5: Open your web browser. Type localhost/YOUR_FILENAME.php. This will open your website on your browser.

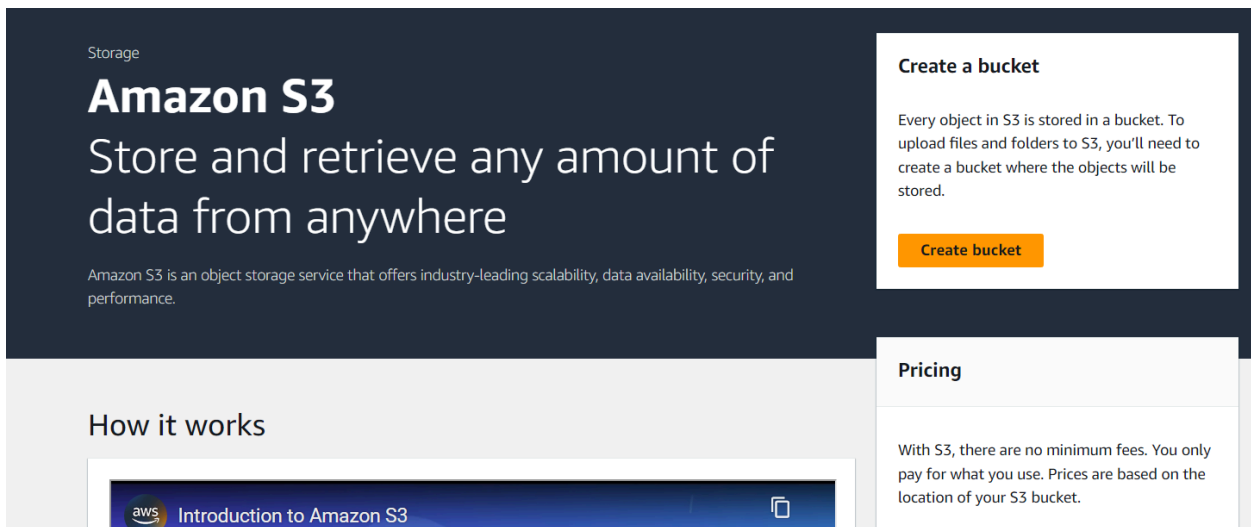


2) AWS S3

Step 1: Login to your AWS account. Go to services and open S3.



Step 2: Click on Create Bucket



Step 3: Give a name to your bucket, keeping other options default, scroll down and click on Create Bucket

[Amazon S3](#) / [BUCKETS](#) / Create bucket

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory - New**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Step 4: Click on the name of your bucket and goto Properties

► **Account snapshot** - updated every 24 hours [All AWS Regions](#)

Storage lens provides visibility into storage usage and activity trends. [Learn more](#)

[View Storage Lens dashboard](#)

General purpose buckets | Directory buckets

General purpose buckets (1) [Info](#) [All AWS Regions](#)

Buckets are containers for data stored in S3.

< 1 > ⚙️

	Name	AWS Region	IAM Access Analyzer	Creation date
☐	bucket-aws34	US East (N. Virginia) us-east-1	View analyzer for us-east-1	August 5, 2024, 01:15:15 (UTC+05:30)

Objects | **Properties** | Permissions | Metrics | Management | Access Points

Bucket overview

AWS Region US East (N. Virginia) us-east-1	Amazon Resource Name (ARN) arn:aws:s3:::bucket-aws34	Creation date August 5, 2024, 01:15:15 (UTC+05:30)
--	--	--

Bucket Versioning [Edit](#)

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

Disabled

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API.

Step 5: Scroll down till you find Static website hosting, click on edit

Requester pays
Edit

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. [Learn more](#)

Requester pays
Disabled

Static website hosting
Edit

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting
Disabled

Step 6: Enable static website hosting, in Index document, write the name of your document and in error document, give name as 404.html. Save your changes.

Edit static website hosting
Info

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting
☐ Disable
☒ Enable

Hosting type
☒ Host a static website
Use the bucket endpoint as the web address. [Learn more](#)
☐ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document
Specify the home or default page of the website.
home.html

Error document - optional
This is returned when an error occurs.
404.html

Redirection rules - optional
Redirection rules, written in JSON, automatically redirect web browser requests for specific content. [Learn more](#)

Step 7: Go to Objects tab and click on upload file.

Objects
Properties
Permissions
Metrics
Management
Access Points

Objects (0)
Info
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload

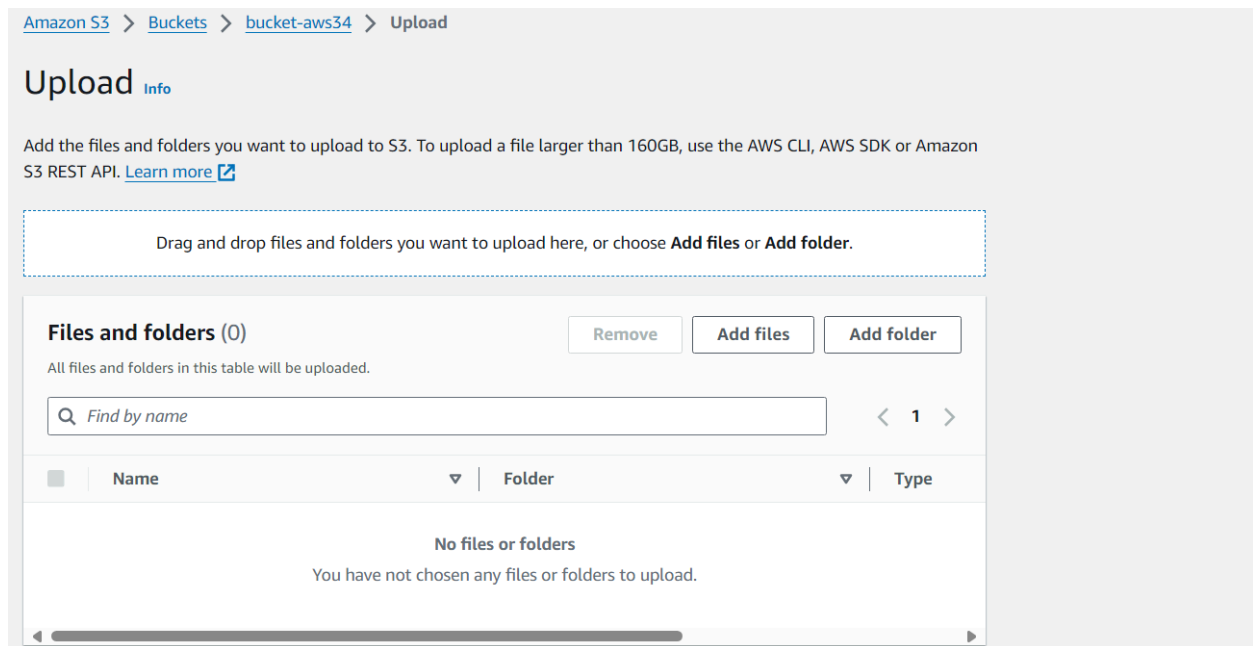
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix
1

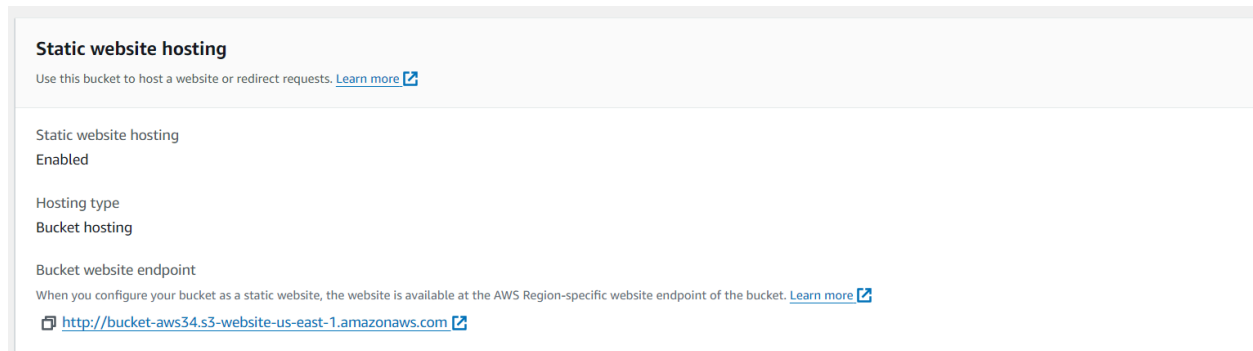
Name
Type
Last modified
Size
Storage class

No objects
You don't have any objects in this bucket.
Upload

Step 8: Click on Add files. Add all the files you want to upload. Then scroll down and click on Upload



Step 9: This will take you to the Objects screen. Switch to Properties, scroll down to Static web hosting. There you would find the link (Bucket website endpoint) to your website.



Step 10: Open the link. It will show a 403 forbidden error screen as the contents of the bucket are not available for the public users. To change this, go to Permissions tab, go to Block public access and click on edit

403 Forbidden

- Code: AccessDenied
- Message: Access Denied
- RequestId: DMNXJYFFWHMDVKEX
- HostId: shEGRFa01y+WK3nJfA/qRrcJSIS/sWWEf6VVQfy1P4v8Kz4WkLrdYN8rUtXgFzVtzbXZkziVxo=

An Error Occurred While Attempting to Retrieve a Custom Error Document

- Code: AccessDenied
- Message: Access Denied

Step 11: Uncheck the Block all public access checkbox and click on save changes

Edit Block public access (bucket settings) [Info](#)

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block *all* public access**

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through *new* access control lists (ACLs)**

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ **Block public access to buckets and objects granted through *any* access control lists (ACLs)**

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects granted through *new* public bucket or access point policies**

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Step 12: Scroll down to bucket policy and click edit

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Edit

Delete

No policy to display.

Copy

Step 13:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
```

```
"Sid": "PublicReadGetObject",
"Effect": "Allow",
"Principal": {
  "AWS": "*"
},
"Action": "s3:GetObject",
"Resource": "arn:aws:s3:::YOUR-BUCKET-NAME-HERE/*"
}
]
}
```

Paste this code snippet in the policy textarea. Replace YOUR-BUCKET-NAME-HERE with the name you have given to your bucket. Save the changes.

Policy

```
1 ▼ {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "PublicReadGetObject",
6       "Effect": "Allow",
7       "Principal": {
8         "AWS": "*"
9       },
10      "Action": "s3:GetObject",
11      "Resource": "arn:aws:s3:::bucket-aws34/*"
12    }
13  ]
14 }
15
```

Edit

Sel

Step 14: Now reload the website. You can see your website

- [About](#)
- [Features](#)
- [Contact](#)

 logo

Login

Sign in



Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod

< >

Select Category:

 Category

 here latest news will come

Music



Book



Book



Book